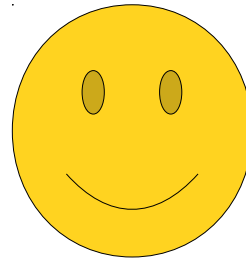
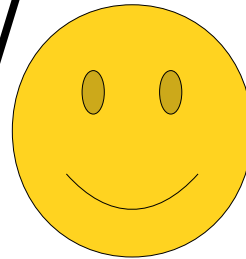


CS101B Debugger Tutorial

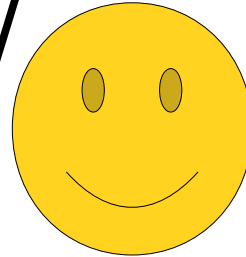
Hi everybody!



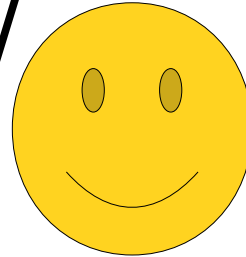
As part of Assignment 0, we'd like you to get a little bit of practice using the debugger in VS Code.



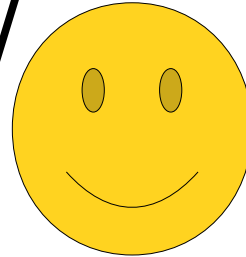
The debugger is a tool you can use to help see what your program is doing as you run it.



It's really useful for helping find errors in your programs, and the more practice you get with it, the easier it'll be to correct mistakes in the programs you write.



Think of this guide as a little tutorial walkthrough to help give you a sense of how to use the debugger and how to make sense of what you're seeing.





EXPLORER

ASSIGNMENT 0 DEBUGGER TUTORIAL

CMakeLists.txt

NameHash.cpp

To start things off, open the Name Hash project you ran in VS Code.



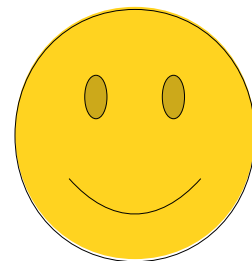
Show All Commands **Ctrl** + **Shift** + **P**

Go to File **Ctrl** + **P**

Open Settings **Ctrl** + **,**

Find in Files **Ctrl** + **Shift** + **F**

Start Debugging **F5**



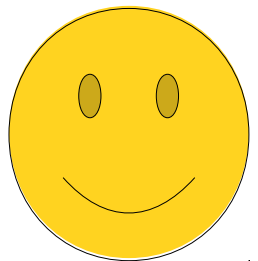
OUTLINE

TIMELINE

CMAKE

- PROJECT STATUS
 - Folder
 - Assignment 0 Debugger Tut...
 - Configure
 - GCC 14.2.0 x86_64-w64-min...
 - Debug
 - Build
 - all
 - Test
 - all
- PROJECT OUTLINE
 - Assignment-0
 - Assignment-0 (Executab...
 - CMakeLists.txt
- PINNED COMMANDS
 - Configure Task
 - Run Task

Configure the project in the CMake panel. Check VS Code Workflow if you don't know how to do it.



- Show All Commands **Ctrl + Shift + P**
- Go to File **Ctrl + P**
- Open Settings **Ctrl + ,**
- Find in Files **Ctrl + Shift + F**
- Start Debugging **F5**

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter CMake/Build

```
[cmake] Not searching for unused variables given on the command line.
[cmake] -- Configuring done (0.4s)
[cmake] -- Generating done (0.1s)
[cmake] -- Build files have been written to: C:/Users/xuehao/Desktop/build-cache/Assignment 0 Debugger Tutorial
```



CMAKE

PROJECT STATUS

PROJECT OUTLINE

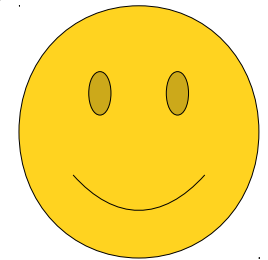
- Assignment-0
 - Assignment-0 (Executab...
 - References
 - CMakeLists.txt
 - NameHash.cpp**
 - CMakeLists.txt

PINNED COMMANDS

NameHash.cpp

NameHash.cpp

Open up the Name Hash program. Scroll down to the nameHash function so that you can see the entire function in your window.



```

1  /**
2   * This program demonstrates how to use the nameHash function.
3   * a program that takes a first and last name and returns a unique hash value.
4   * The hash value is used to index into an array of student records.
5   * In this program, we will use the nameHash function to calculate the hash value.
6   * is the first name and last is the last name.
7   * is the first name and last is the last name.
8   * While the program is running, you will see the hash value for each name.
9   * unique hashes (different names can hash to the
10  * same value), there is a relatively low possibility
11  * of a "collision" with a class of roughly 325 students.
12  *
13  * We will learn more about hashing later this quarter!
14  */
15
16  #include <iostream>
17  #include <string>
18  #include "console.h"
19  #include "simpio.h" // for getLine
20  using namespace std;
21
22  /* Prototype for the nameHash function. This lets us use the function
23   * in main and then define it later in the program.
24   */
25  int nameHash(string first, string last);
26
27  int main() {
28      string first = getLine("What is your first name? ");
29      string last = getLine("What is your last name? ");
30
31      int hashValue = nameHash(first, last);
32  
```

CMAKE

> PROJECT STATUS

> PROJECT OUTLINE

Assignment-0

Assignment-0 (Executab...

References

CMakeLists.txt

NameHash.cpp

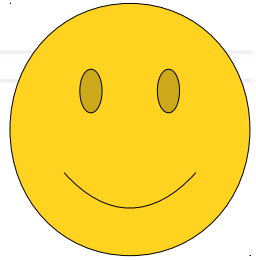
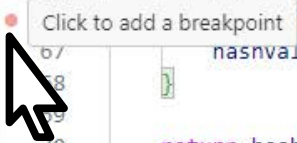
CMakeLists.txt

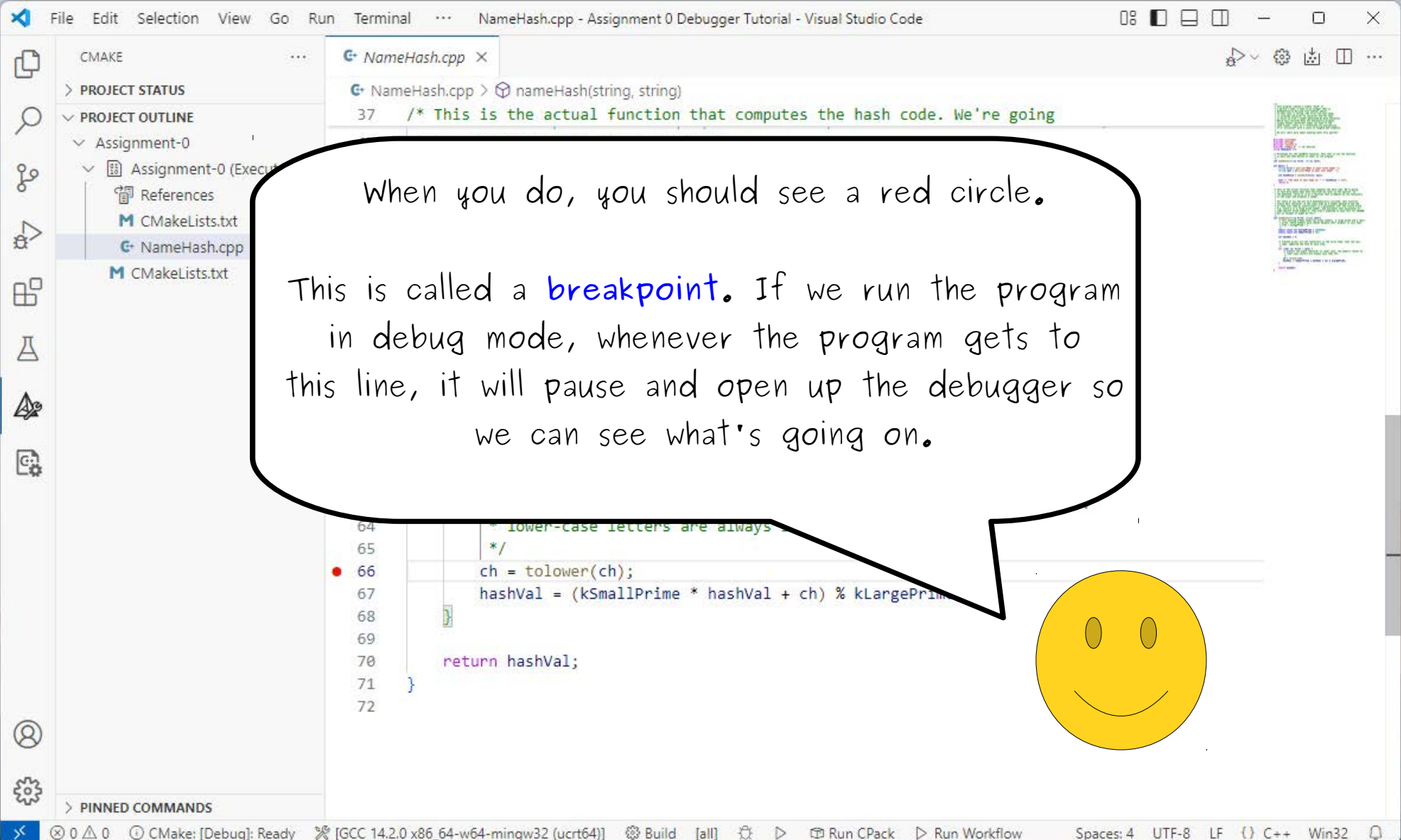
PINNED COMMANDS

```
NameHash.cpp x
NameHash.cpp > nameHash(string, string)
37  /* This is the actual function that computes the hash code. We're going
47  * but we thought it might be fun!)
48  */
49  int nameHash(string first, string last){
50
62      for (char ch: first + last) {
63          /* Convert the input character to lower case.
64          * lower-case letters are always less than 127.
65          */
66          lower(ch);
67          hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68
69
70      return hashVal;
71  }
72  }
```

Move your mouse cursor so that it's in the space right before the line number for line 66.

Now, click the mouse!

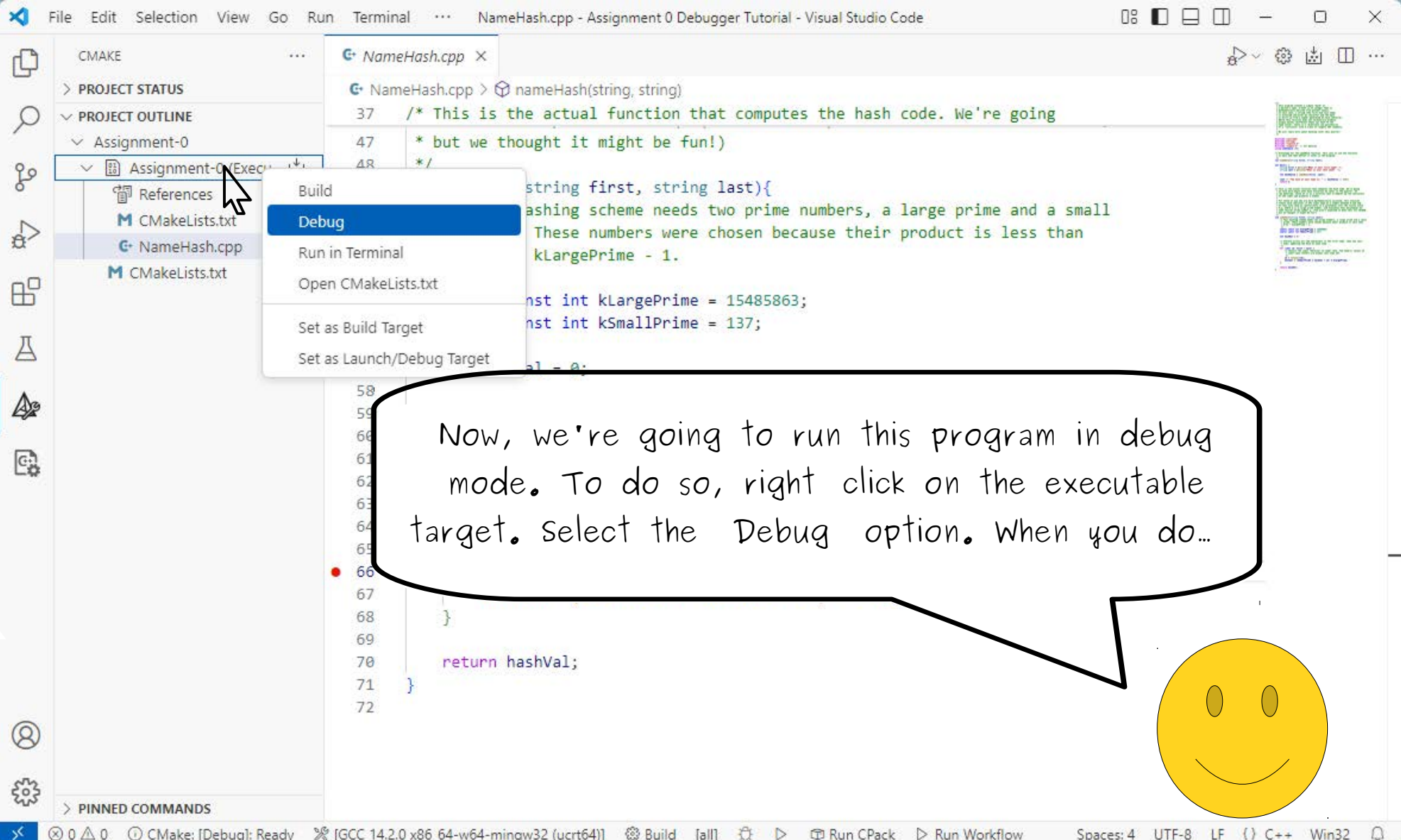


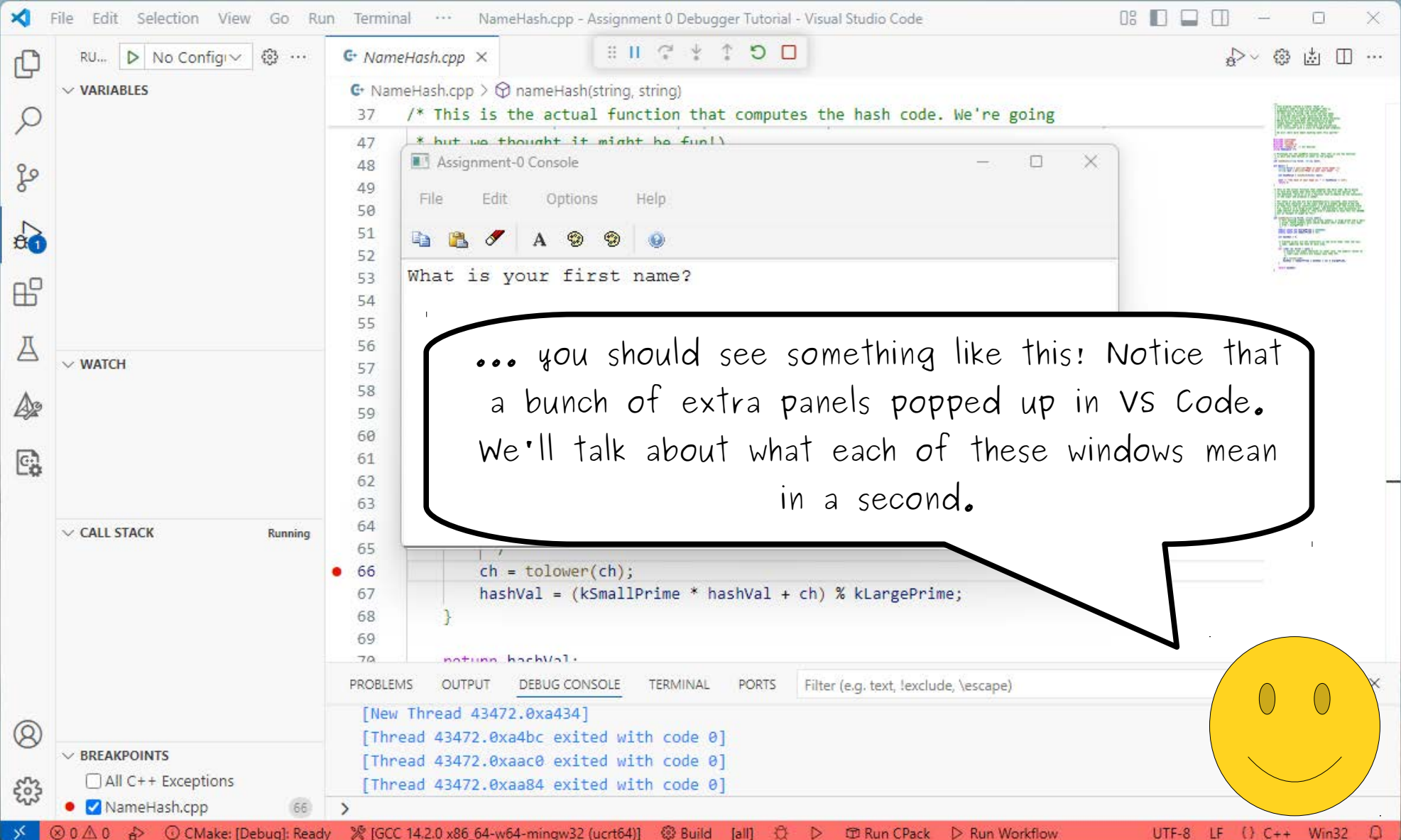


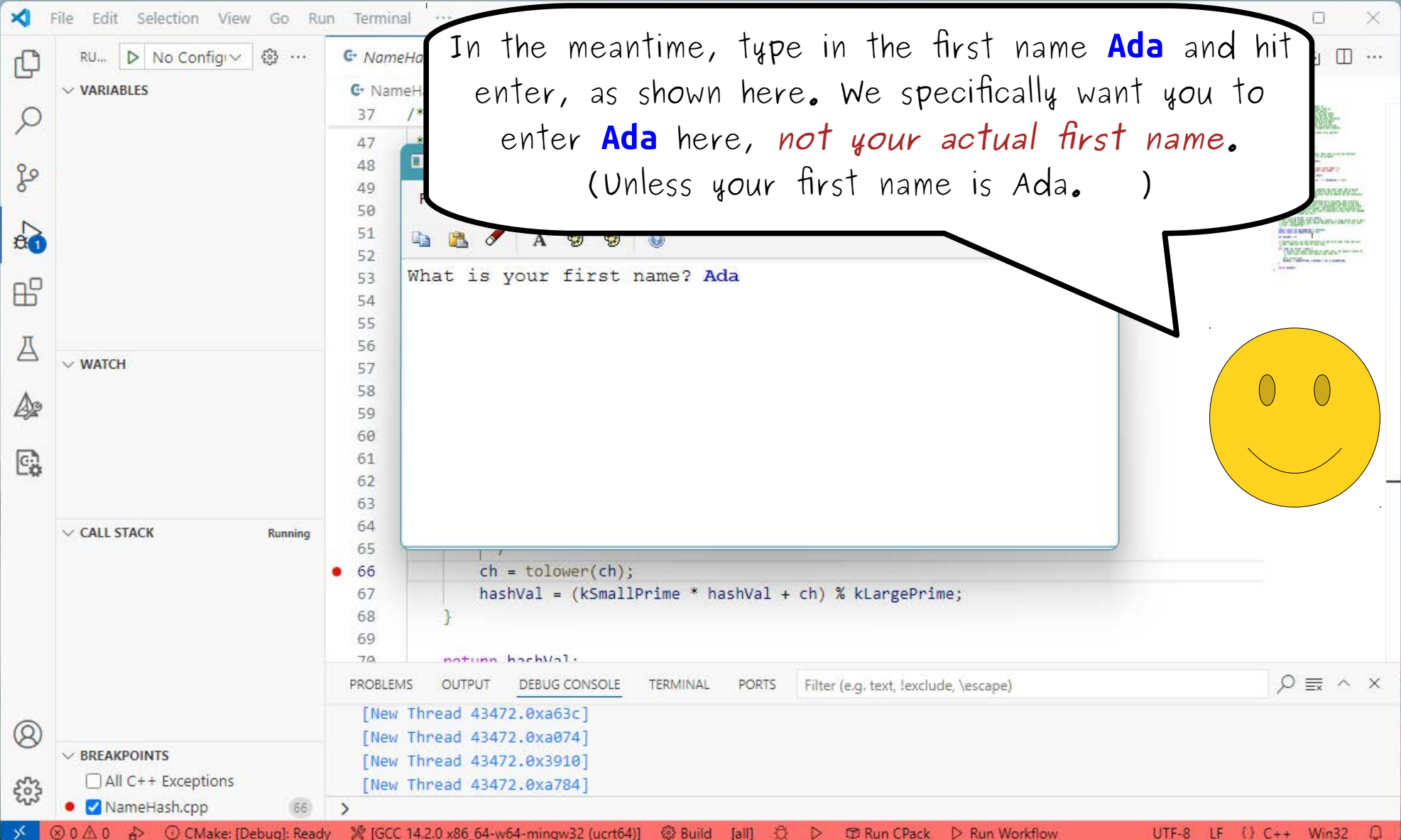
When you do, you should see a red circle.

This is called a **breakpoint**. If we run the program in debug mode, whenever the program gets to this line, it will pause and open up the debugger so we can see what's going on.









In the meantime, type in the first name **Ada** and hit enter, as shown here. We specifically want you to enter **Ada** here, *not your actual first name*. (Unless your first name is Ada.)



What is your first name? Ada

```
ch = tolower(ch);  
hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;  
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !exclude, \escape)

[New Thread 43472.0xa63c]
[New Thread 43472.0xa074]
[New Thread 43472.0x3910]
[New Thread 43472.0xa784]

RU... No Config...

VARIABLES

WATCH

CALL STACK

Running

BREAKPOINTS

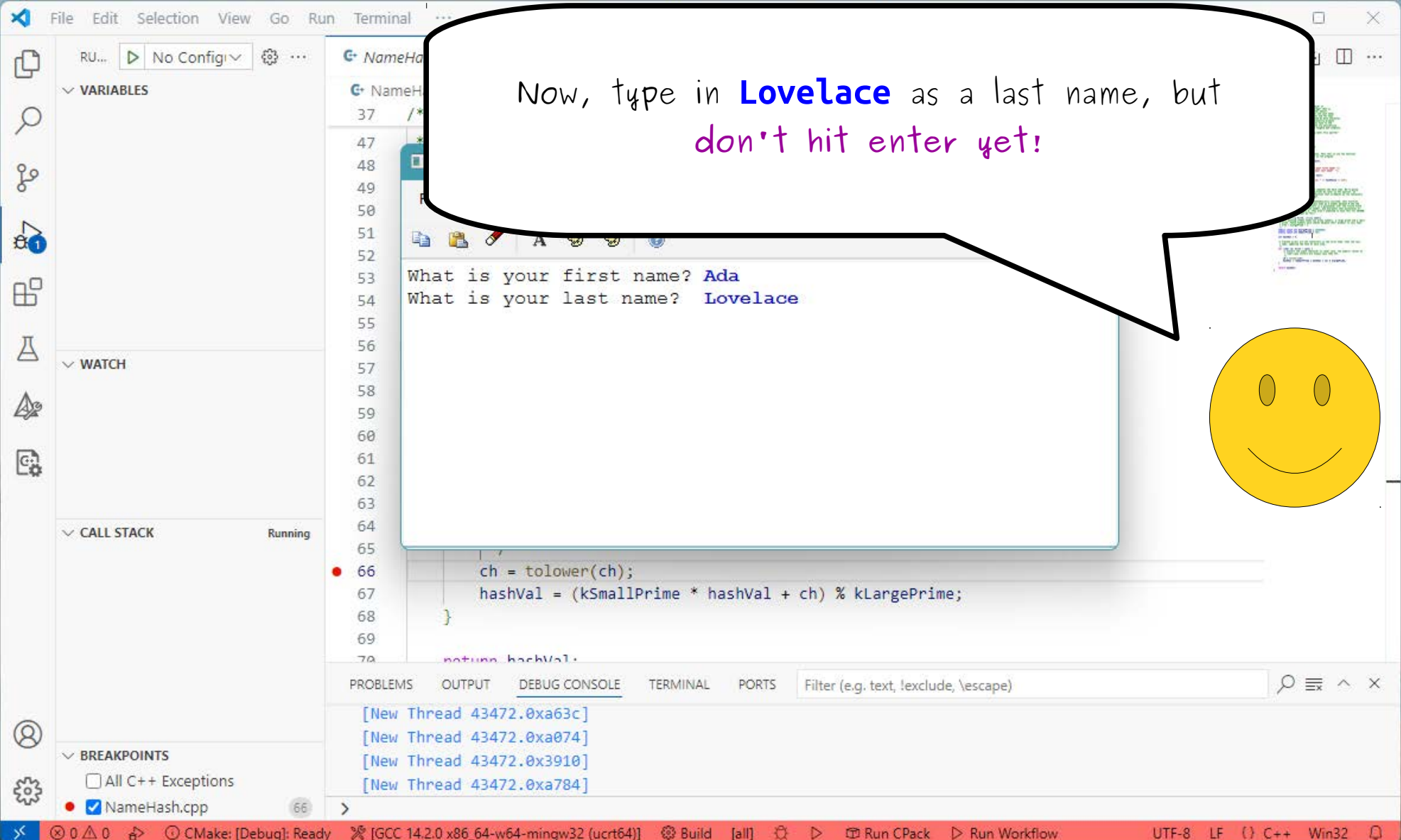
☐ All C++ Exceptions

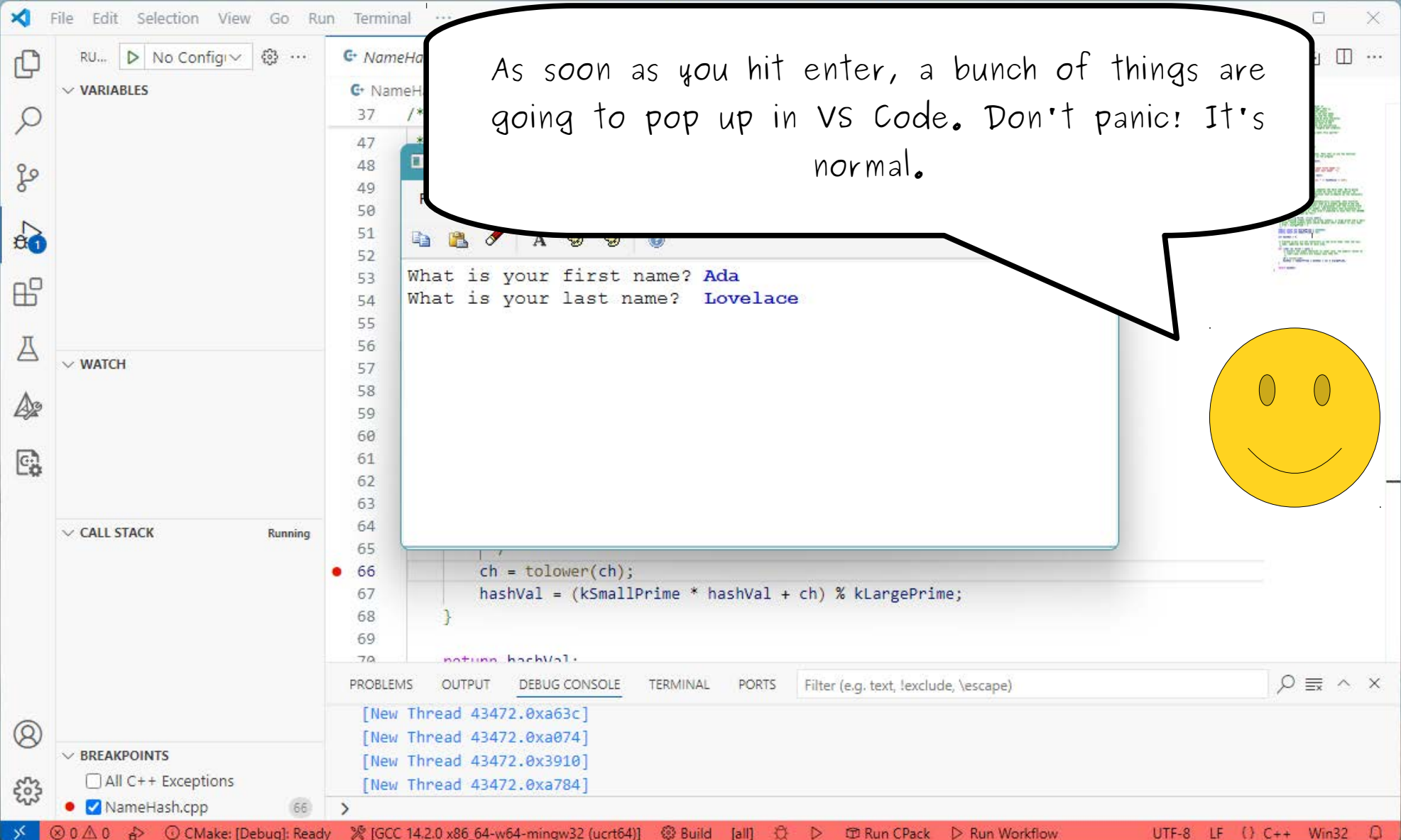
☒ NameHash.cpp

66

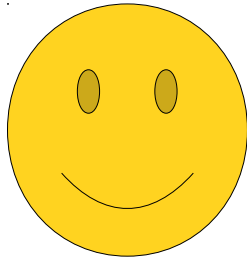
CMake: [Debug]: Ready [GCC 14.2.0 x86_64-w64-mingw32 (ucrt64)] Build [all] Run CPack Run Workflow

UTF-8 LF C++ Win32





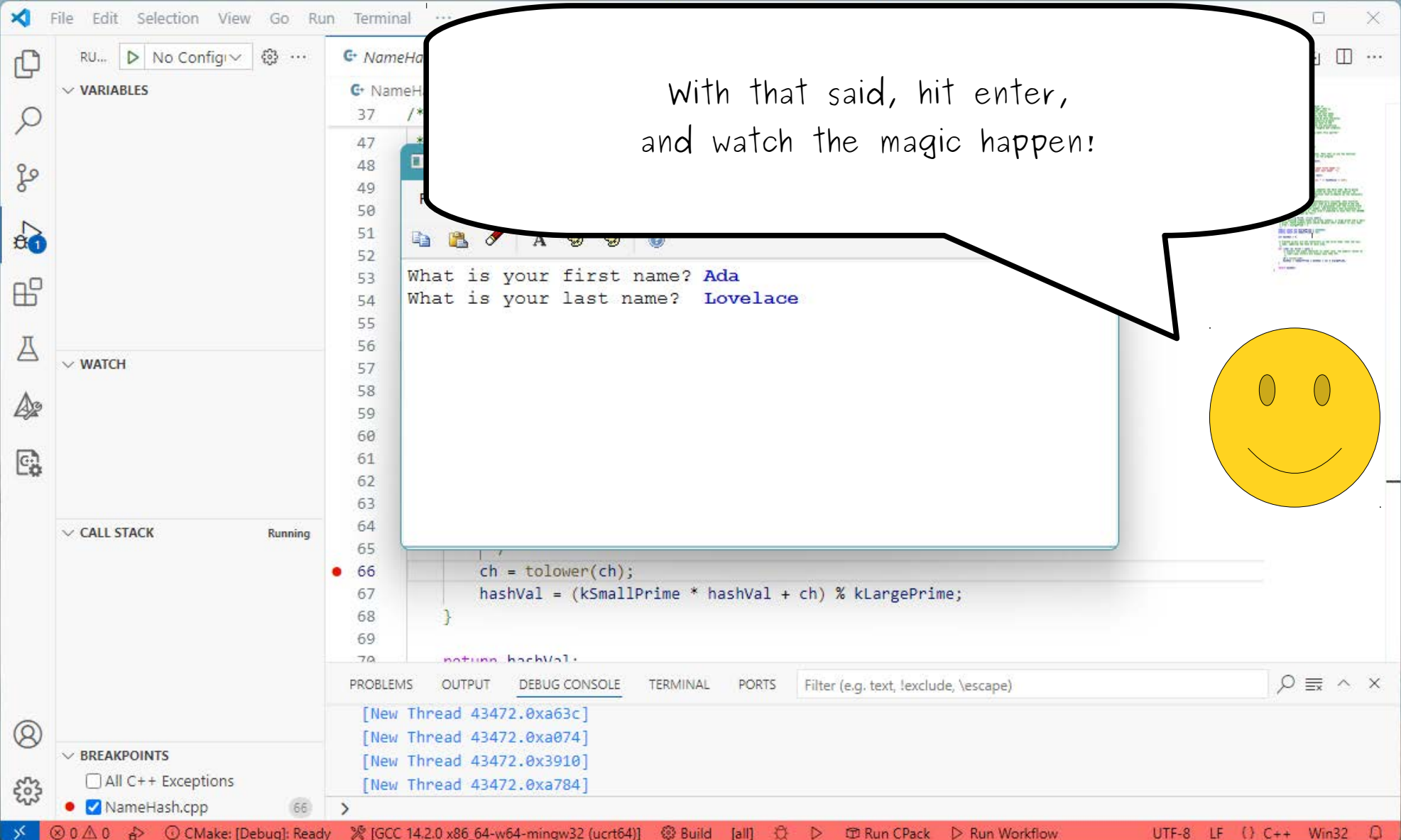
As soon as you hit enter, a bunch of things are going to pop up in VS Code. Don't panic! It's normal.



```
37 /*  
47  
48  
49  
50  
51  
52  
53 What is your first name? Ada  
54 What is your last name? Lovelace  
55  
56  
57  
58  
59  
60  
61  
62  
63  
64  
65  
66 ch = tolower(ch);  
67 hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;  
68  
69  
70 return hashVal;
```

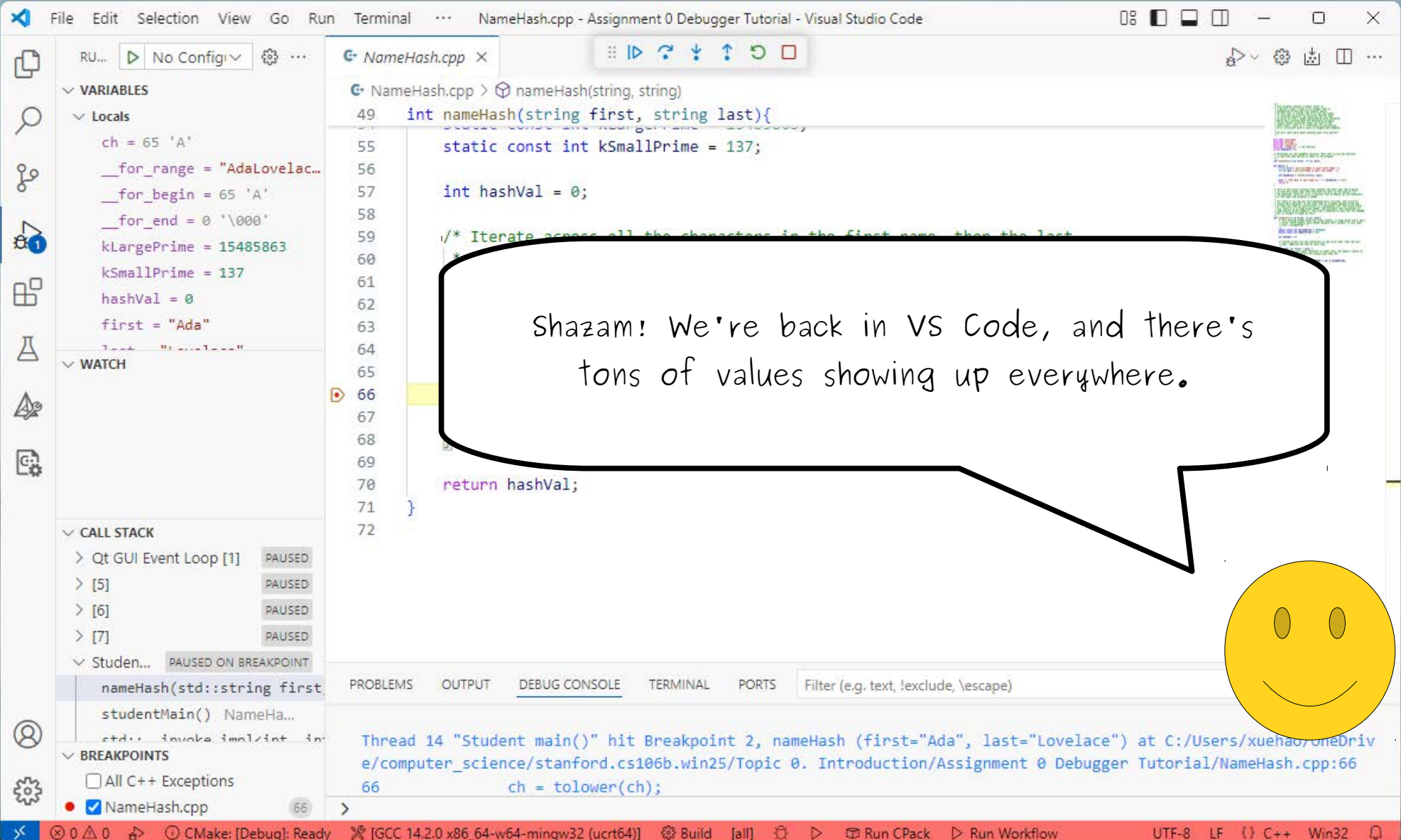
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !exclude, \escape)

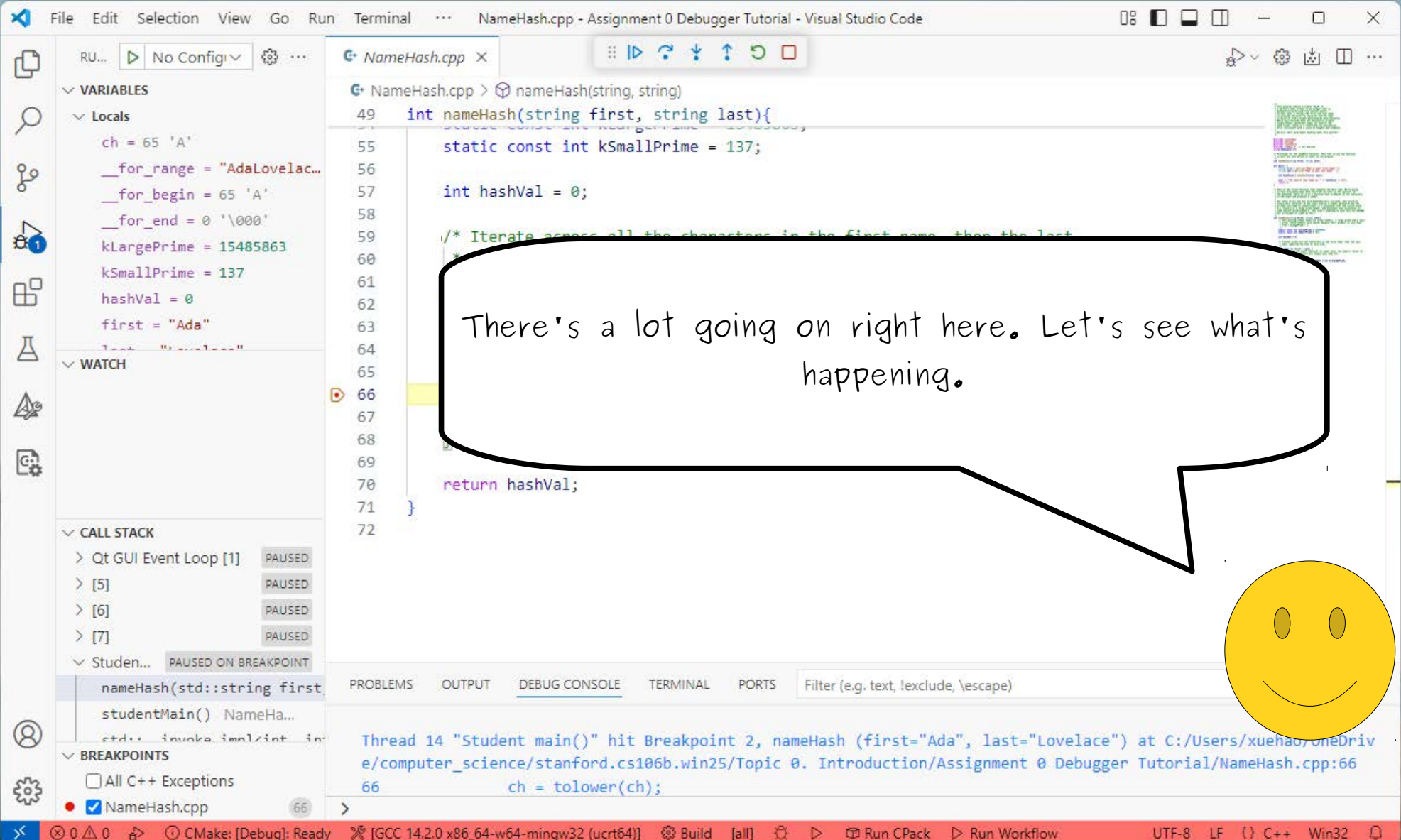
```
[New Thread 43472.0xa63c]  
[New Thread 43472.0xa074]  
[New Thread 43472.0x3910]  
[New Thread 43472.0xa784]
```

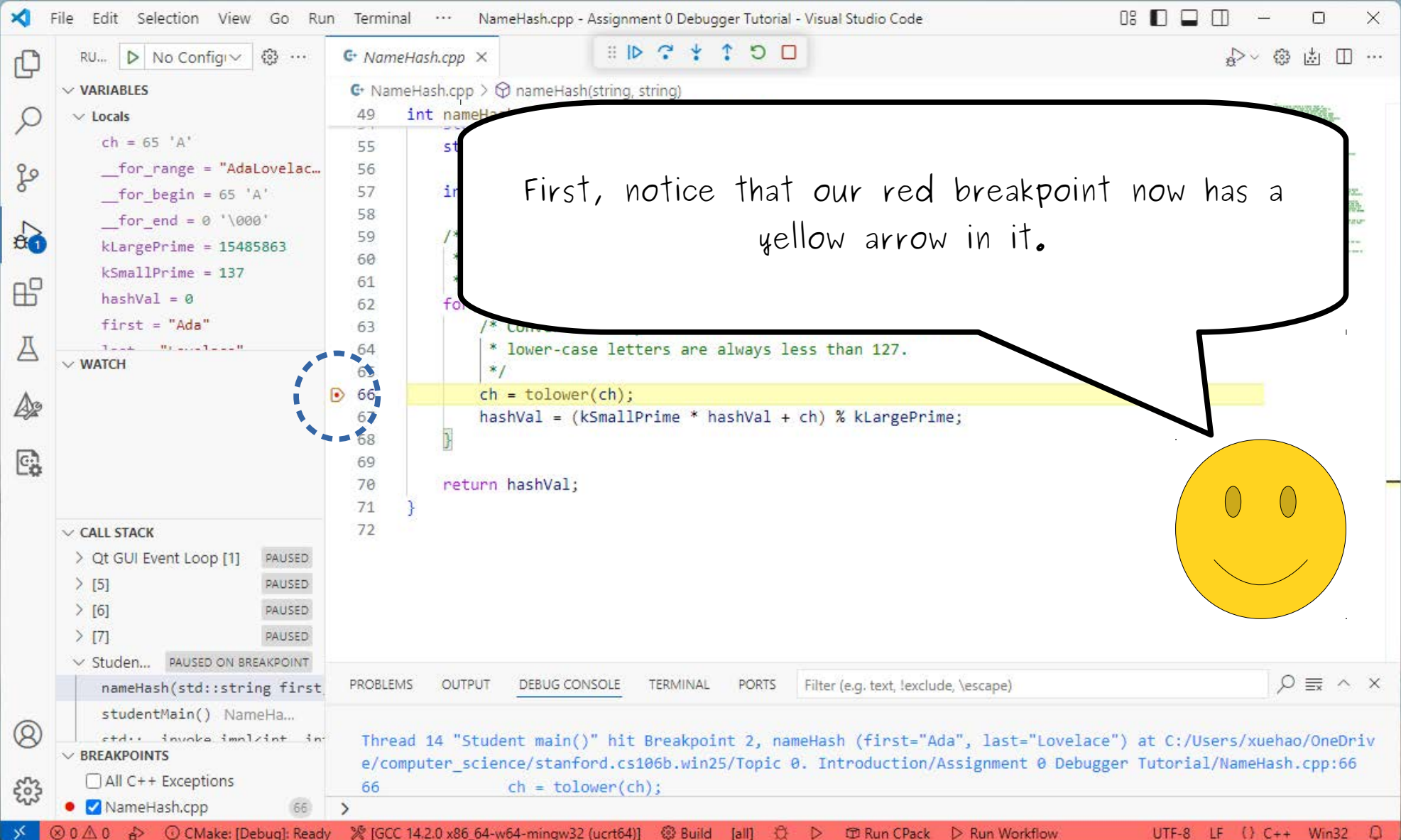


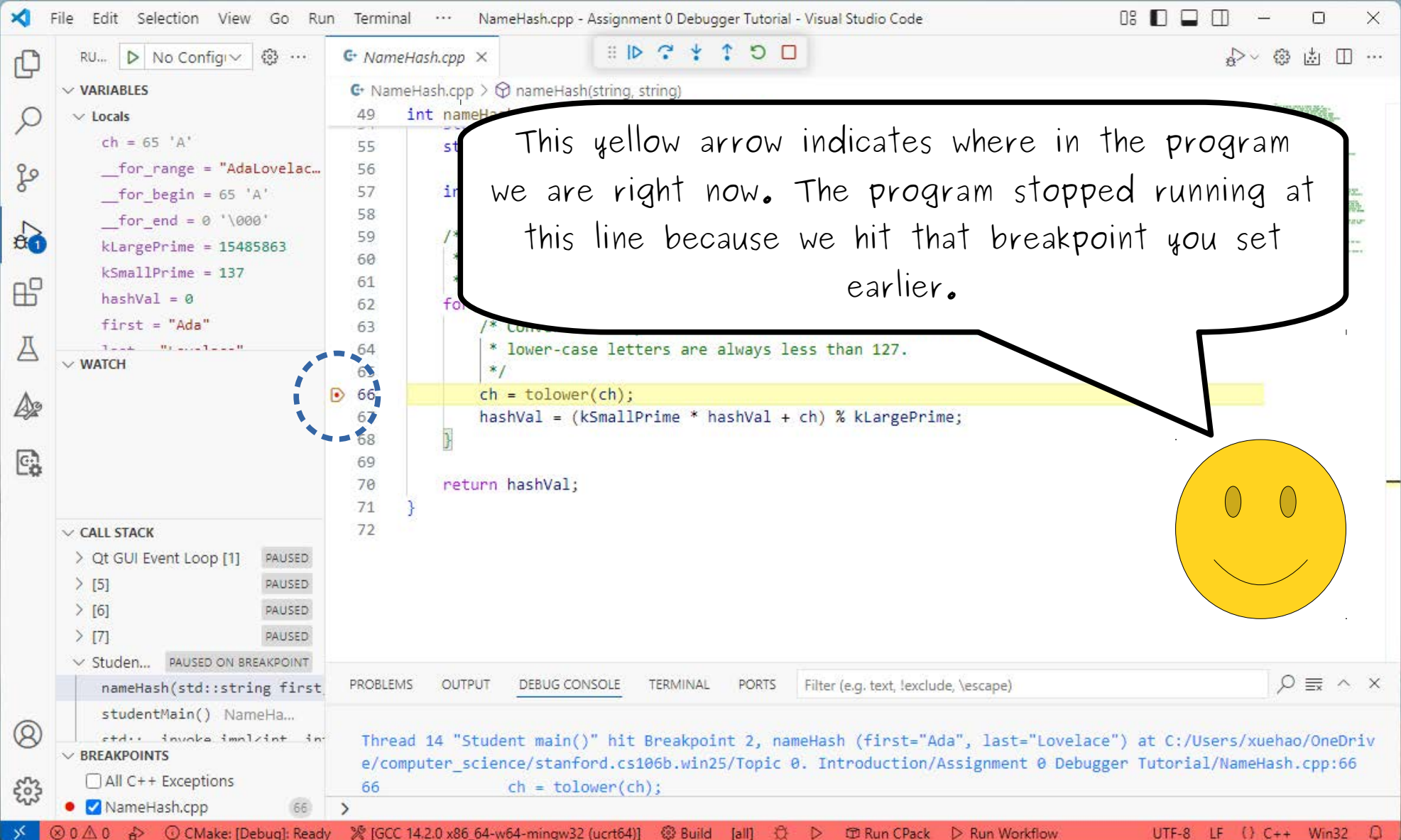
With that said, hit enter,
and watch the magic happen!



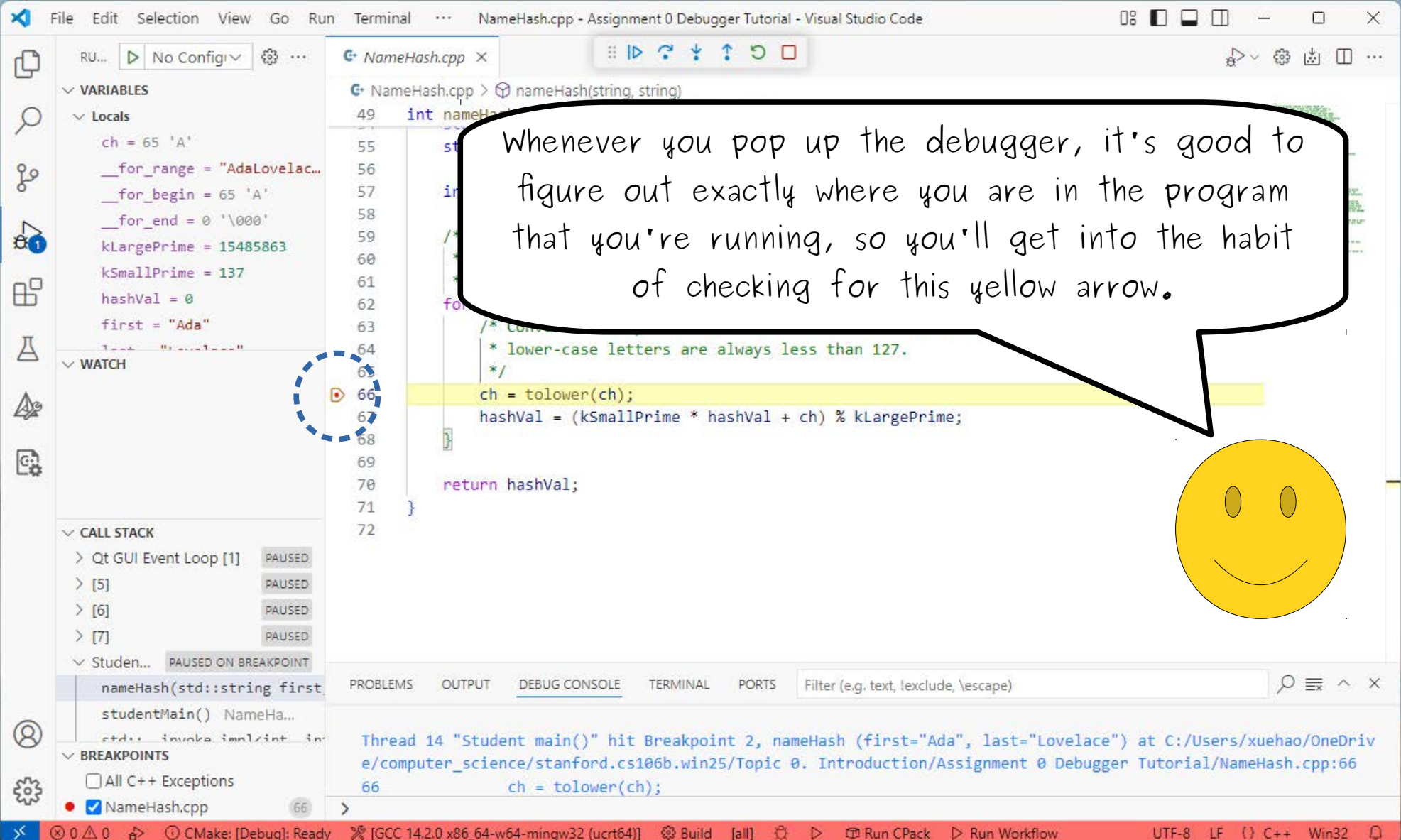








This yellow arrow indicates where in the program we are right now. The program stopped running at this line because we hit that breakpoint you set earlier.



RU... No Config

VARIABLES

Locals

- ch = 65 'A'
- __for_range = "AdaLovelace..."
- __for_begin = 65 'A'
- __for_end = 0 '\000'
- kLargePrime = 15485863
- kSmallPrime = 137
- hashVal = 0
- first = "Ada"
- last = "Lovelace"

Registers

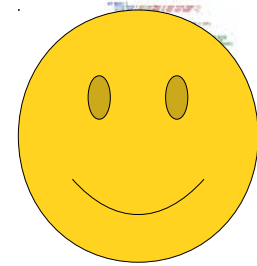
WATCH

CALL STACK

- Qt GUI Event Loop [1] PAUSED
- [5] PAUSED
- [6] PAUSED
- [7] PAUSED
- StudentMain() PAUSED ON BREAKPOINT
- nameHash(std::string first, studentMain() NameHash...
- std::__invoke_impl<int, in...
- std::__invoke_r<int, int (...
- std::_Function_handler<int...
- std::function<int()>::oper...
- operator()(const struct {...
- std::__invoke_impl<int, Qt...

BREAKPOINTS

```
NameHash.cpp x
NameHash.cpp > nameHash(string, string)
49 int nameHash(string first, string last){
55     static const int kSmallPrime = 137;
56
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first name, then the last
60      * name, updating the hash at each step.
61      */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case. The numeric values of
64          * lower-case letters are always less than 127.
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69
70     return hashVal;
71 }
72
```



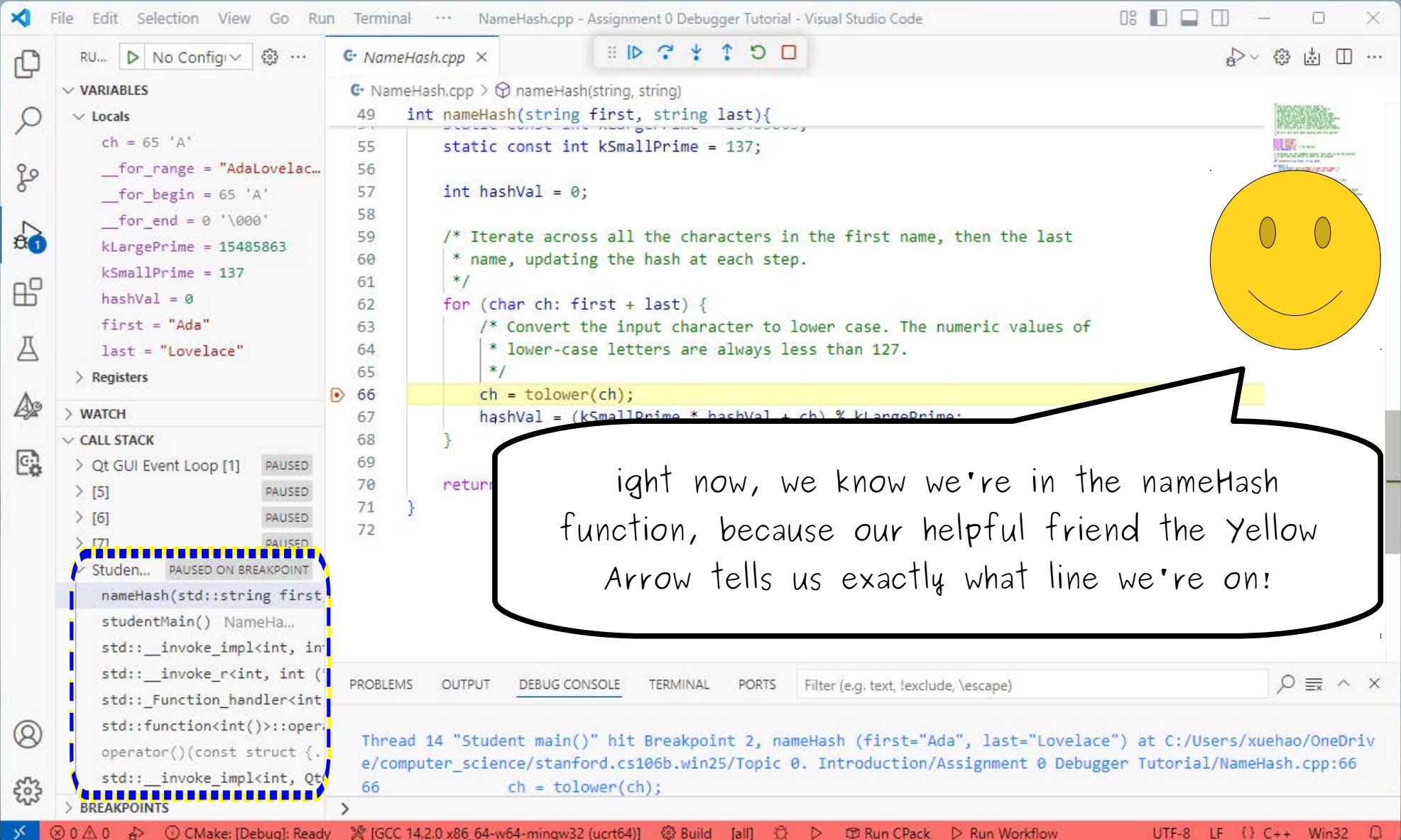
Next, let's take a look at this panel.
This is called the **call stack**.

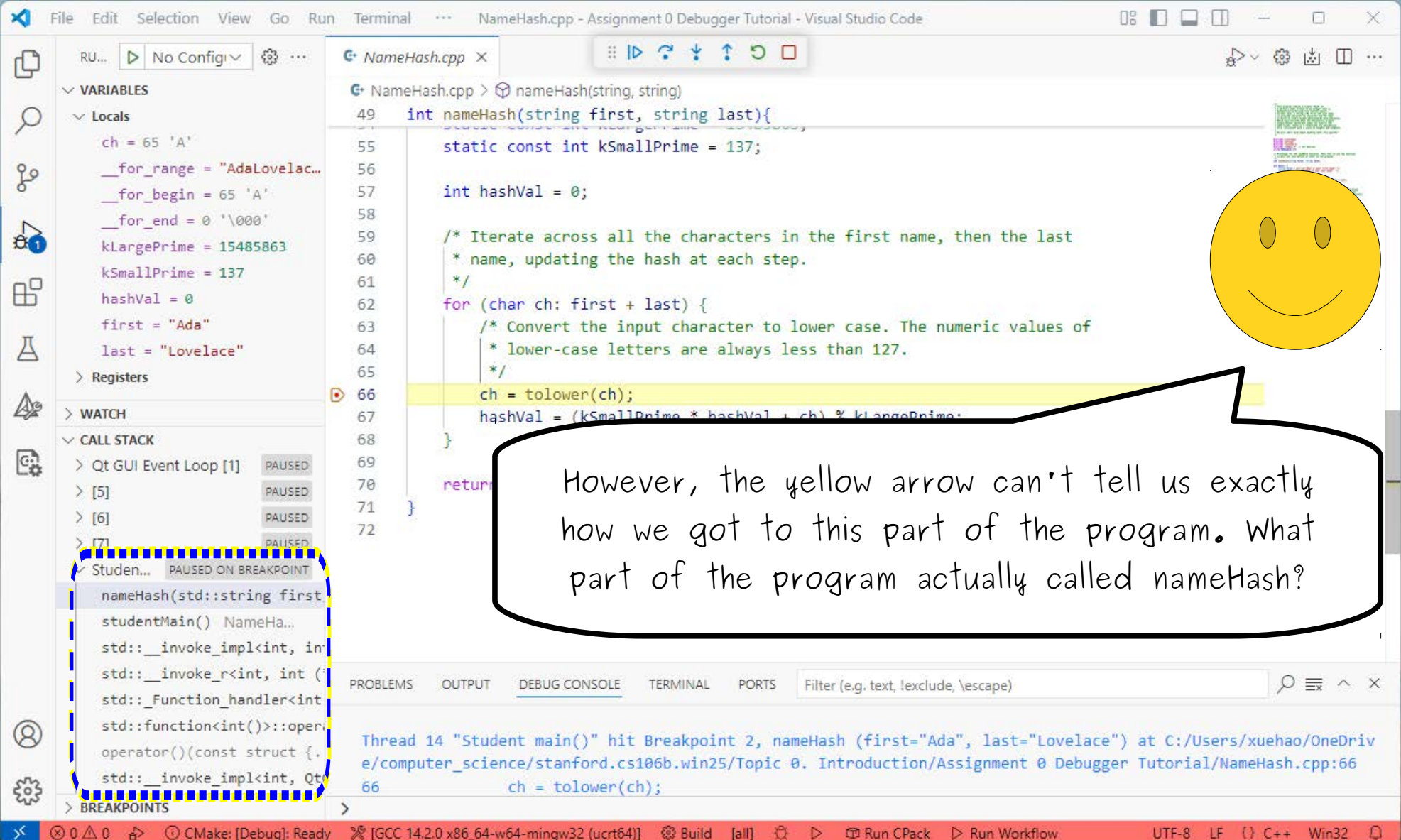
PROBLEMS OUTPUT **DEBUG CONSOLE** TERMINAL PORTS

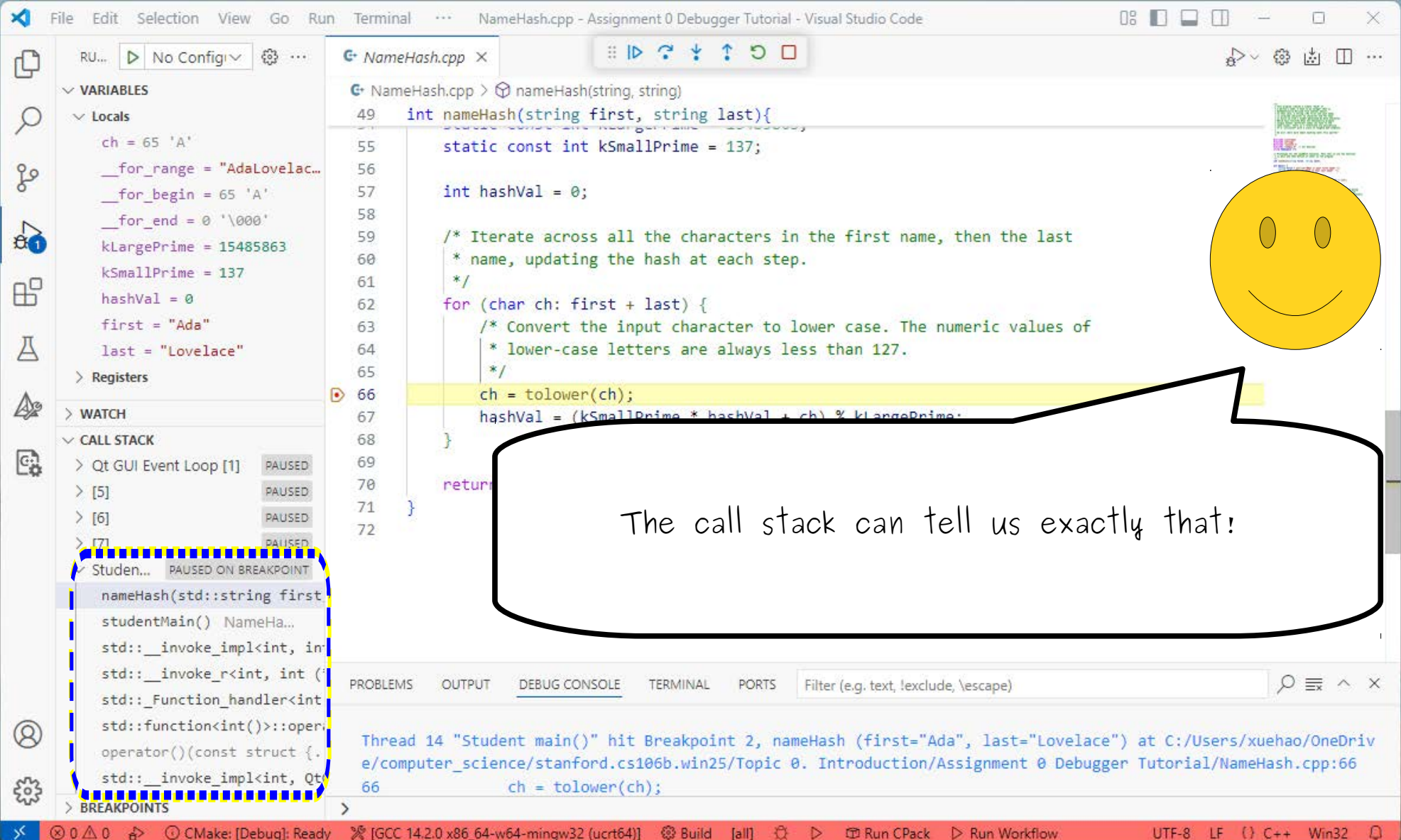
Filter (e.g. text, !exclude, \escape)

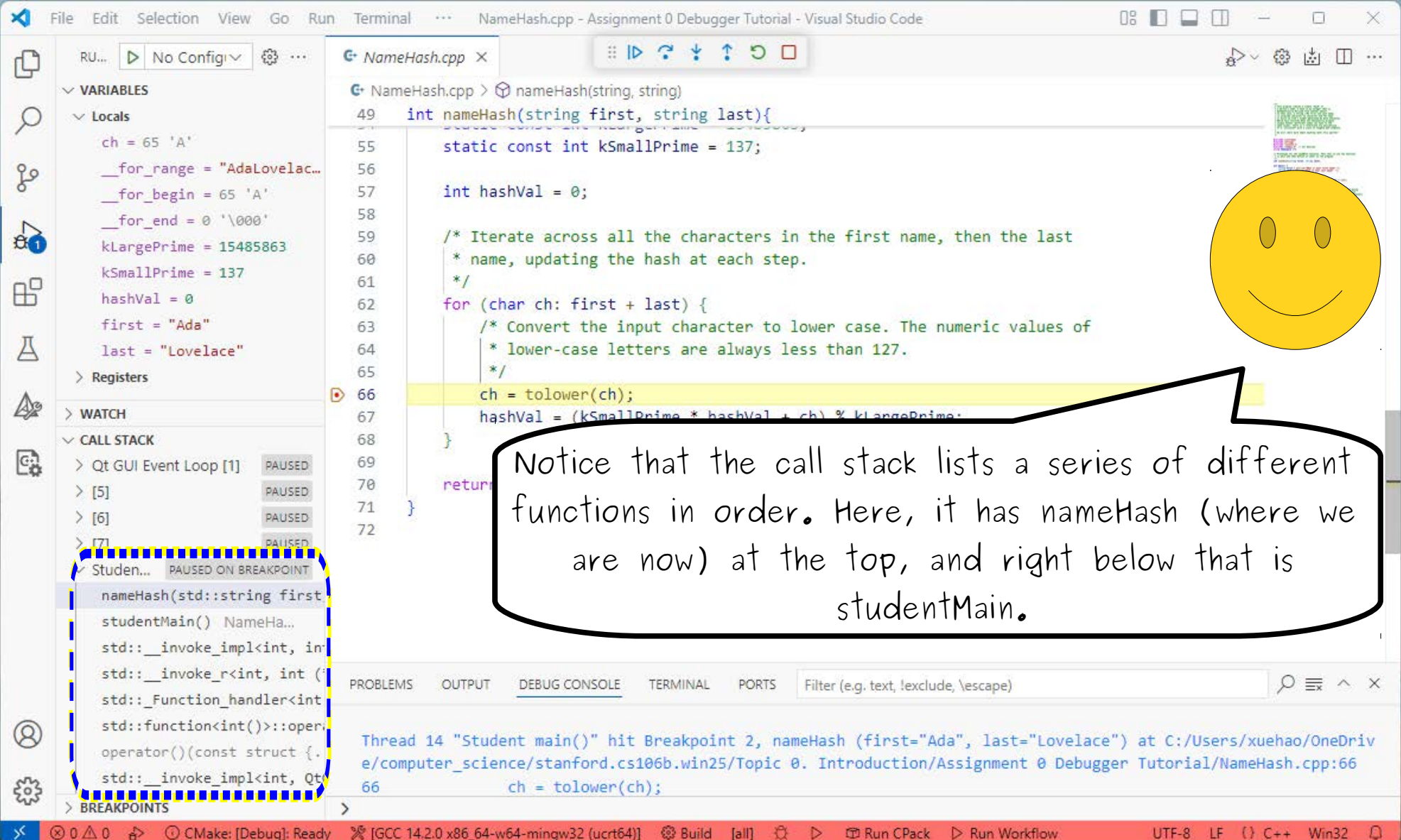
Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66

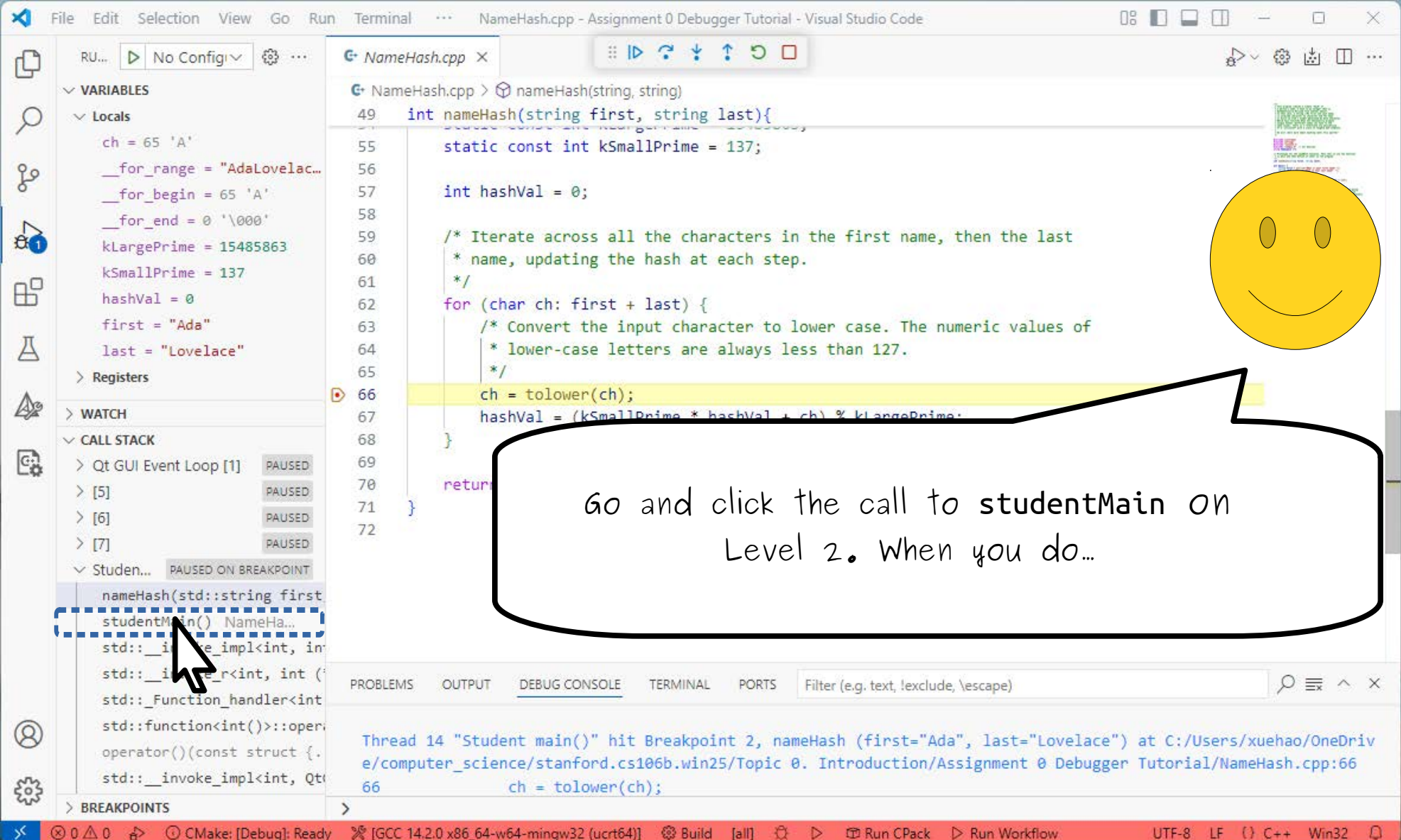
```
66 ch = tolower(ch);
```

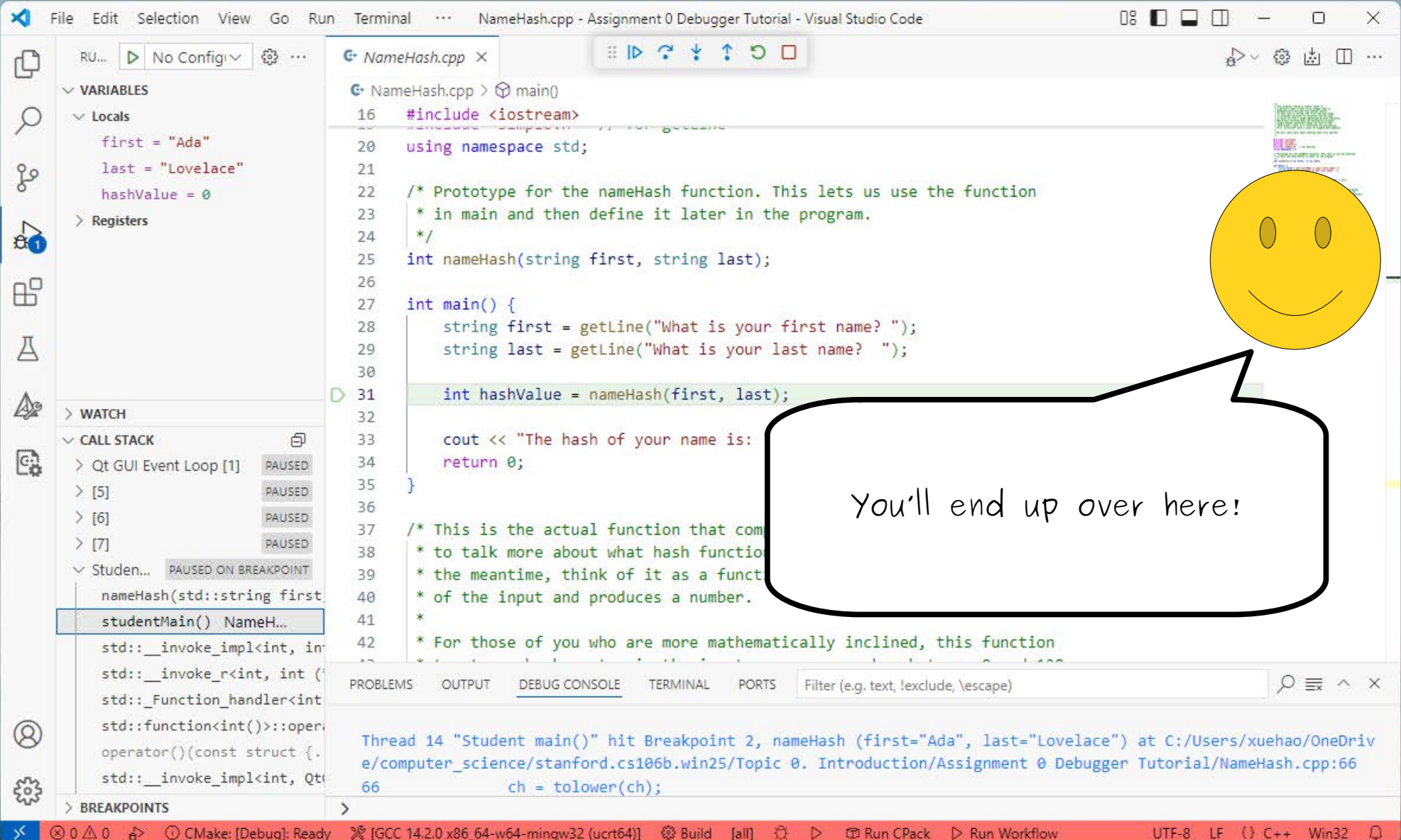



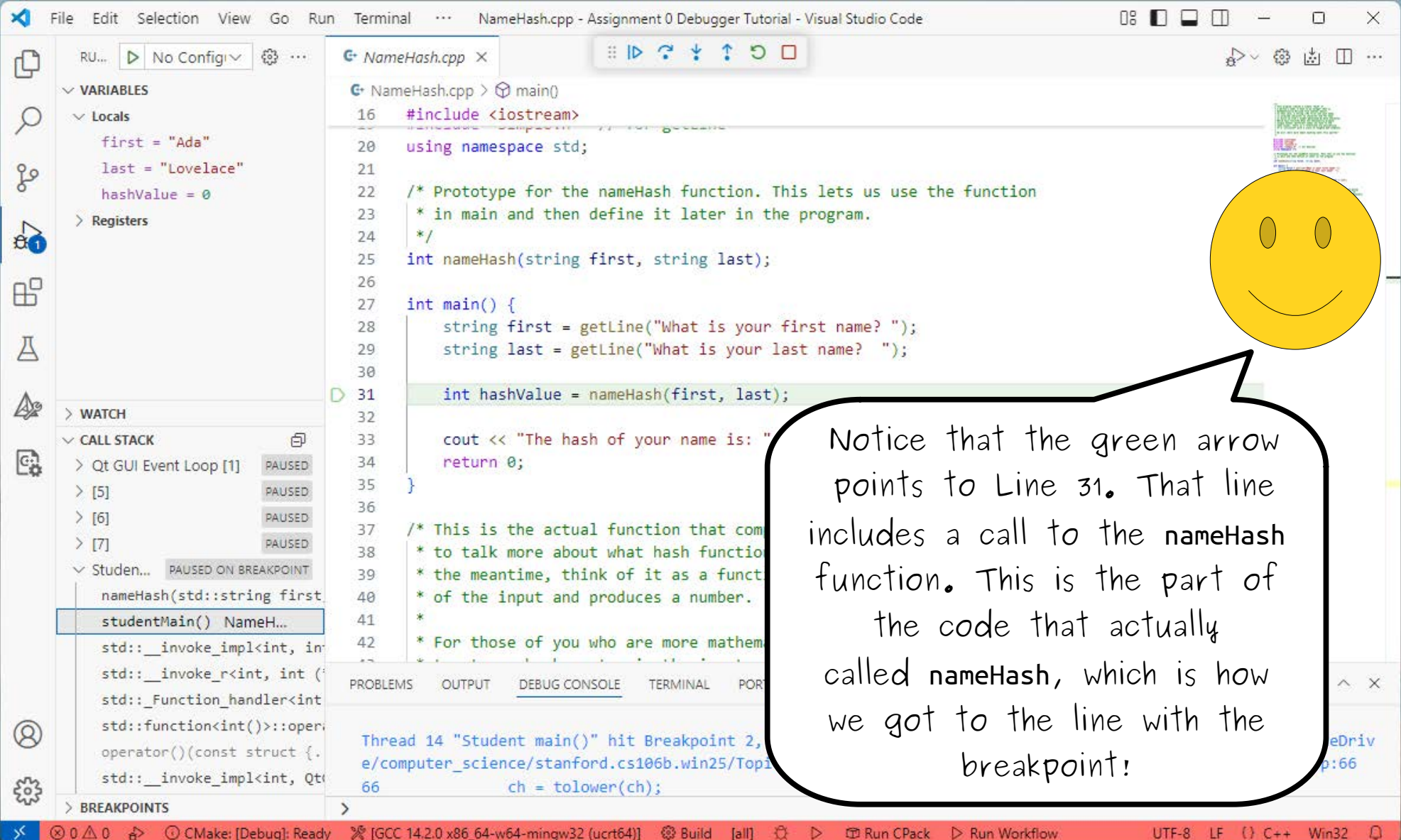


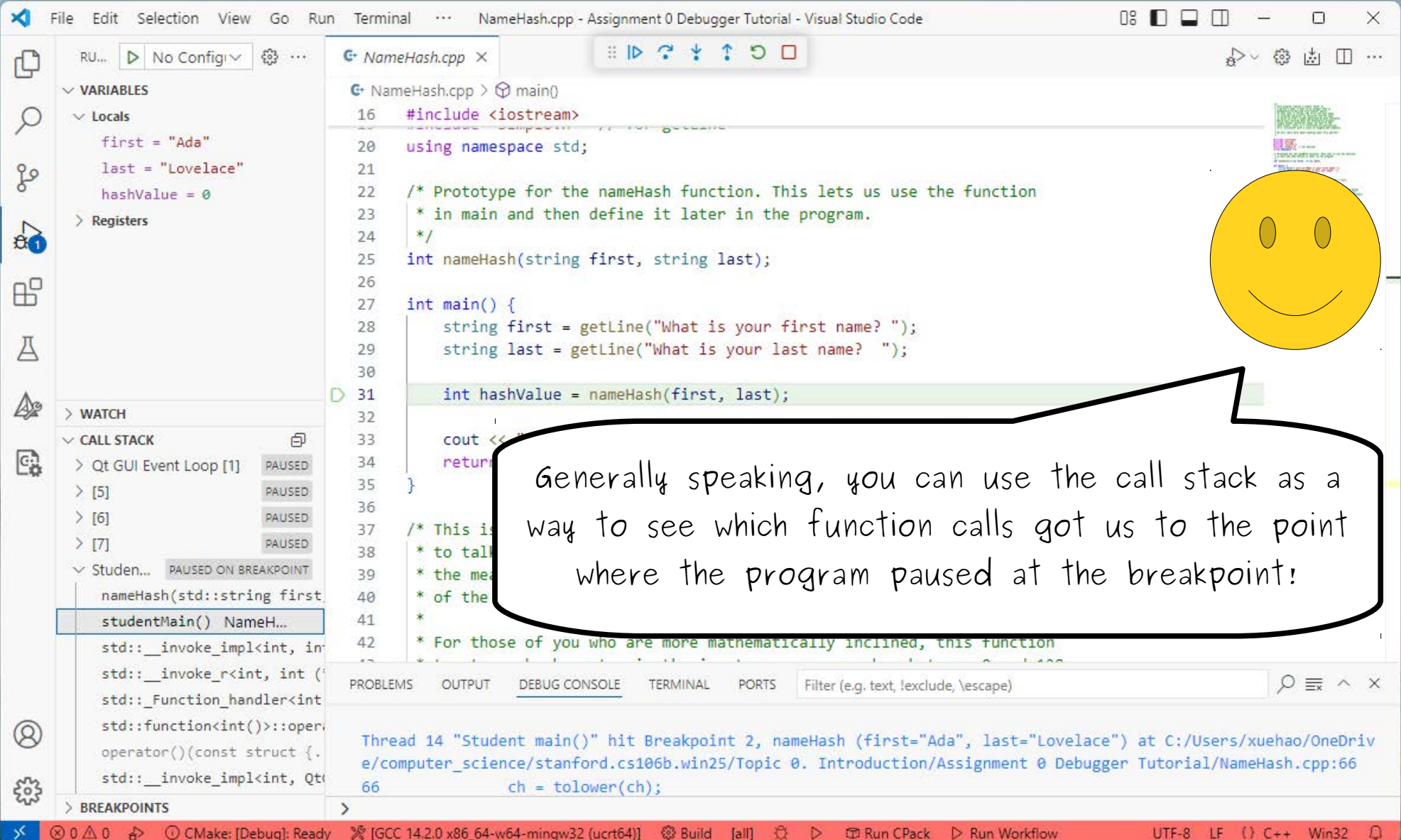


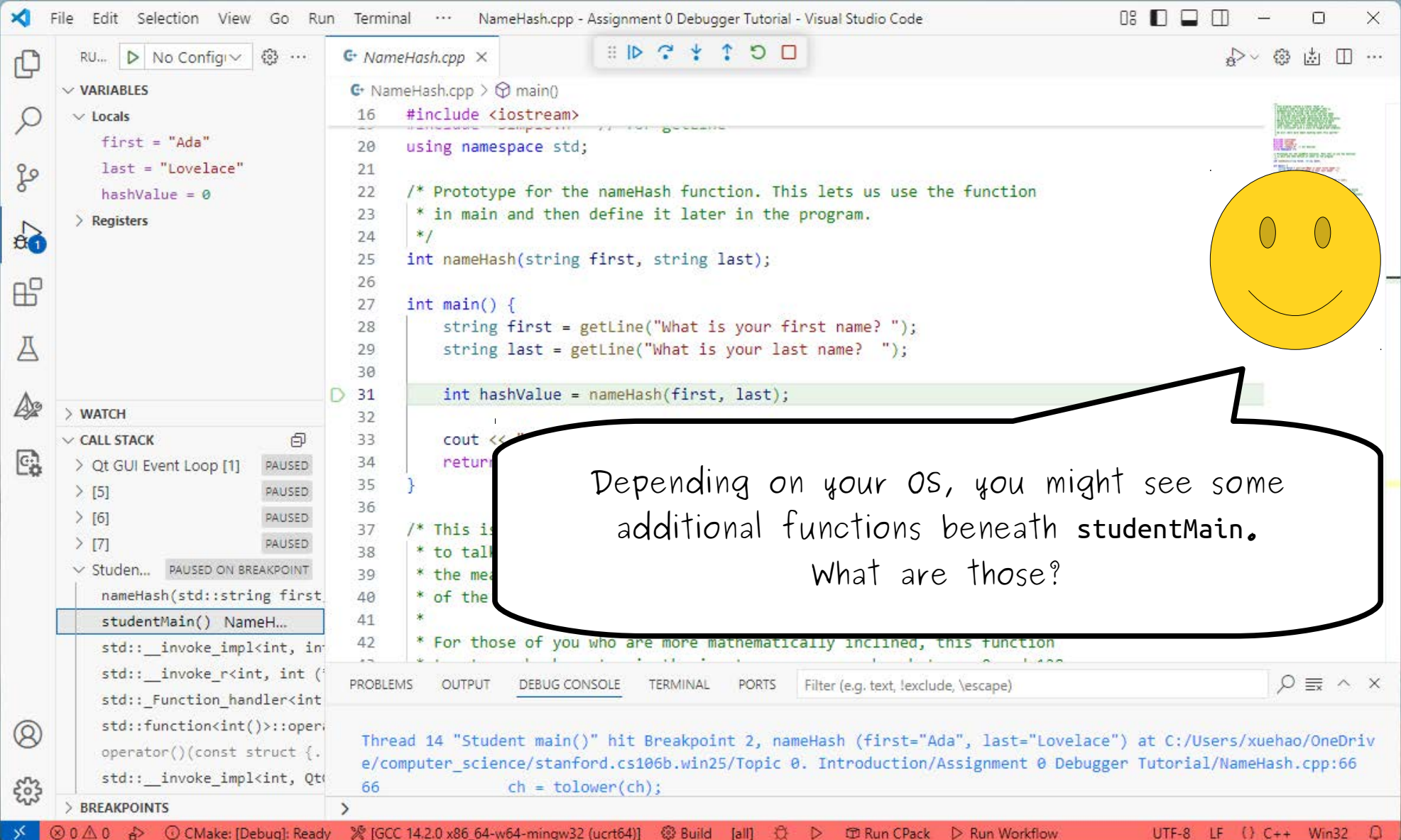


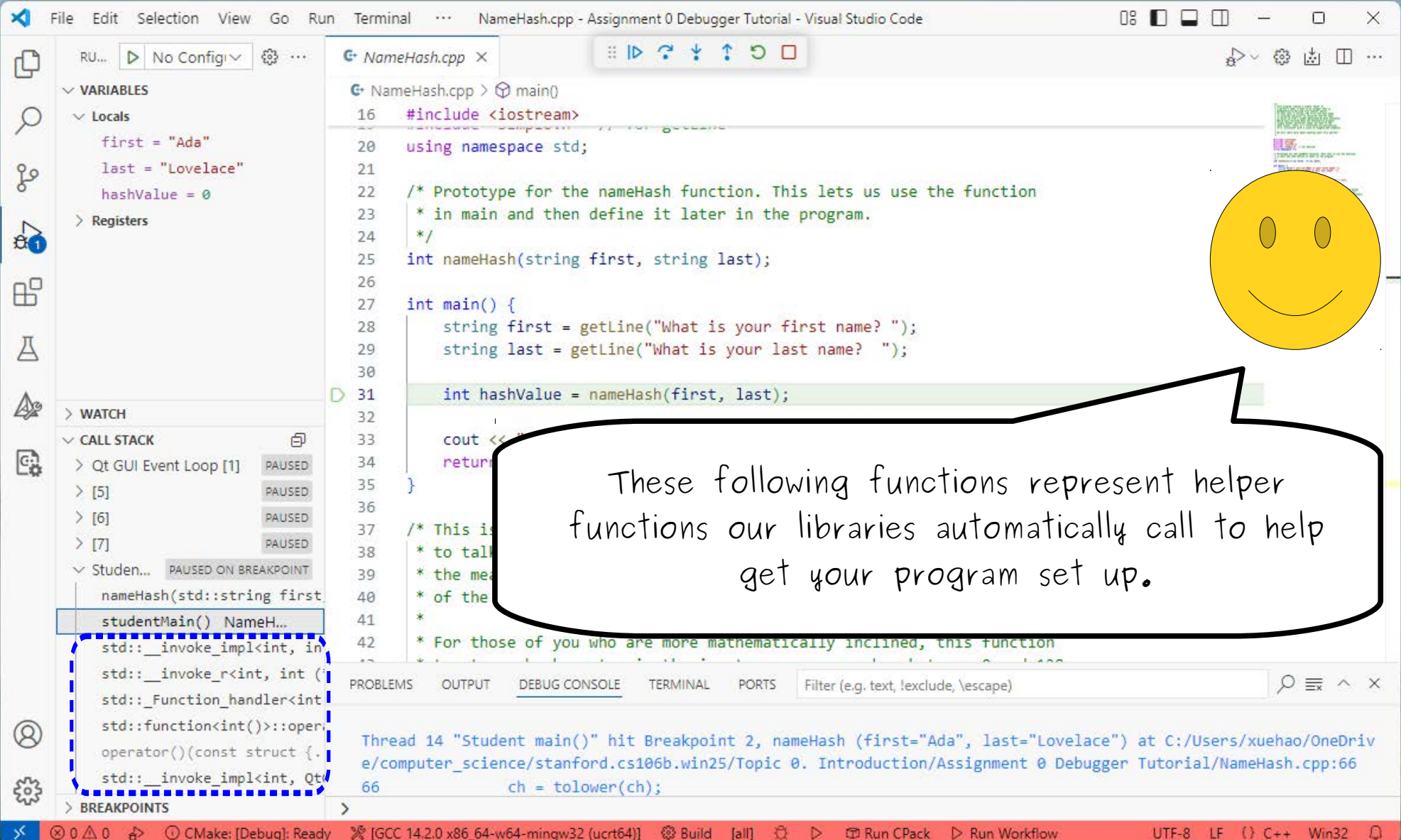


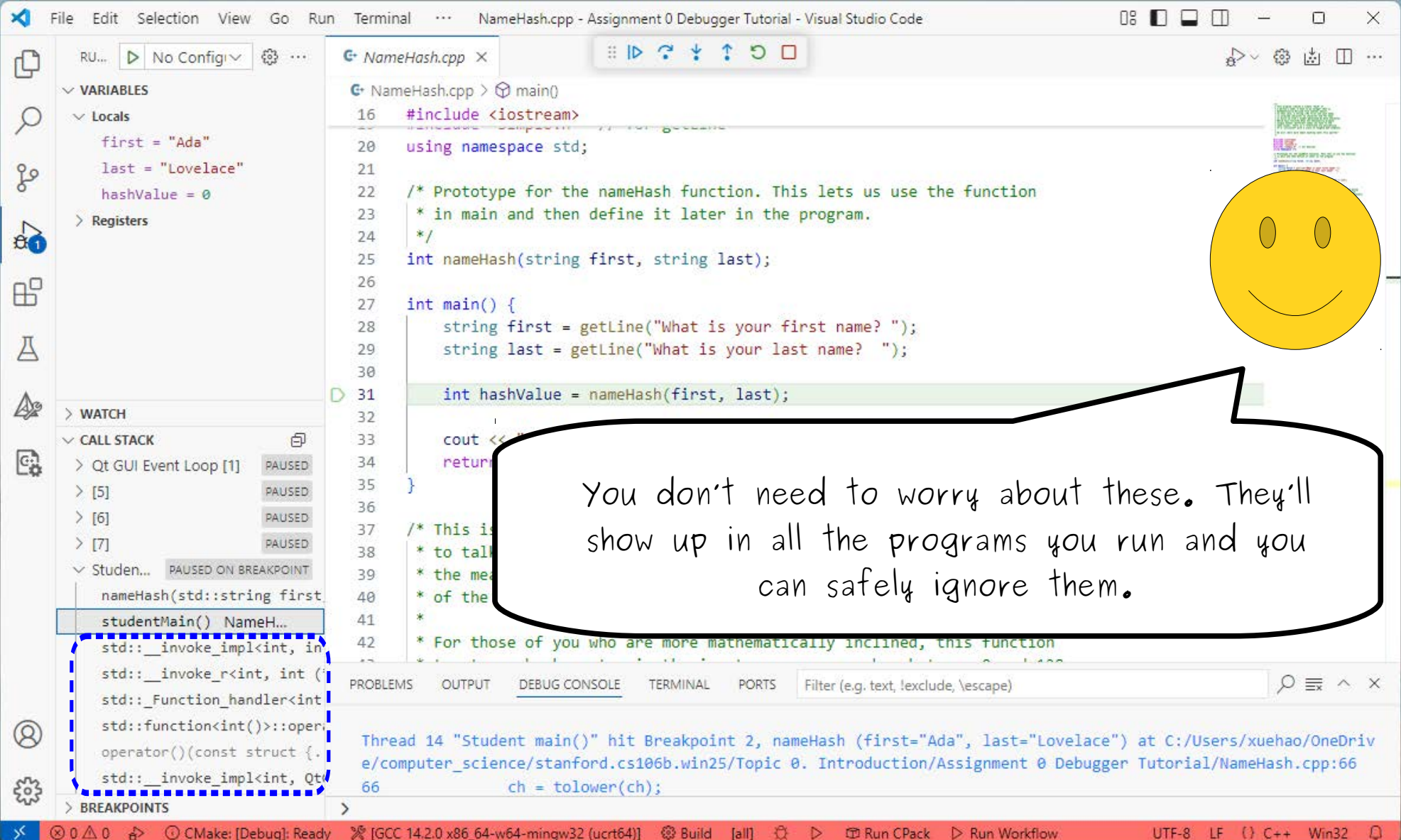


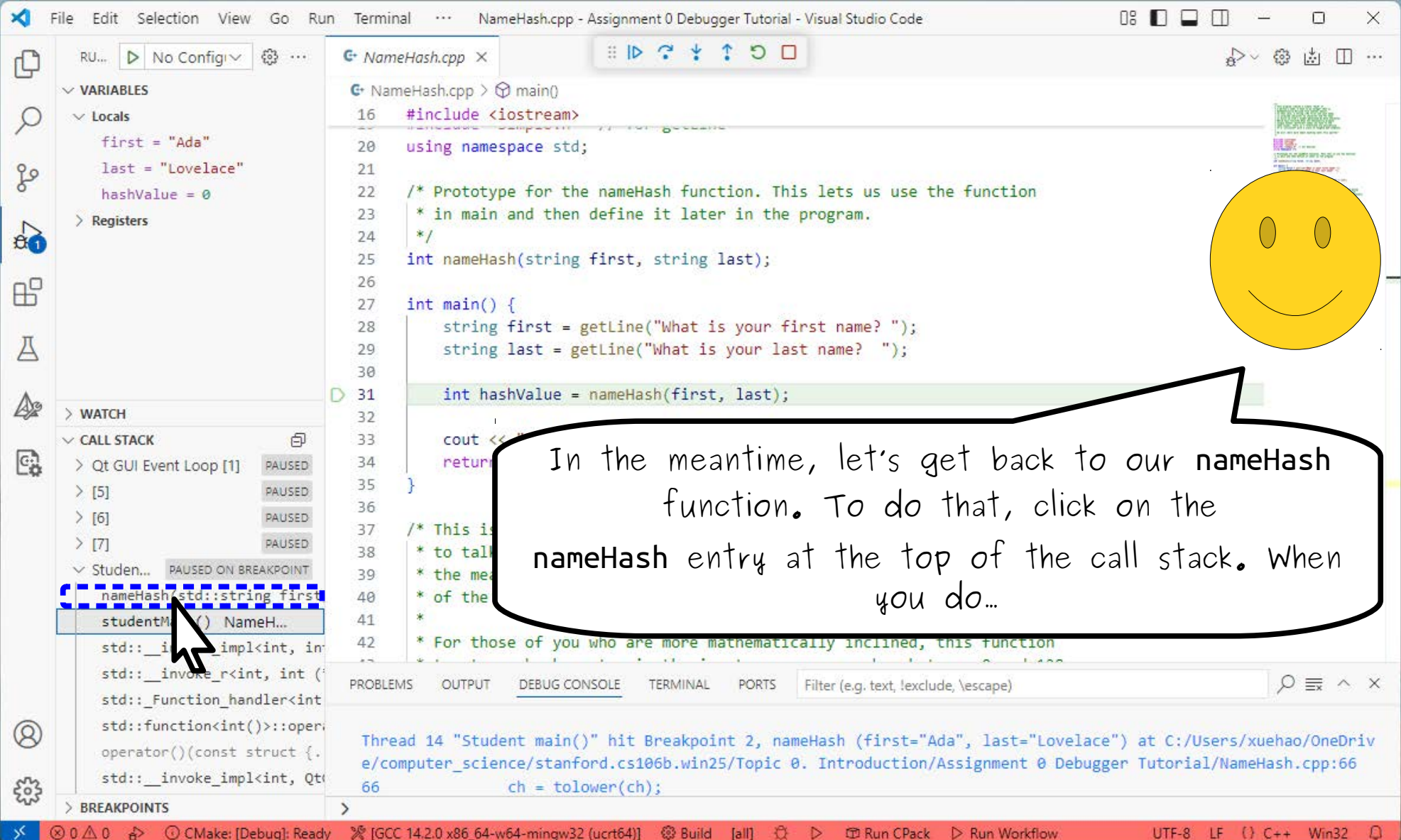


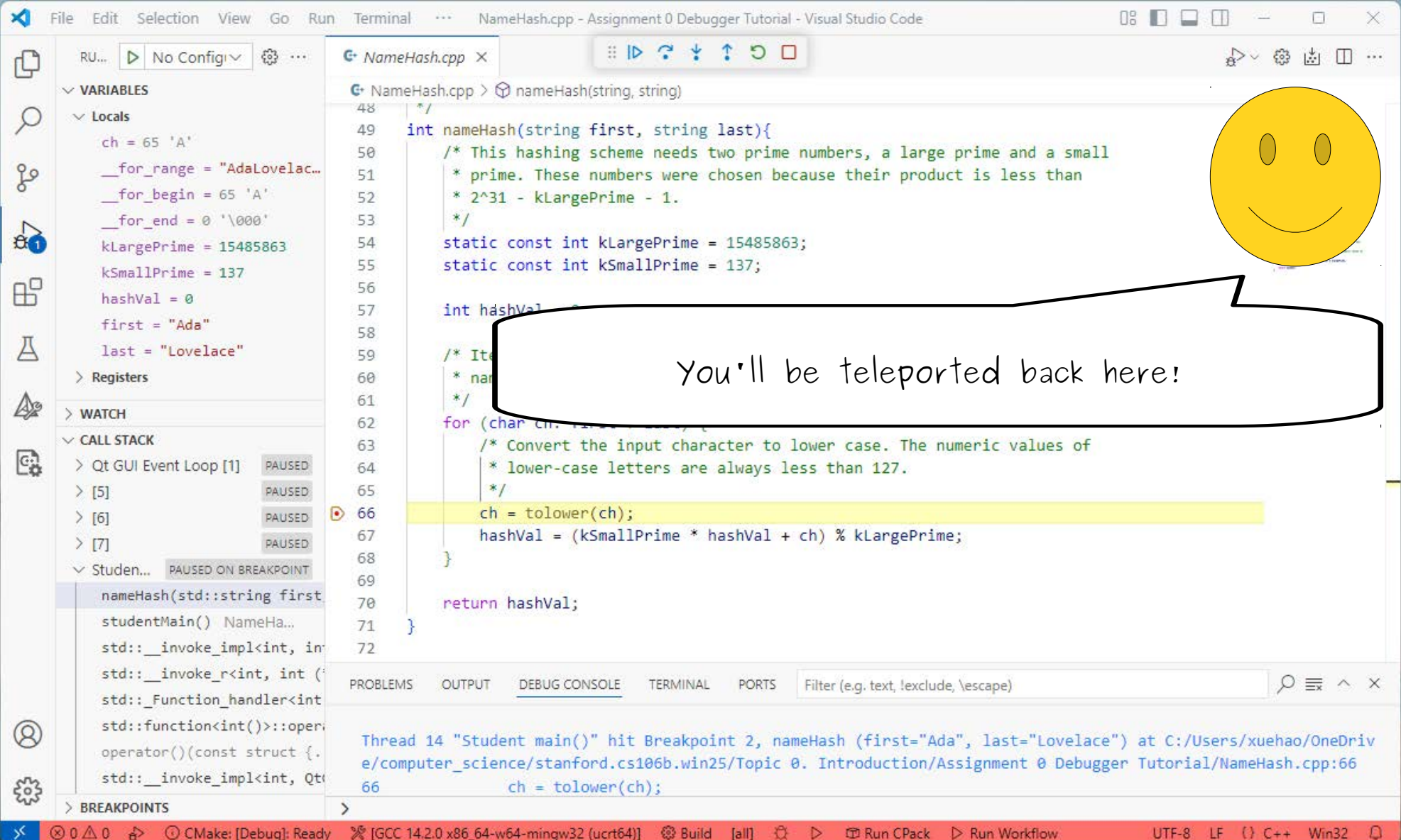


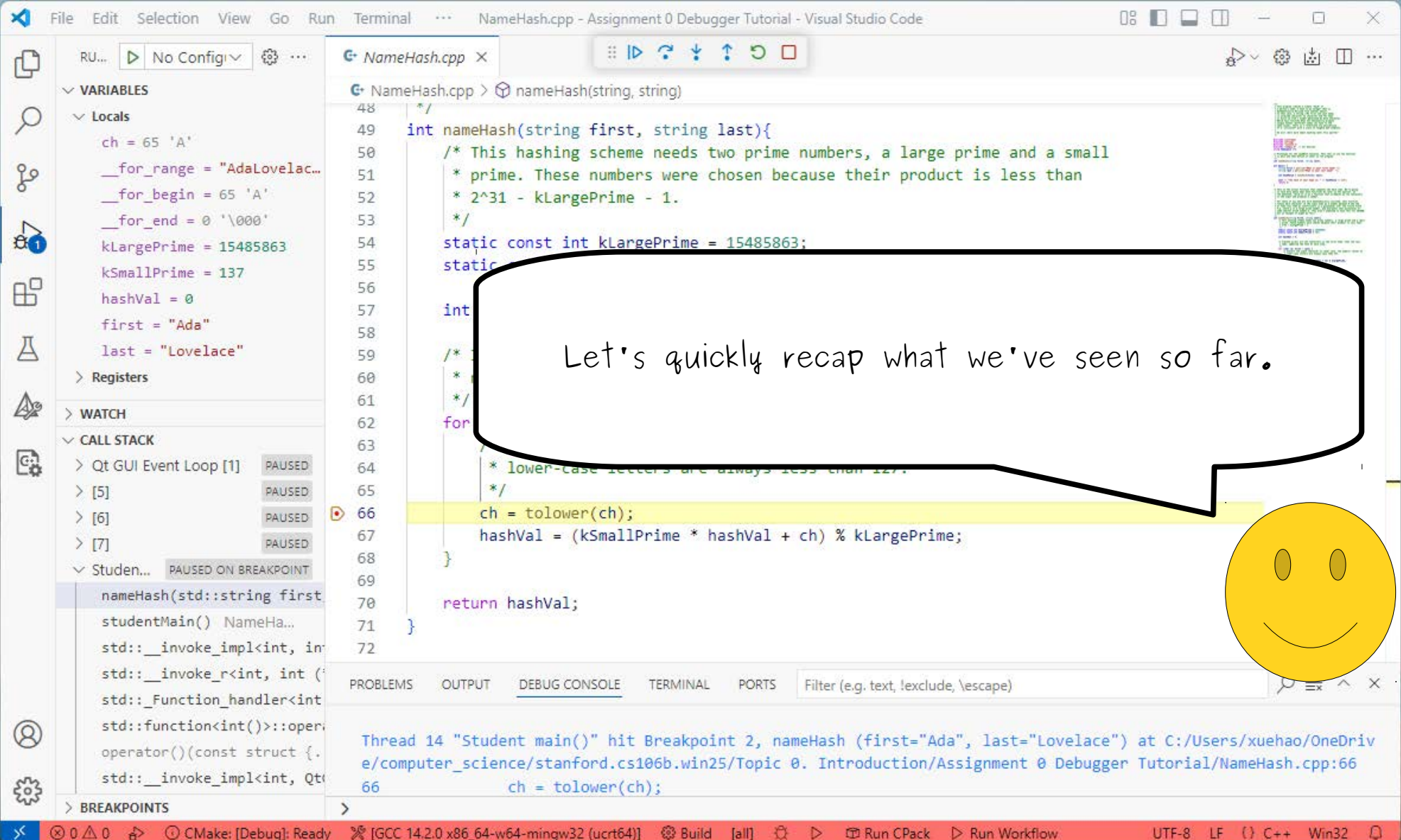






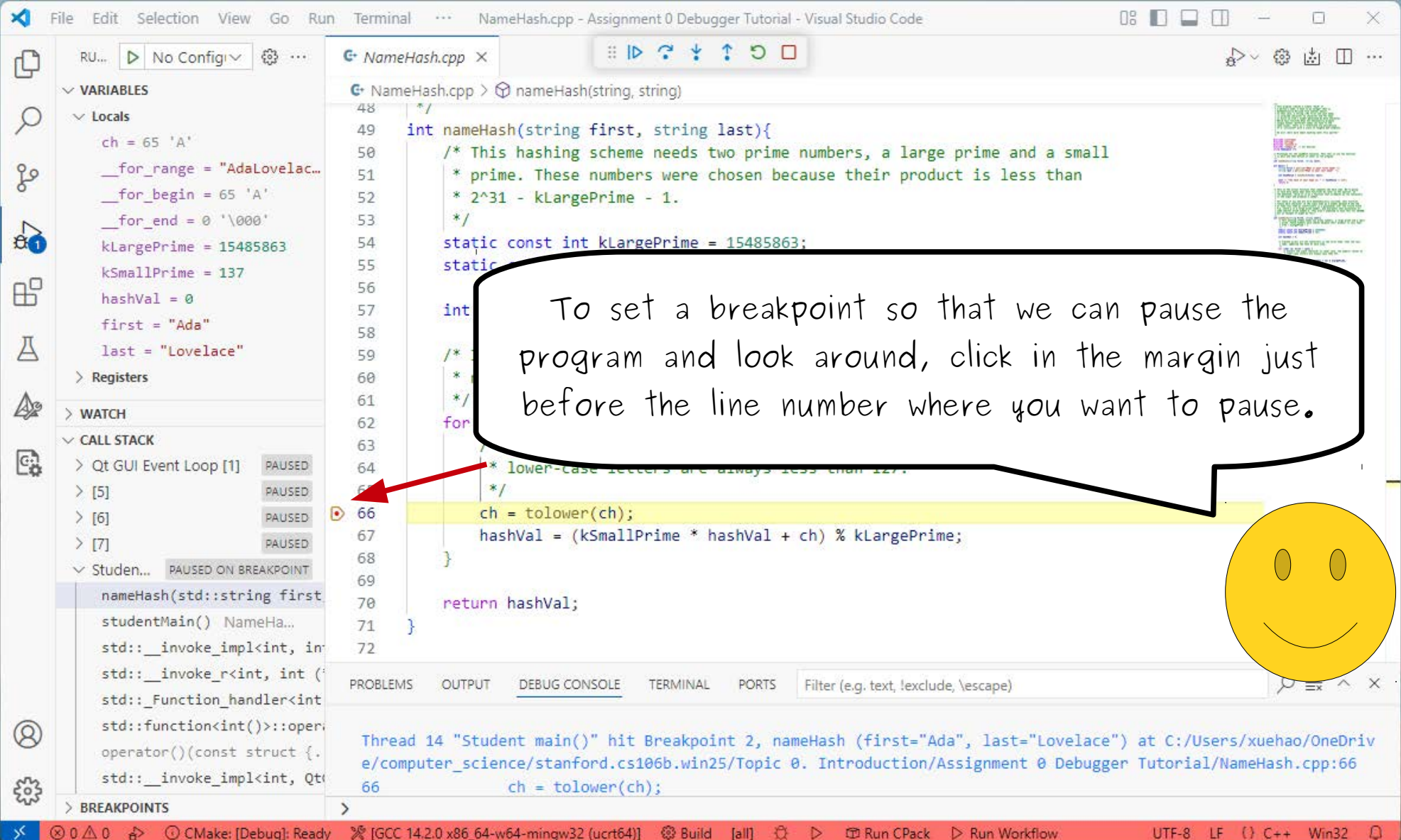


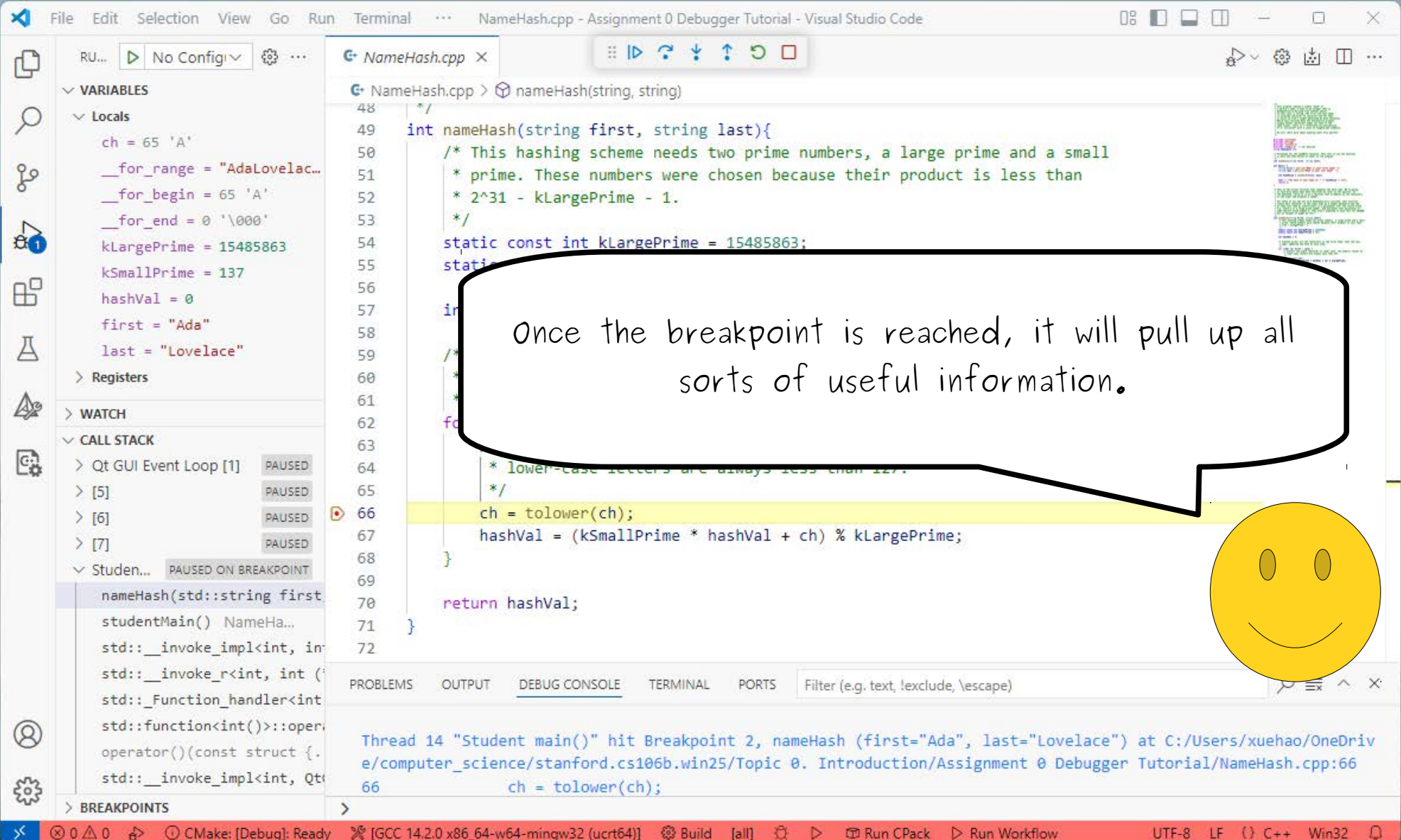




Let's quickly recap what we've seen so far.







RU... No Config

VARIABLES

Locals

ch = 65 'A'

__for_range = "AdaLovelace..."

__for_begin = 65 'A'

__for_end = 0 '\000'

kLargePrime = 15485863

kSmallPrime = 137

hashVal = 0

first = "Ada"

last = "Lovelace"

Registers

WATCH

CALL STACK

Qt GUI Event Loop [1] PAUSED

[5] PAUSED

[6] PAUSED

[7] PAUSED

Student... PAUSED ON BREAKPOINT

nameHash(std::string first

studentMain() NameHa...

std::__invoke_impl<int, in

std::__invoke_r<int, int (

std::_Function_handler<int

std::function<int()>::oper

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std::__invoke_impl<int, Qt

BREAKPOINTS

NameHash.cpp

NameHash.cpp > nameHash(string, string)

48 */

49 int nameHash(string first, string last){

50 /* This hashing scheme needs two prime numbers, a large prime and a small

51 * prime. These numbers were chosen because their product is less than

52 * 2^31 - kLargePrime - 1.

53 */

54 static const int kLargePrime = 15485863;

55 stati

56

57 in

58

59 /*

60 *

61 *

62 fo

63

64 * lower-case letters are always less than 127.

65 */

66 ch = tolower(ch);

67 hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;

68 }

69

70 return hashVal;

71 }

72

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, !exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66

66 ch = tolower(ch);

The yellow arrow points out where we are right now.



0 0 0 0

CMake: [Debug]: Ready

[GCC 14.2.0 x86_64-w64-mingw32 (ucrt64)]

Build [all]

Run CPack

Run Workflow

UTF-8 LF C++ Win32

RU... No Config

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Locals

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__for_begin = 65 'A'

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PROBLEMS

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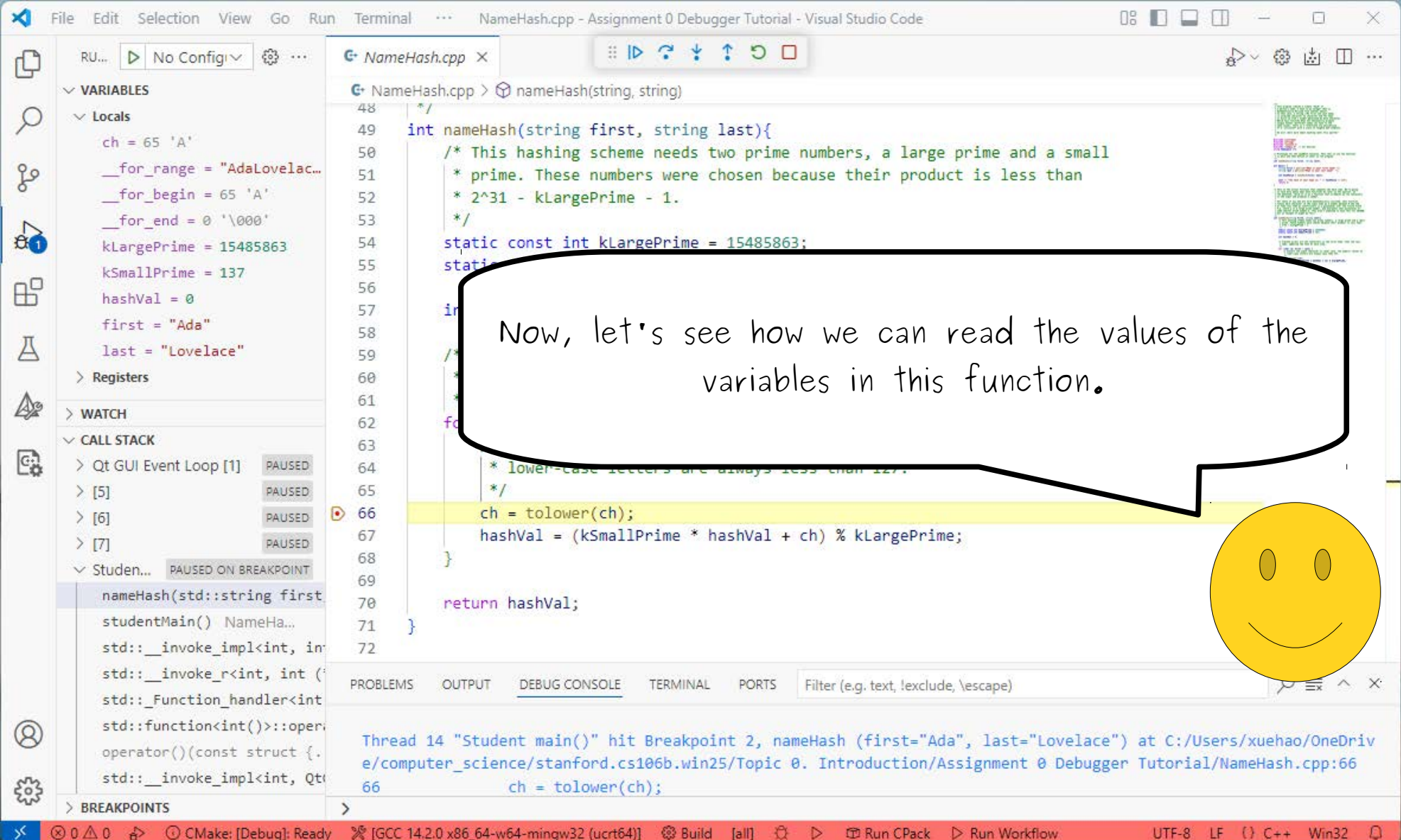
66 ch = tolower(ch);

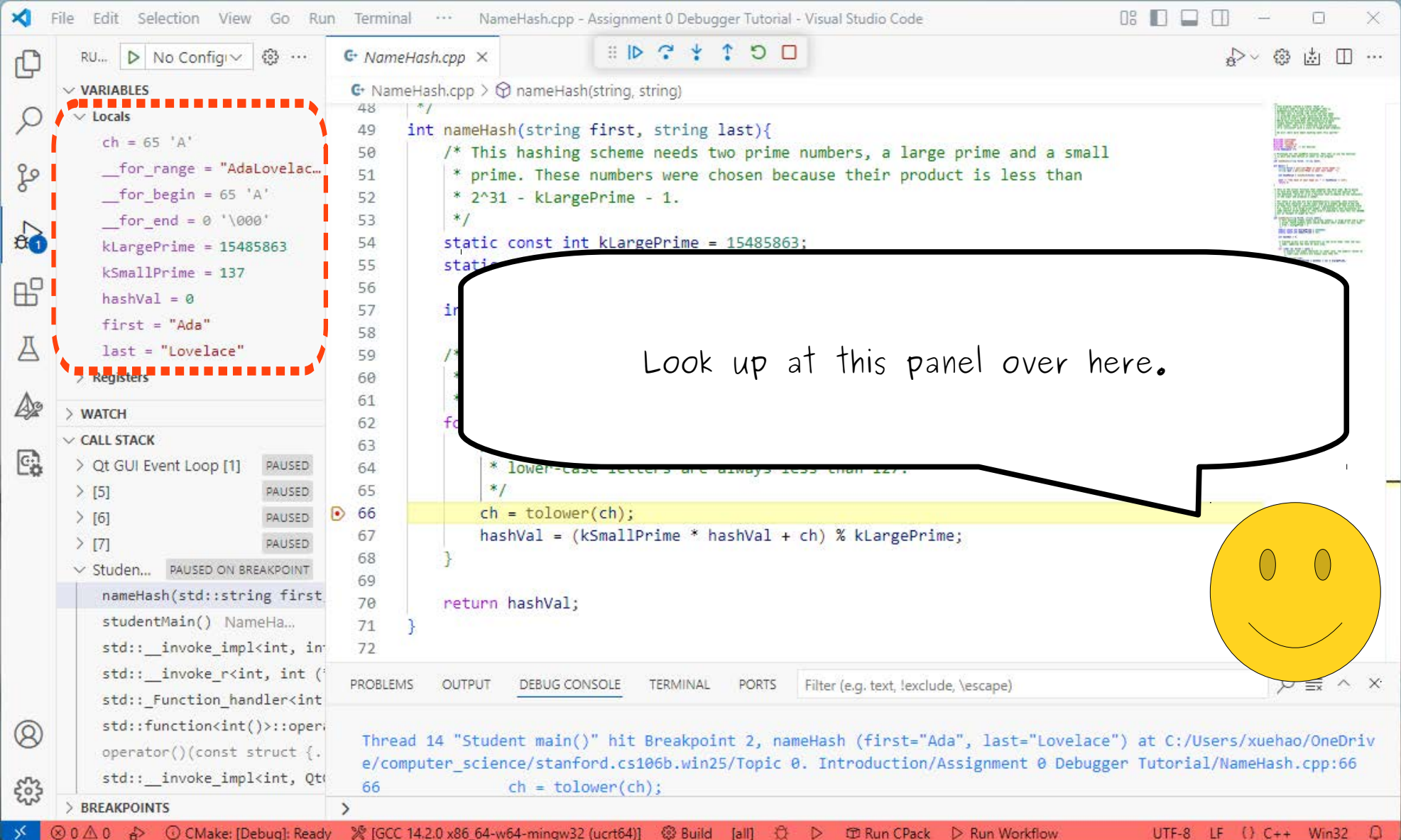
The call stack shows us how we got into the current function.

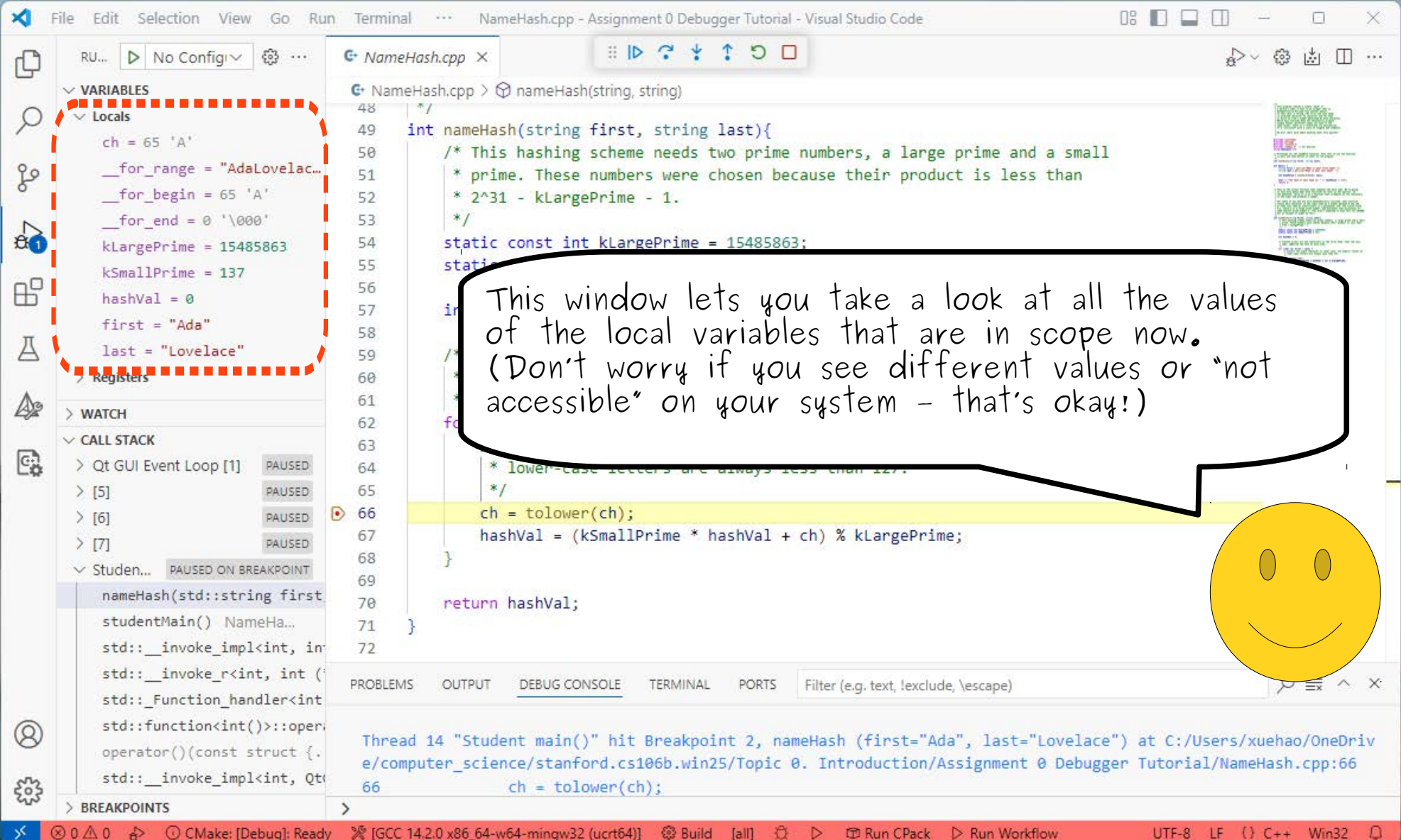


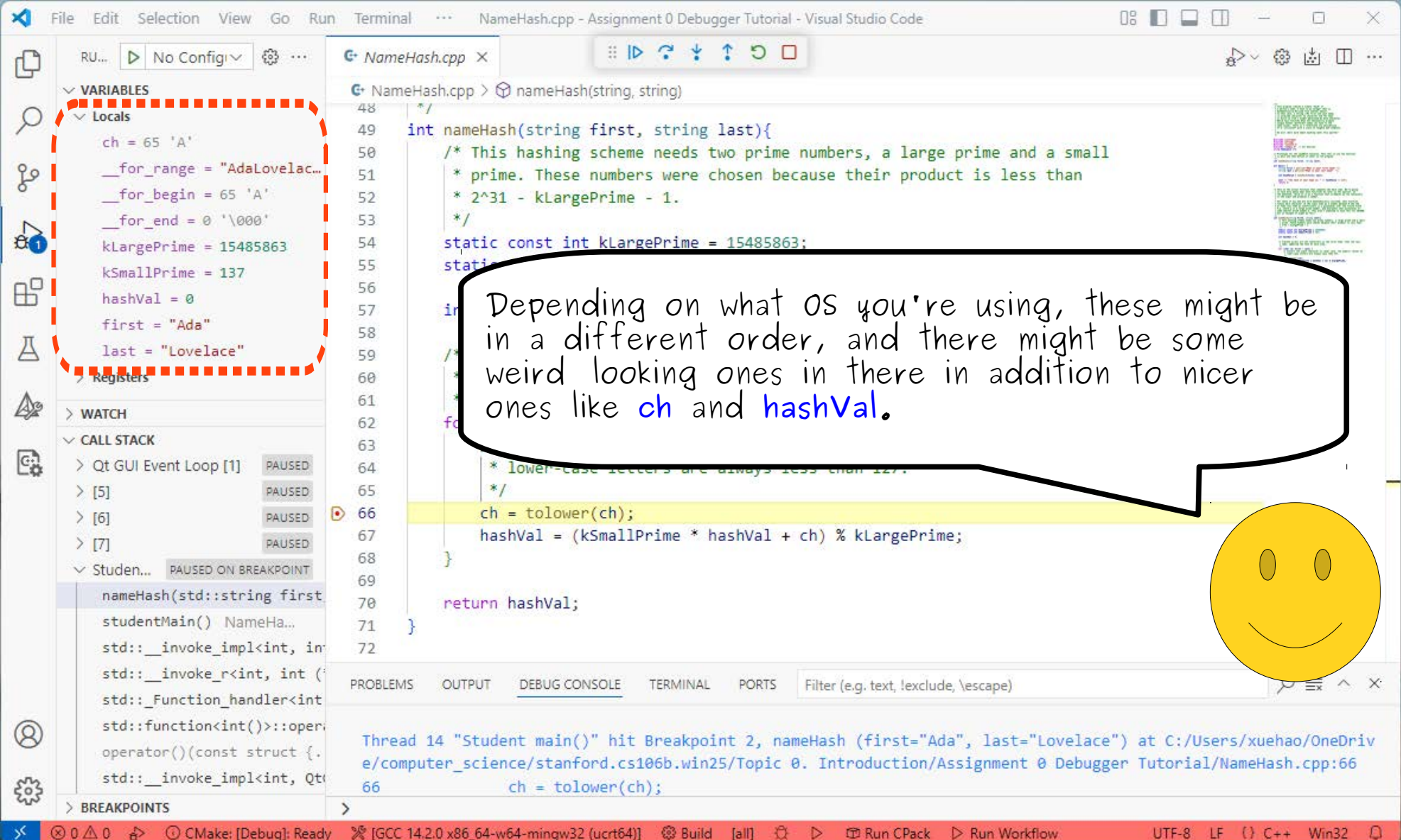
File Edit Selection View Go Run Terminal ... NameHash.cpp - Assignment 0 Debugger Tutorial - Visual Studio Code

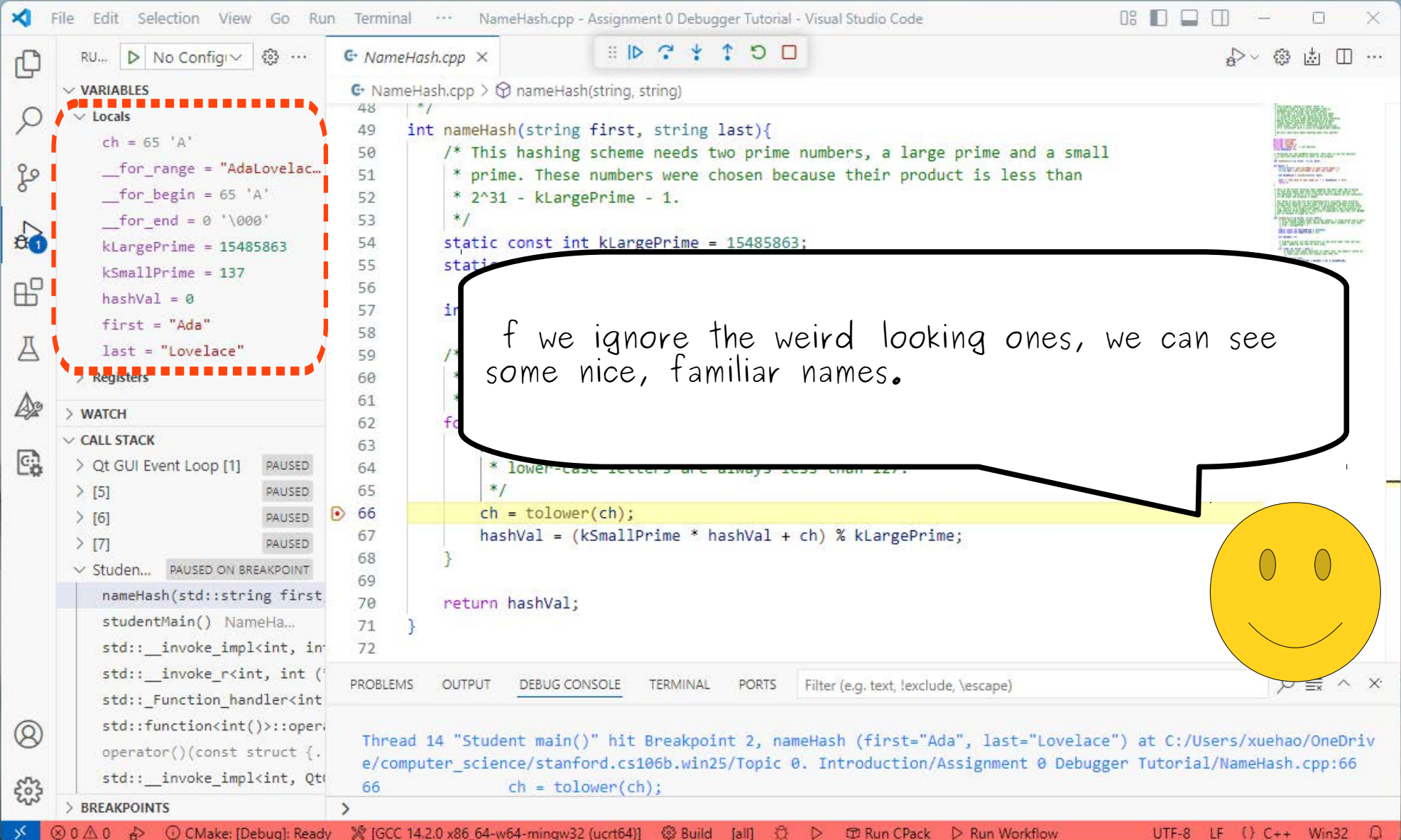
UTF-8 LF C++ Win32

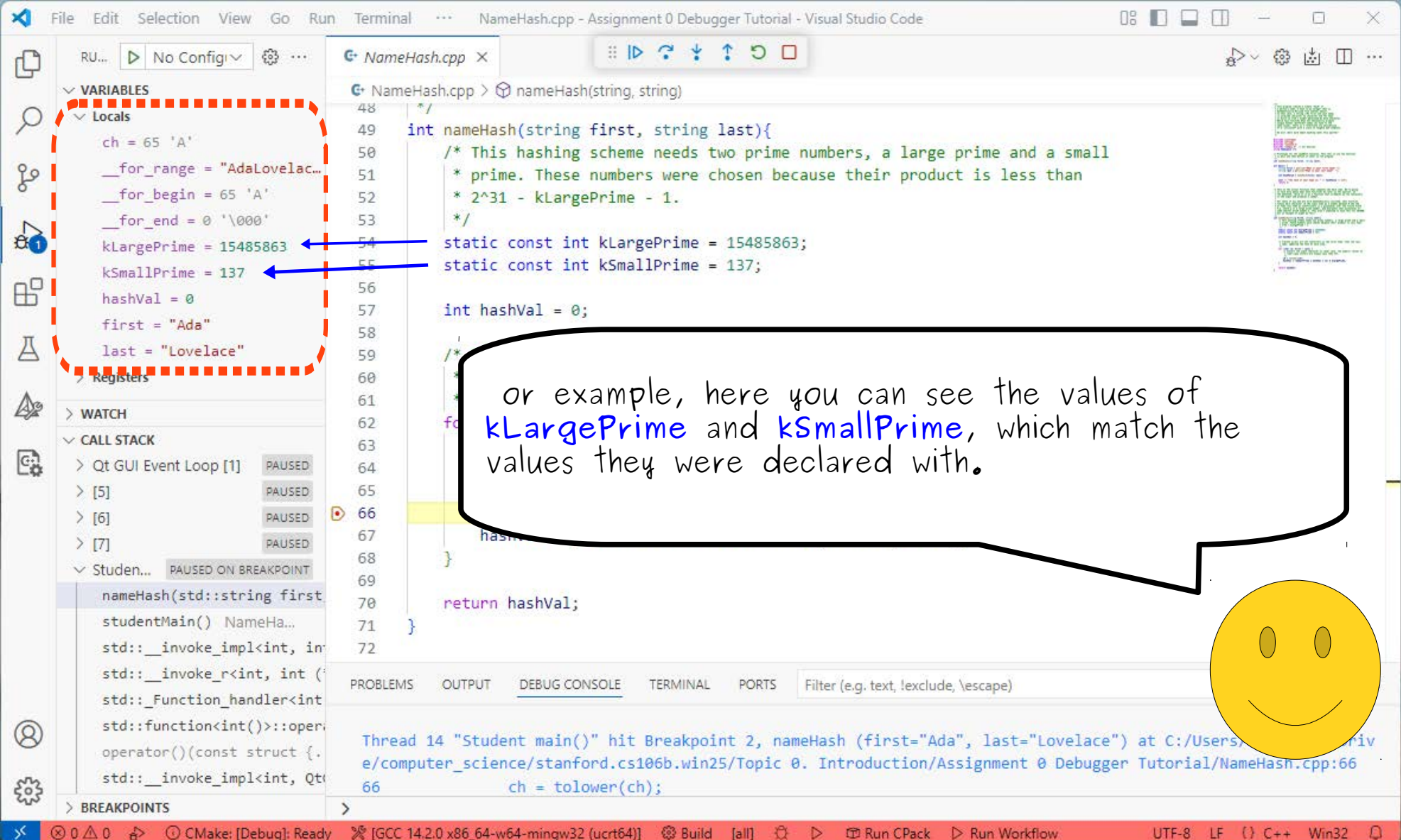


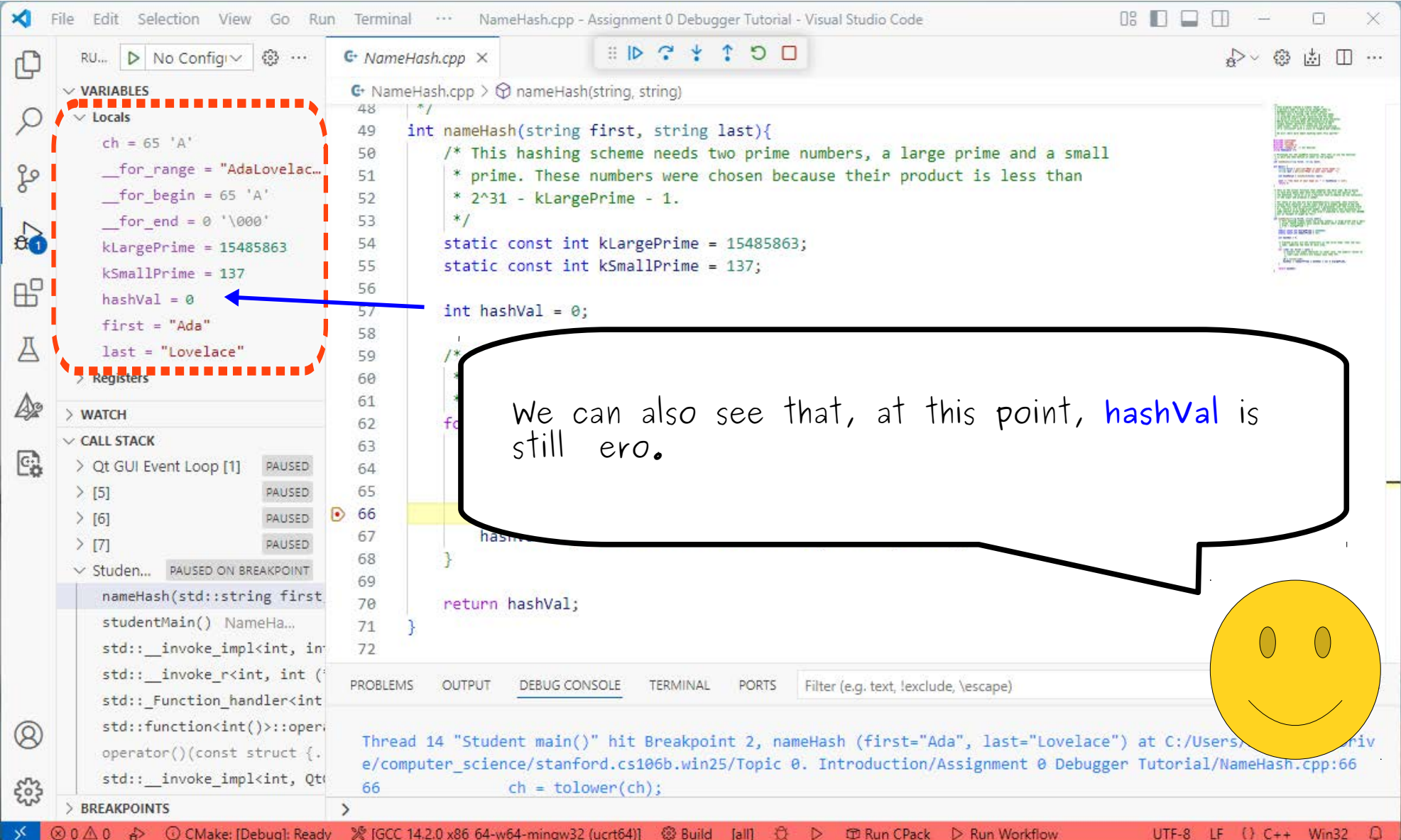












RU... No Config

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52 * primes is greater than the maximum possible value of the hashVal.

53 */

54 static const int kLargePrime = 15485863;

55 static const int kSmallPrime = 137;

56

57 int

58

59 /*

60 * name,

61 */

62 for (char ch: first + last) {

63 /* Convert the input character to lower case. The numeric values of

64 * lower-case letters are always less than 127.

65 */

66 ch = tolower(ch);

67 hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;

68 }

69

70 return hashVal;

71 }

72

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, !exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66

66 ch = tolower(ch);

As we walk through the program one step at a time, we'll see these values change.



File Edit Selection View Go Run Terminal ...

NameHash.cpp - Assignment 0 Debugger Tutorial - Visual Studio Code

UTF-8 LF C++ Win32

RU... No Config

VARIABLES

Locals

ch = 65 'A'

__for_range = "AdaLovelace"

__for_begin = 65 'A'

__for_end = 0 '\000'

kLargePrime = 15485863

kSmallPrime = 137

hashVal = 0

first = "Ada"

last = "Lovelace"

Registers

WATCH

CALL STACK

Qt GUI Event Loop [1] PAUSED

[5] PAUSED

[6] PAUSED

[7] PAUSED

Student... PAUSED ON BREAKPOINT

nameHash(std::string first

studentMain() NameHa...

std::__invoke_impl<int, in

std::__invoke_r<int, int (

std::_Function_handler<int

std::function<int()>::oper

operator()(const struct {.

std::__invoke_impl<int, Qt

BREAKPOINTS

NameHash.cpp

NameHash.cpp > nameHash(string, string)

48 */

49 int nameHash(string first, string last){

50 /* This hashing scheme needs two prime numbers, a large prime and a small

51 * prime

52 * prime

53 * prime

54 sta

55 sta

56

57 int

58

59 /* Iterate across all the characters in the first name, then the last

60 * name, updating the hash at each step.

61 */

62 for (char ch: first + last) {

63 /* Convert the input character to lower case. The numeric values of

64 * lower-case letters are always less than 127.

65 */

66 ch = tolower(ch);

67 hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;

68 }

69

70 return hashVal;

71 }

72

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, !exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66

66 ch = tolower(ch);

Now, let's take a look at this for loop.



RU... No Config

VARIABLES

Locals

ch = 65 'A'

__for_range = "AdaLovelac...

__for_begin = 65 'A'

__for_end = 0 '\000'

kLargePrime = 15485863

kSmallPrime = 137

hashVal = 0

first = "Ada"

last = "Lovelace"

Registers

WATCH

CALL STACK

Qt GUI Event Loop [1] PAUSED

[5] PAUSED

[6] PAUSED

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Student... PAUSED ON BREAKPOINT

nameHash(std::string first

studentMain() NameHa...

std::__invoke_impl<int, in

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std::_Function_handler<int

std::function<int()>::oper

operator()(const struct {.

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BREAKPOINTS

NameHash.cpp

NameHash.cpp > nameHash(string, string)

48 */

49 int nameHash(string first, string last){

50 /* This hashing scheme needs two prime numbers, a large prime and a small

51 * prime.

52 * prime.

53 * prime.

54 sta

55 sta

56

57 int

58

59 /* Iterate across all the characters in the first name, then the last

60 * name, updating the hash at each step.

61 */

62 for (char ch: first + last) {

63 /* Convert the input character to lower case. The numeric values of

64 * lower-case letters are always less than 127.

65 */

66 ch = tolower(ch);

67 hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;

68 }

69

70 return hashVal;

71 }

72

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, !exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66

66 ch = tolower(ch);

This loop is a range-based for loop. † says "for each character in the string first + last, do something with that character."

0 0 0 0 0

CMake: [Debug]: Ready

[GCC 14.2.0 x86_64-w64-mingw32 (ucrt64)]

Build [all]

Run CPack

Run Workflow

UTF-8 LF C++ Win32

RU... No Config

VARIABLES

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Registers

WATCH

CALL STACK

Qt GUI Event Loop [1] PAUSED

[5] PAUSED

[6] PAUSED

[7] PAUSED

Student... PAUSED ON BREAKPOINT

nameHash(std::string first

studentMain() NameHa...

std::__invoke_impl<int, in

std::__invoke_r<int, int (

std::_Function_handler<int

std::function<int()>::oper

operator()(const struct {.

std::__invoke_impl<int, Qt

BREAKPOINTS

NameHash.cpp

NameHash.cpp > nameHash(string, string)

48 /*

49 int nameHash(string first, string last){

50 /* This hashing scheme needs two prime numbers, a large prime and a small

51 * prime

52 * prime

53 * prime

54 sta

55 sta

56

57 int

58

59 /* Iterate across all the characters in the first name, then the last

60 * name, updating the hash at each step.

61 */

62 for (char ch: first + last) {

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64 * lower-case letters are always less than 127.

65 */

66 ch = tolower(ch);

67 hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;

68 }

69

70 return hashVal;

71 }

72

emember (from a while back) that we entered the name Ada Lovelace?

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66

66 ch = tolower(ch);

0 0 0 0 0

CMake: [Debug]: Ready

[GCC 14.2.0 x86_64-w64-mingw32 (ucrt64)]

Build [all]

Run CPack

Run Workflow

UTF-8 LF C++ Win32

RU... No Config

VARIABLES

Locals

ch = 65 'A'

__for_range = "AdaLovelac...

__for_begin = 65 'A'

__for_end = 0 '\000'

kLargePrime = 15485863

kSmallPrime = 137

hashVal = 0

first = "Ada"

last = "Lovelace"

Registers

WATCH

CALL STACK

Qt GUI Event Loop [1] PAUSED

[5] PAUSED

[6] PAUSED

[7] PAUSED

Student... PAUSED ON BREAKPOINT

nameHash(std::string first

studentMain() NameHa...

std::__invoke_impl<int, in

std::__invoke_r<int, int (

std::_Function_handler<int

std::function<int()>::oper

operator()(const struct {.

std::__invoke_impl<int, Qt

BREAKPOINTS

NameHash.cpp

NameHash.cpp > nameHash(string, string)

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54 sta

55 sta

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58

59 /* Iterate across all the characters in the first name, then the last

60 * name, updating the hash at each step.

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62 for (char ch: first + last) {

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72

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, !exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66

66 ch = tolower(ch);

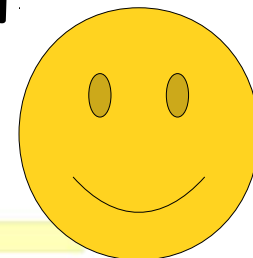
f we take a look at the current value of the variable **ch**, we can see that it has the value **A**. That's the first letter of the name Ada Lovelace.





```
string, string)
```

So now we know where we are (line 10), how we got there (main called `nameHash`), and the values in the program at this point.



```
Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66
66          ch = tolower(ch);
```

RU... No Config

VARIABLES

Locals

ch = 65 'A'

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__for_begin = 65 'A'

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kLargePrime = 15485863

kSmallPrime = 137

hashVal = 0

first = "Ada"

last = "Lovelace"

Registers

WATCH

CALL STACK

Qt GUI Event Loop [1] PAUSED

[5] PAUSED

[6] PAUSED

[7] PAUSED

Student... PAUSED ON BREAKPOINT

nameHash(std::string first

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BREAKPOINTS

NameHash.cpp

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49 int nameHash(string first, string last){

50 /* This hashing scheme needs two prime numbers, a large prime and a small

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52 * prime

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56

57 int

58

59 /* Iterate across all the characters in the first name, then the last

60 * name, updating the hash at each step.

61 */

62 for (char ch: first + last) {

63 /* Convert the input character to lower case. The numeric values of

64 * lower-case letters are always less than 127.

65 */

66 ch = tolower(ch);

67 hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;

68 }

69

70 return hashVal;

71 }

72

Now, let's do something really cool - we're going to run this program one line at a time, watching what happens at each step!



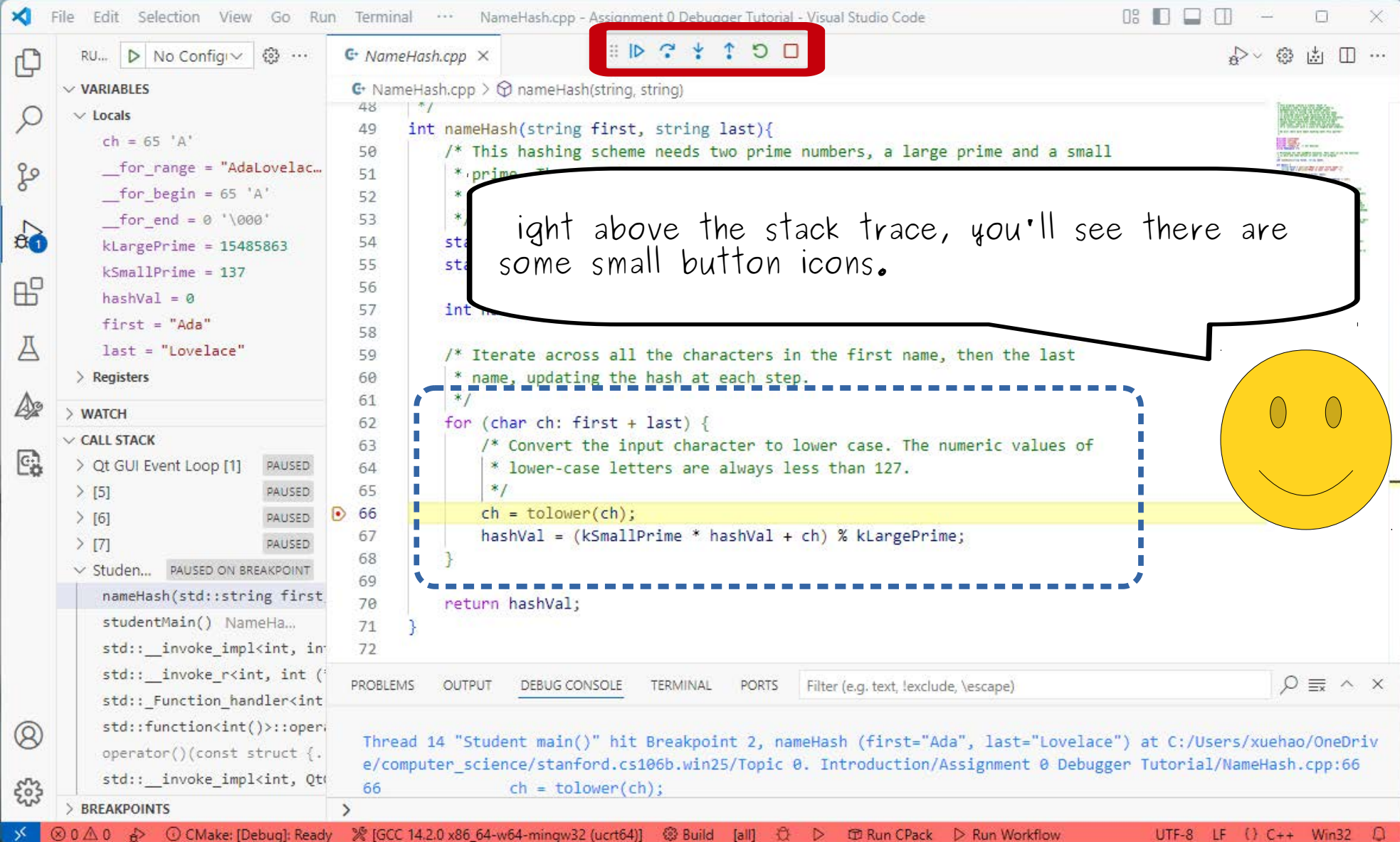
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

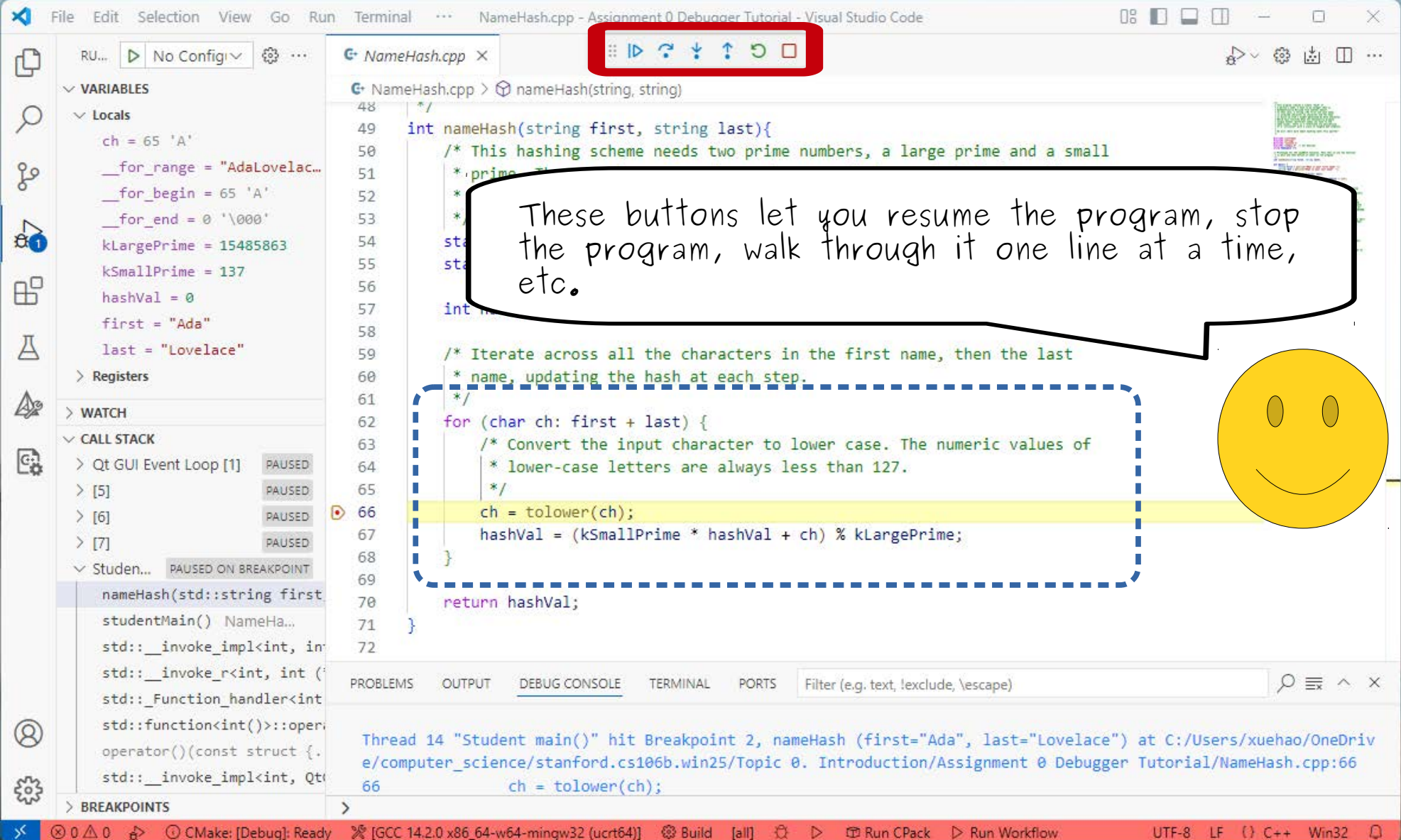
Filter (e.g. text, !exclude, \escape)

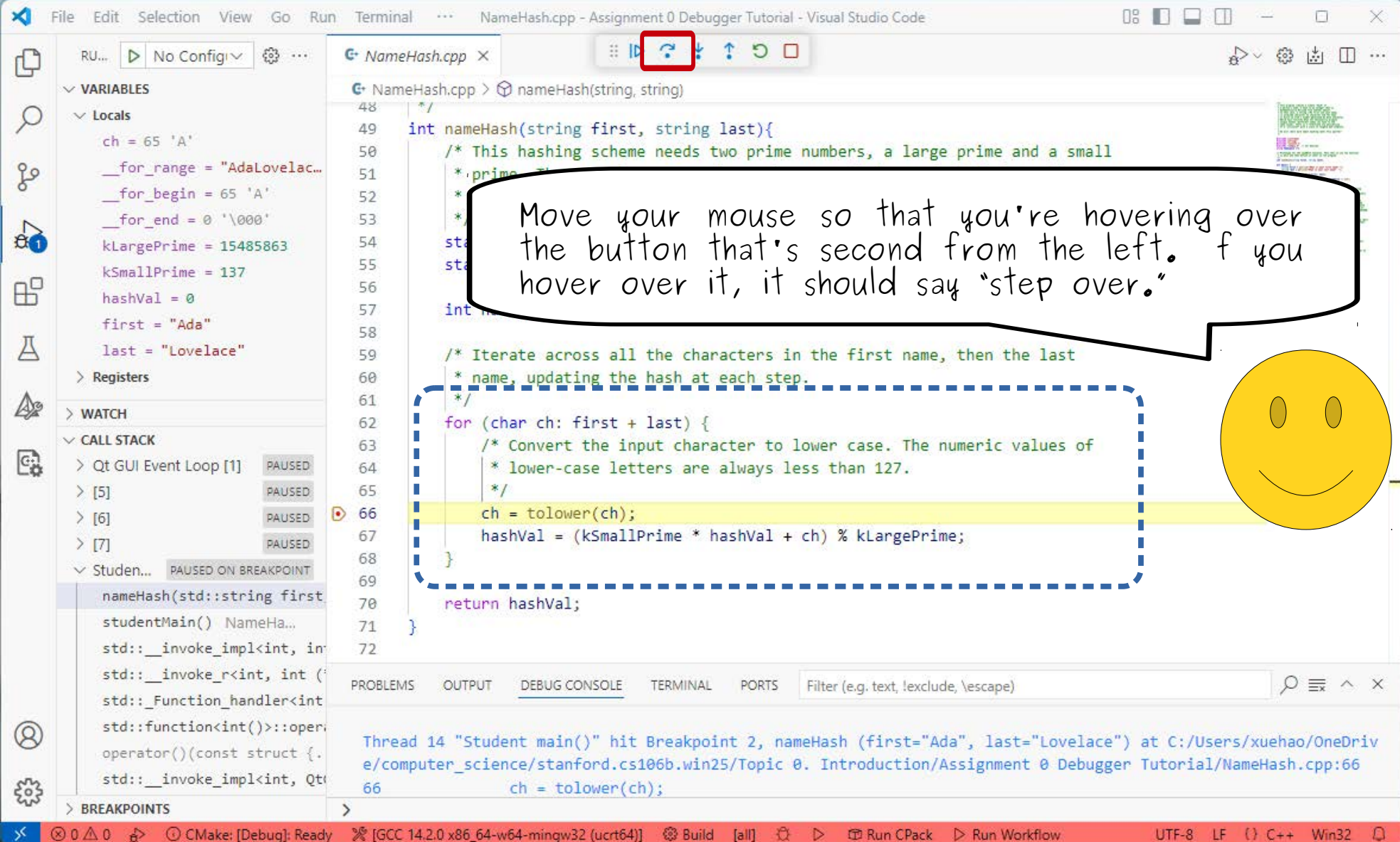
Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66

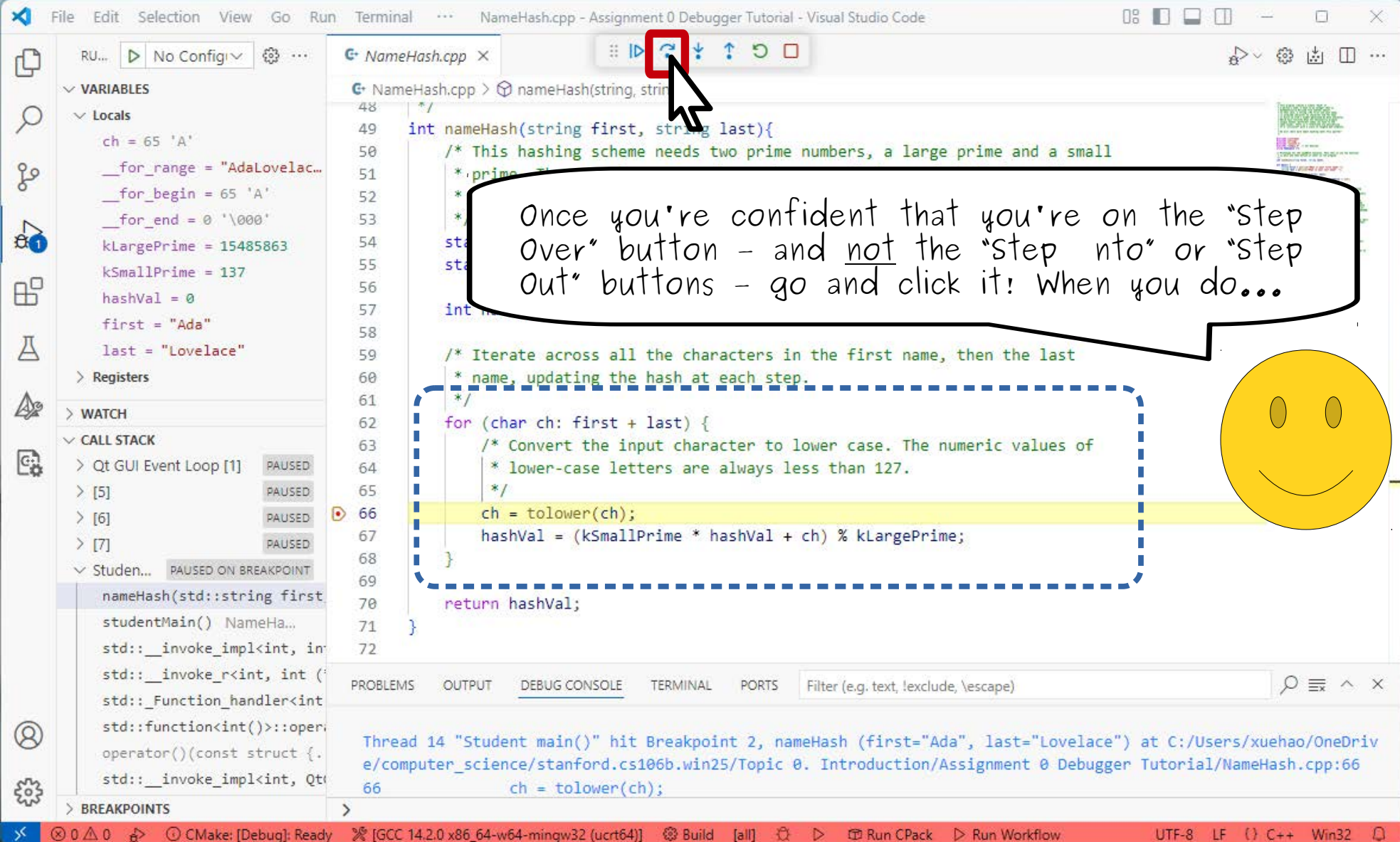
66 ch = tolower(ch);

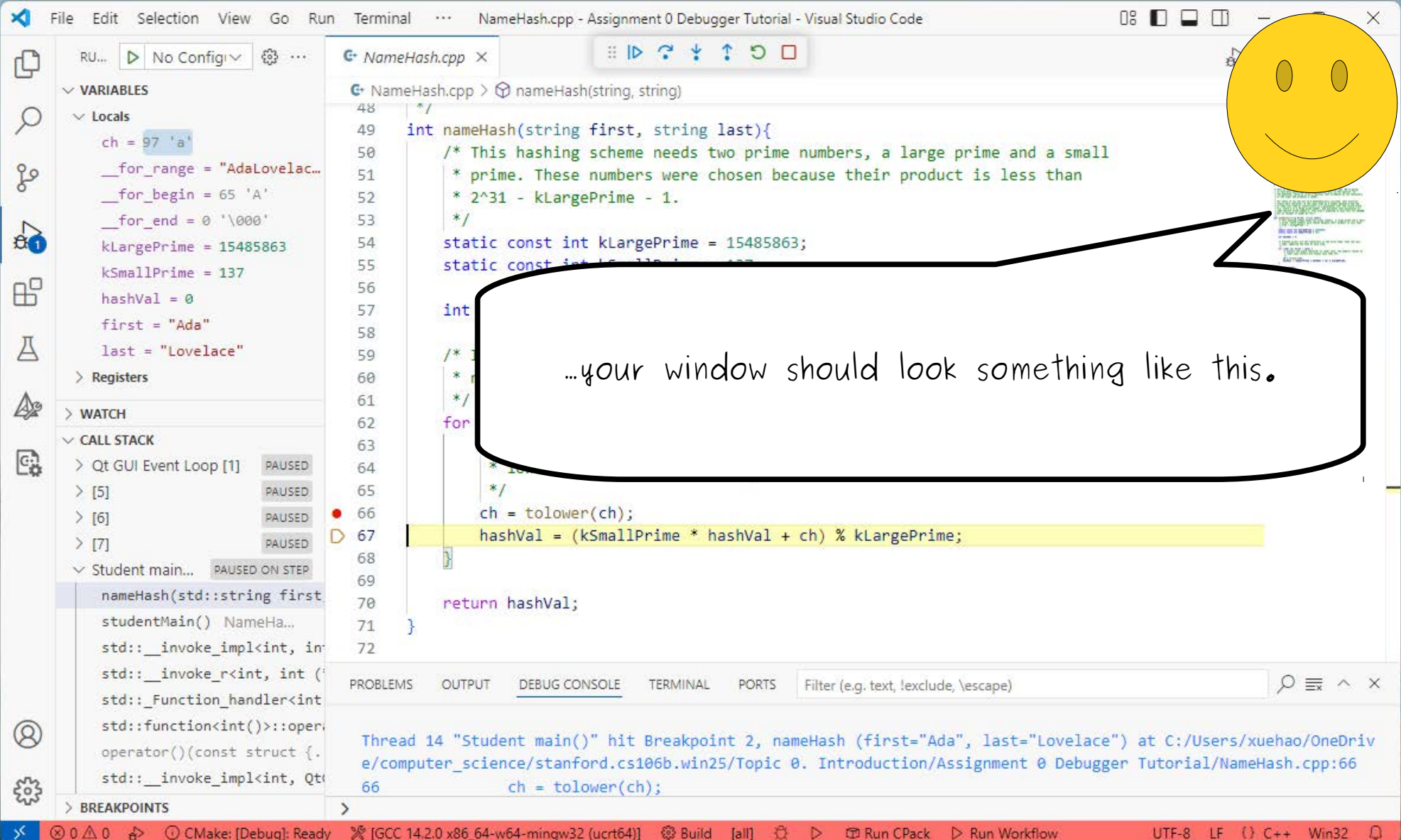
CMake: [Debug]: Ready [GCC 14.2.0 x86_64-w64-mingw32 (ucrt64)] Build [all] Run CPack Run Workflow UTF-8 LF C++ Win32

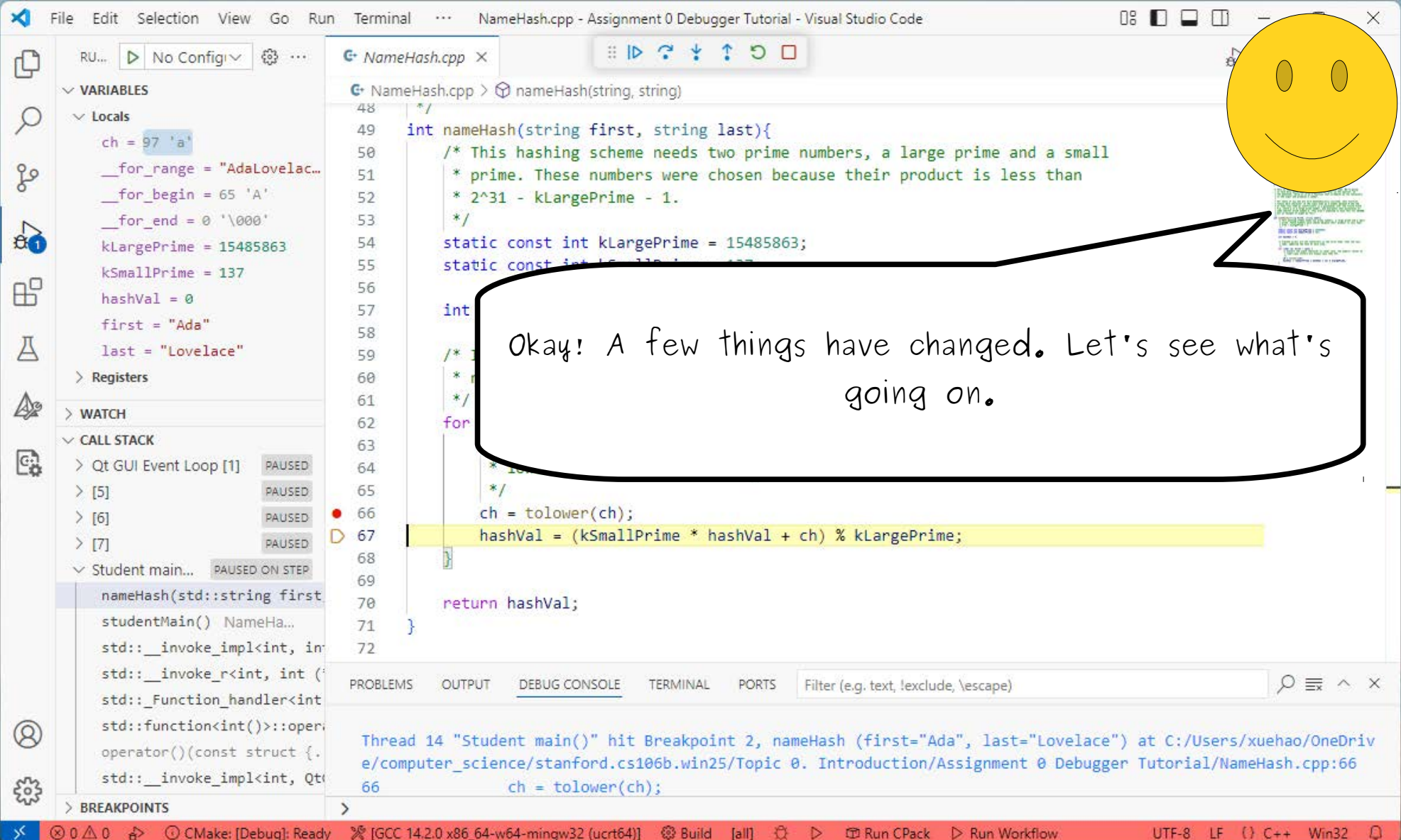












Okay! A few things have changed. Let's see what's going on.

NameHash.cpp > nameHash(string, string)

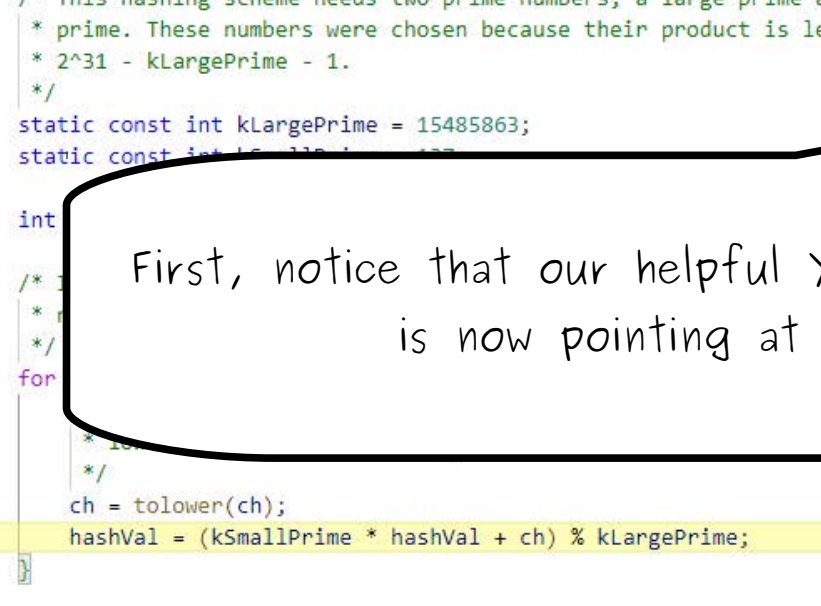
```
48  */
49  int nameHash(string first, string last){
50      /* This hashing scheme needs two prime numbers, a large prime and a small
51       * prime. These numbers were chosen because their product is less than
52       * 2^31 - kLargePrime - 1.
53       */
54      static const int kLargePrime = 15485863;
55      static const int kSmallPrime = 137;
56
57      int
58
59      /*
60       *
61       */
62      for
63
64      * 10
65      */
66      ch = tolower(ch);
67      hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68  }
69
70      return hashVal;
71  }
72  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66
66 ch = tolower(ch);



First, notice that our helpful Yellow Arrow friend is now pointing at line 67.



The image shows a C++ code snippet for a hash function. A handwritten note in a black box points to line 67, which is highlighted in yellow. The code uses two prime numbers, `kSmallPrime` and `kLargePrime`, to calculate a hash value. The note explains that the yellow highlight is now pointing to line 67.

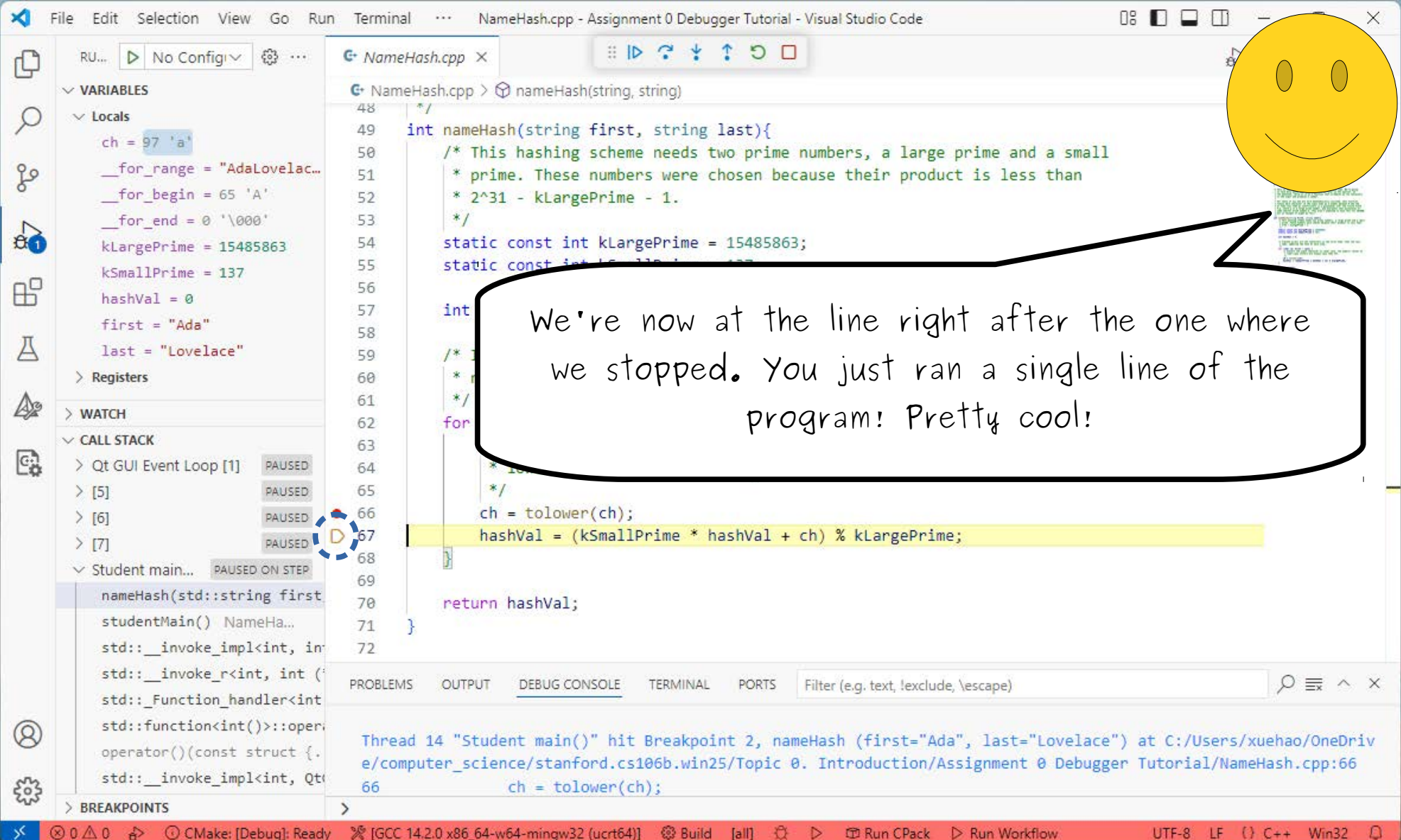
```

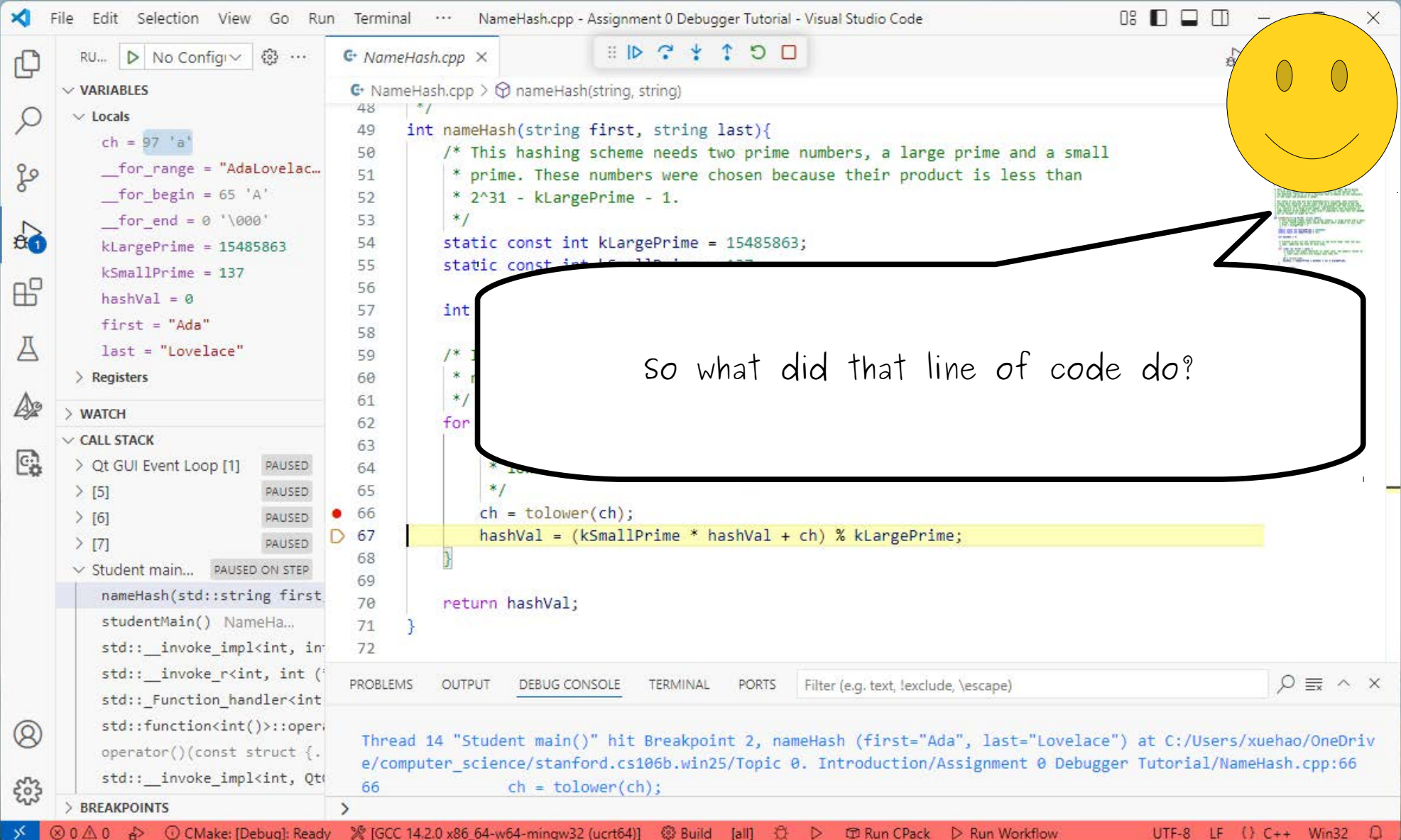
48  */
49  int nameHash(string first, string last){
50      /* This hashing scheme needs two prime numbers, a large prime and a small
51       * prime. These numbers were chosen because their product is less than
52       * 2^31 - kLargePrime - 1.
53       */
54      static const int kLargePrime = 15485863;
55      static const int kSmallPrime = 1013;
56
57      int
58
59      /*
60       *
61       */
62      for
63
64
65
66      ch = tolower(ch);
67      hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68  }
69
70      return hashVal;
71  }
72

```

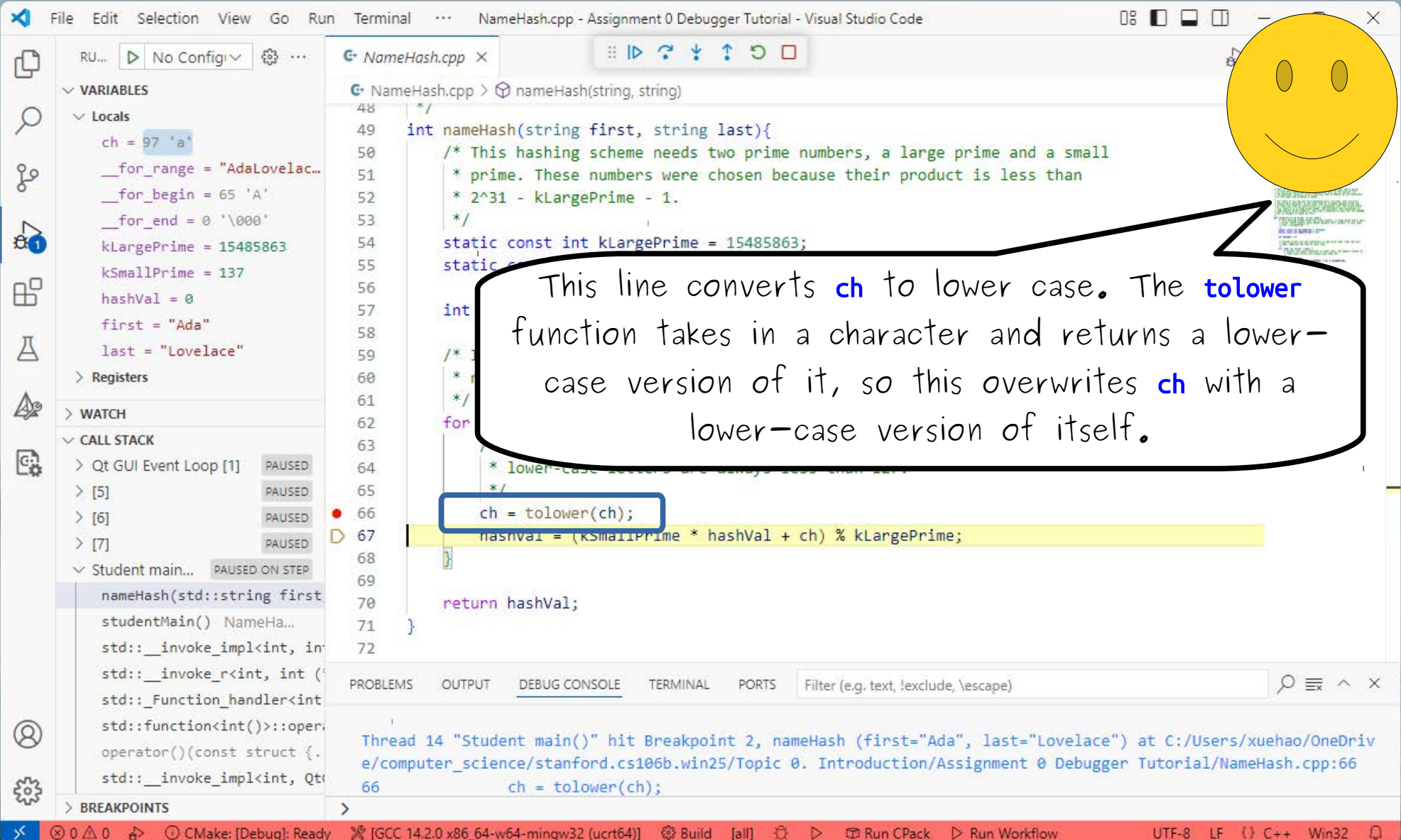
First, notice that our helpful yellow is now pointing at line 67

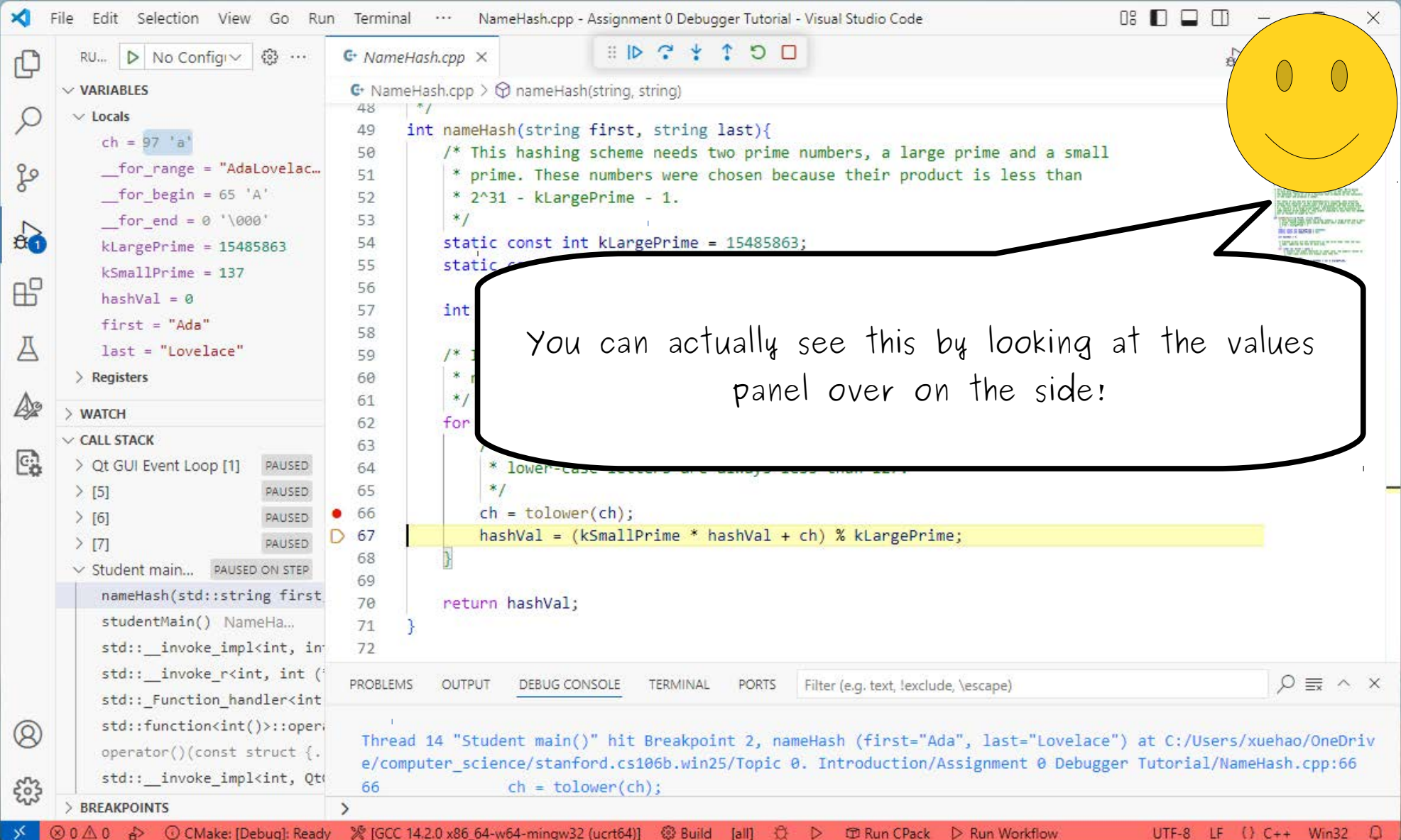
⌕ ≡ ^ ×





So what did that line of code do?





You can actually see this by looking at the values panel over on the side!

NameHash.cpp > nameHash(string, string)

```
48  /*
49  int nameHash(string first, string last){
50      /* This hashing scheme needs two prime numbers, a large prime and a small
51      * prime. These numbers were chosen because their product is less than
52      * 2^31 - kLargePrime - 1.
53      */
54      static const int kLargePrime = 15485863;
55      static const int kSmallPrime = 137;
56
57      int hashVal = 0;
58
59      for (int i = 0; i < first.length(); i++)
60      {
61          *
62          /*
63          * lower-case letters are always less than 26
64          */
65
66          ch = tolower(ch);
67          hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68      }
69
70      return hashVal;
71  }
72  */
```

VARIABLES

Locals

ch = 97 'a'
__for_range = "AdaLovelace"
__for_begin = 65 'A'
__for_end = 0 '\000'
kLargePrime = 15485863
kSmallPrime = 137
hashVal = 0
first = "Ada"
last = "Lovelace"

Registers

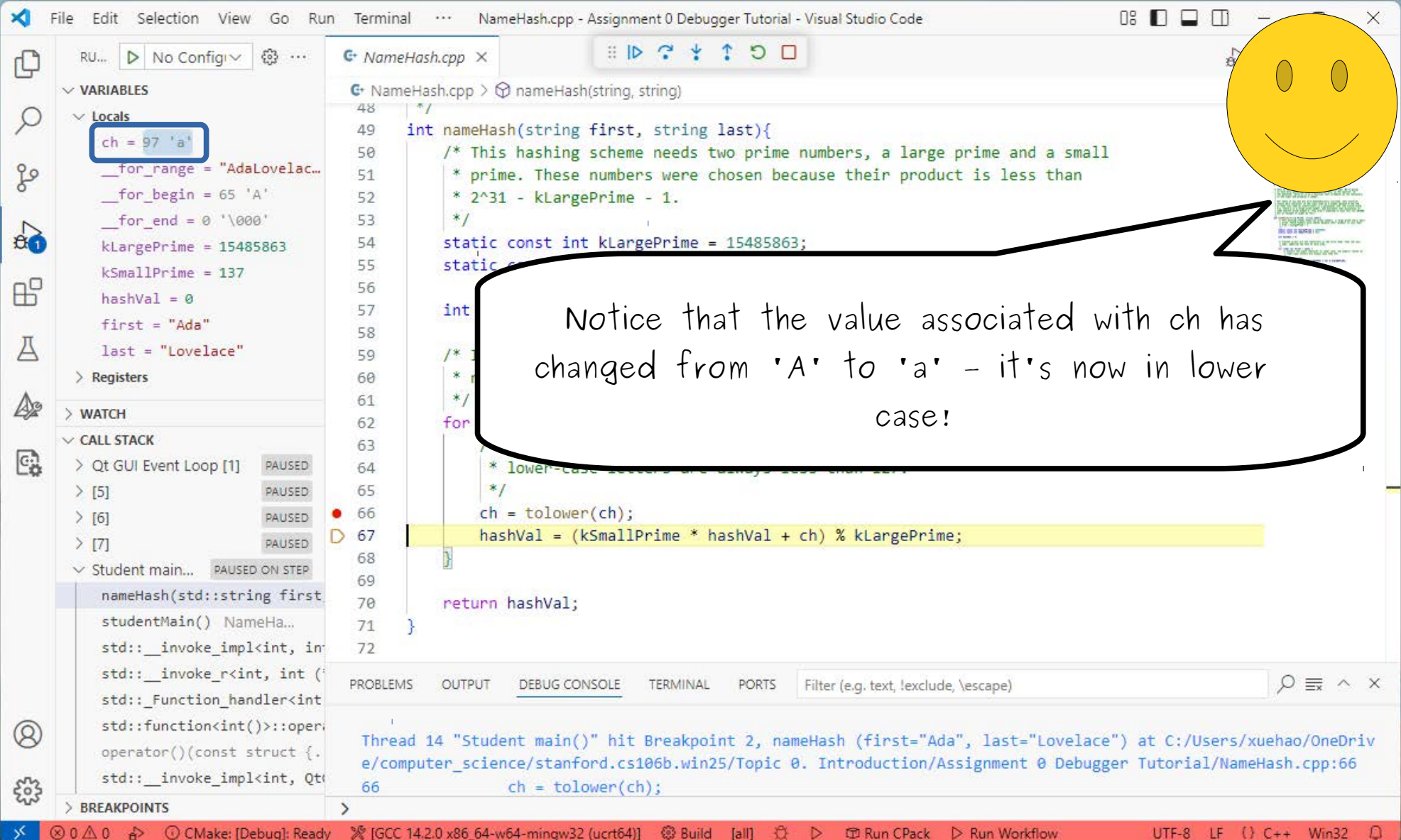
WATCH

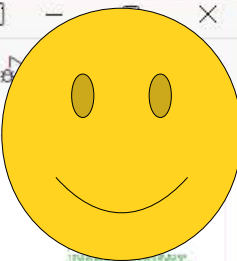
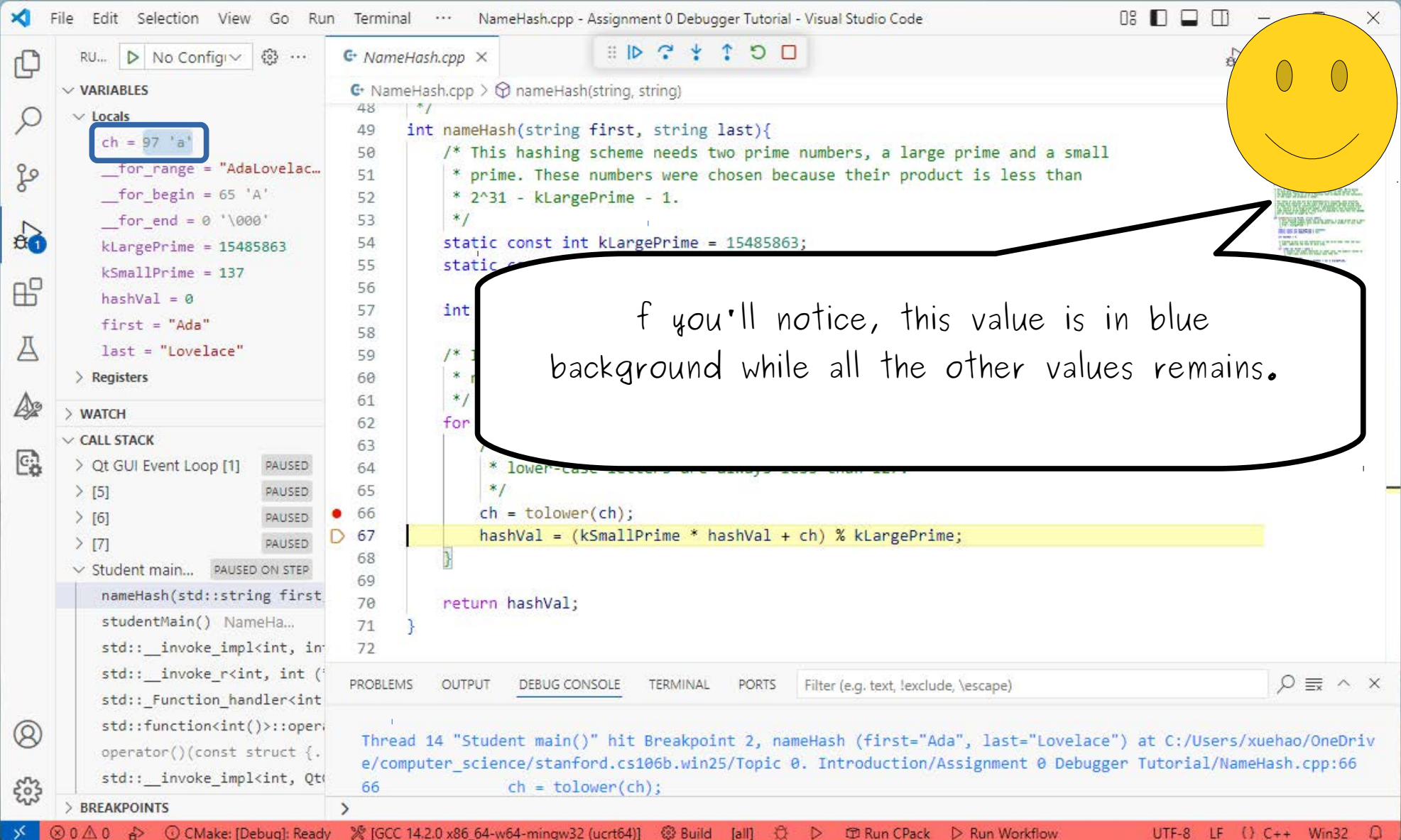
CALL STACK

> Qt GUI Event Loop [1] PAUSED
> [5] PAUSED
> [6] PAUSED
> [7] PAUSED
Student main... PAUSED ON STEP
nameHash(std::string first
studentMain() NameHa...
std::__invoke_impl<int, in
std::__invoke_r<int, int (<
std::_Function_handler<int
std::function<int()>::oper
operator()(const struct {
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66
66 ch = tolower(ch);





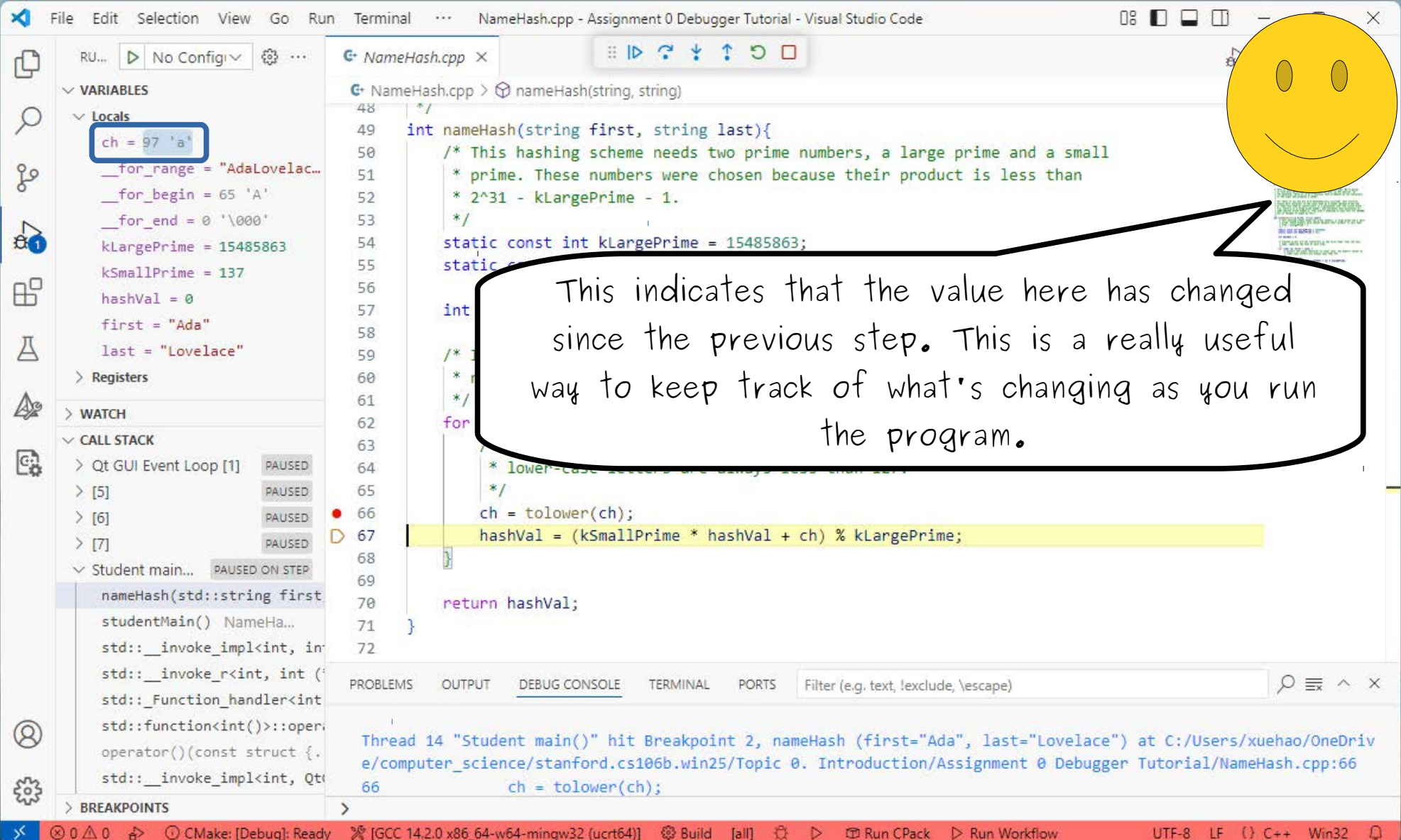
f you'll notice, this value is in blue background while all the other values remains.

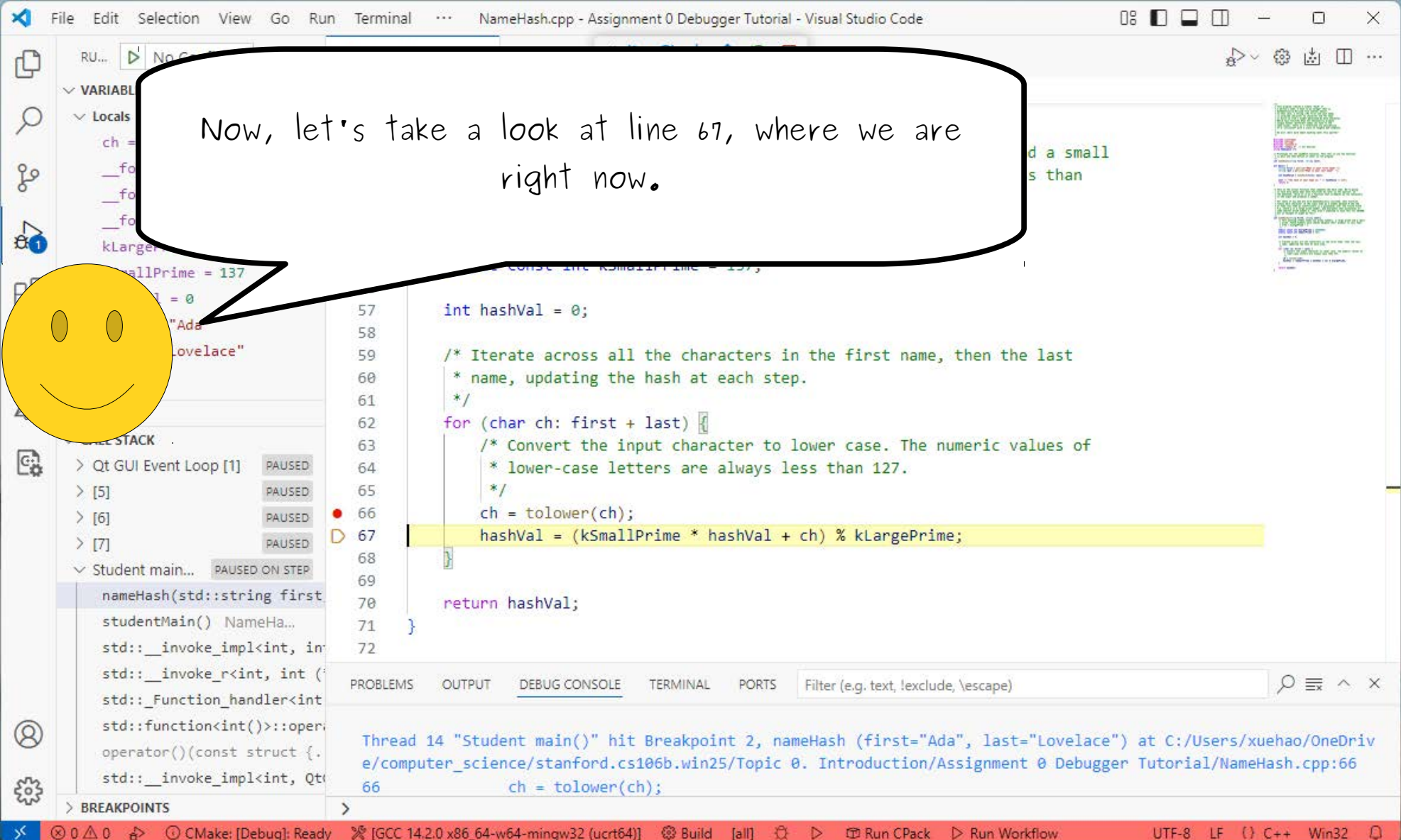
NameHash.cpp > nameHash(string, string)

```
48  /*
49  int nameHash(string first, string last){
50      /* This hashing scheme needs two prime numbers, a large prime and a small
51      * prime. These numbers were chosen because their product is less than
52      * 2^31 - kLargePrime - 1.
53      */
54      static const int kLargePrime = 15485863;
55      static const int kSmallPrime = 137;
56
57      int hashVal = 0;
58      first = "Ada";
59      last = "Lovelace";
60
61      for (int i = 0; i < first.length(); i++)
62      {
63          /* lower-case letters are always less than 26
64          */
65          ch = tolower(first[i]);
66          hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
67      }
68
69      return hashVal;
70  }
71
72  }
```

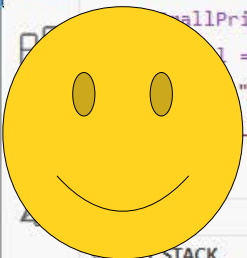
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66
66 ch = tolower(ch);





Now, let's take a look at line 67, where we are right now.

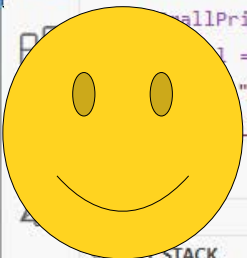


```
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first name, then the last
60      * name, updating the hash at each step.
61      */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case. The numeric values of
64          * lower-case letters are always less than 127.
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69
70     return hashVal;
71 }
72
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66
66 ch = tolower(ch);

Not gonna lie, this is a pretty dense line of code. It performs some weird sort of mathematical calculation on a bunch of different values.

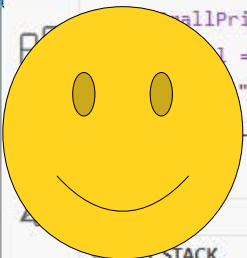


```
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first name, then the last
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61      */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case. The numeric values of
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65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69
70     return hashVal;
71 }
72
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66
66 ch = tolower(ch);

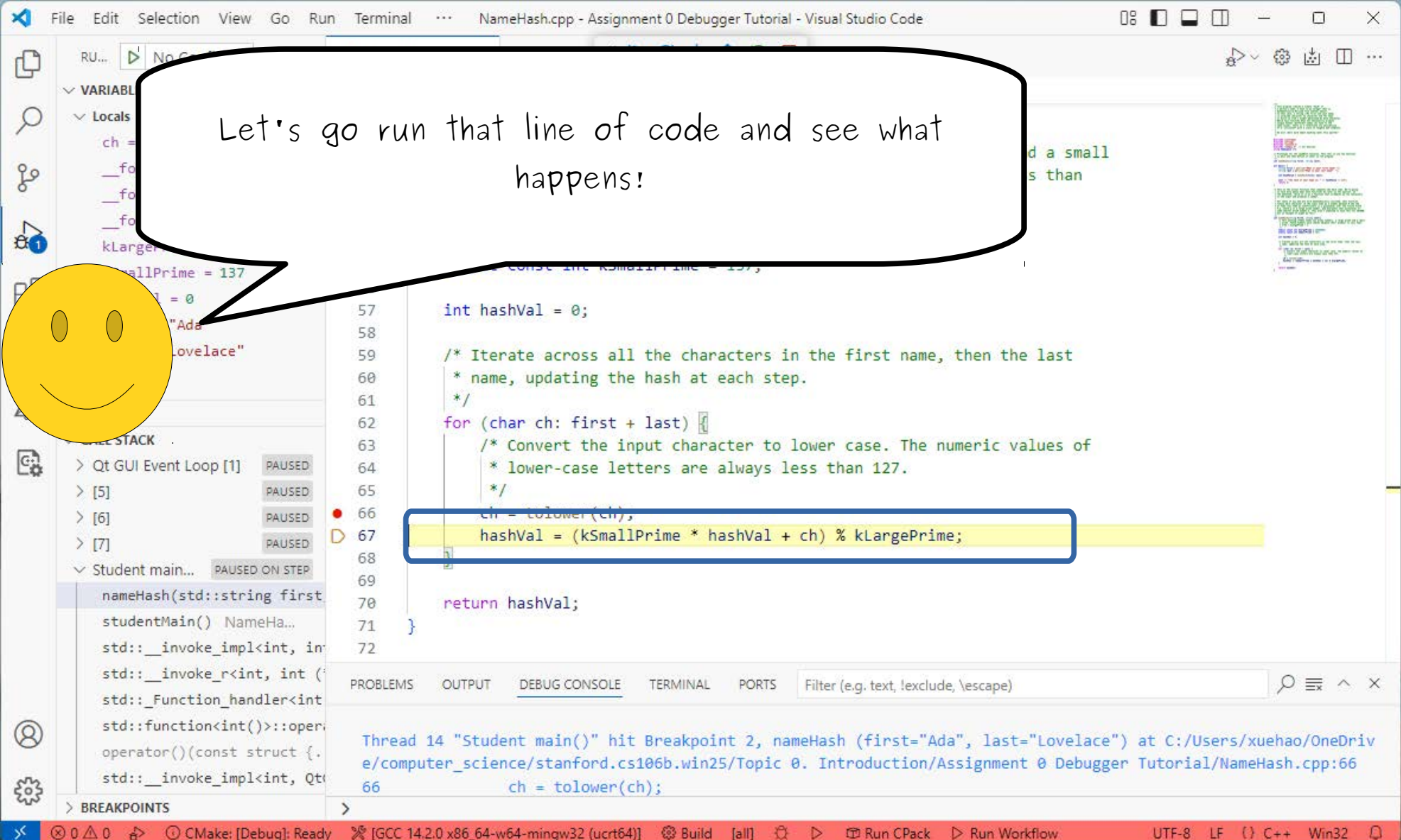
Fundamentally, though, it's just computing some weird function of some values and stashing it into **hashVal**.

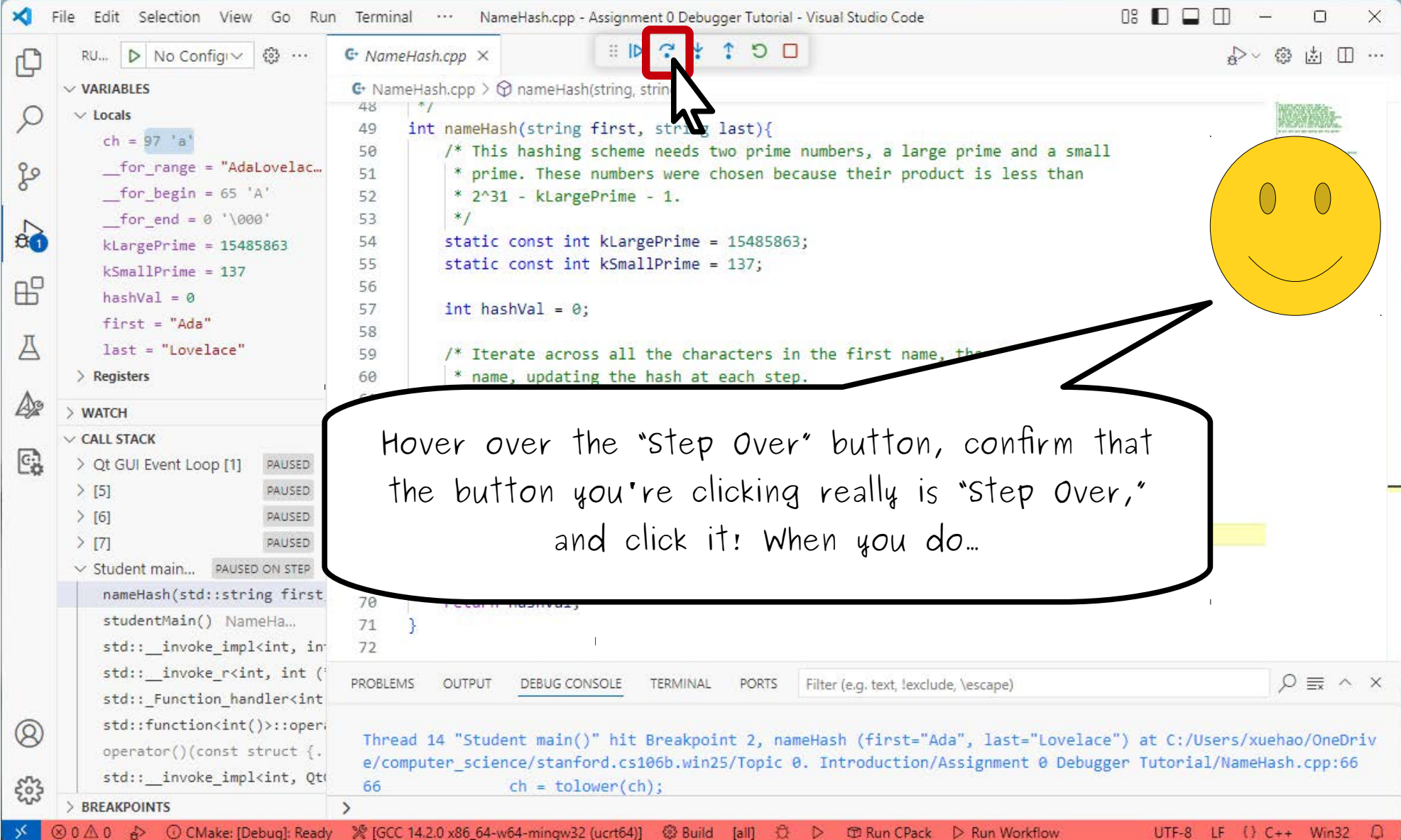


```
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first name, then the last
60      * name, updating the hash at each step.
61      */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case. The numeric values of
64          * lower-case letters are always less than 127.
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69
70     return hashVal;
71 }
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```

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66 ch = tolower(ch);





Hover over the "Step Over" button, confirm that the button you're clicking really is "Step Over," and click it! When you do...

RU...

No Config

Variables

Locals

Registers

WATCH

CALL STACK

BREAKPOINTS

ch = 97 'a'

__for_range = "AdaLovelace..."

__for_begin = 65 'A'

__for_end = 0 '\000'

kLargePrime = 15485863

kSmallPrime = 137

hashVal = 97

first = "Ada"

last = "Lovelace"

Qt GUI Event Loop [1]

[5]

[6]

[7]

Student main...

nameHash(std::string first

studentMain() NameHa...

std::__invoke_impl<int, in

std::__invoke_r<int, int (

std::_Function_handler<int

std::function<int()>::oper

operator()(const struct {.

std::__invoke_impl<int, Qt

NameHash.cpp

nameHash(string, string)

48 */

49 int nameHash(string first, string last){

50 /* This hashing scheme needs two prime numbers, a large prime and a small

51 * prime. These numbers were chosen because their product is less than

52 * 2^31 - kLargePrime - 1.

53 */

54 static const int kLargePrime = 15485863;

55 static const int kSmallPrime = 137;

56

57 int hashVal = 0;

58

59 /* Iterate across all the characters in the first name, then the last

60 * name, updating the hash at each step.

61 */

62 for (char ch: first + last) {

63 /* Convert the input character to lower case. The numeric values of

64 * lower-case letters are always less than 127.

65 */

66 ch = tolower(ch);

67

68 }

69

70 ret

71 }

72

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, !exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66

66 ch = tolower(ch);

0 0 0 0

CMake: [Debug]: Ready

[GCC 14.2.0 x86_64-w64-mingw32 (ucrt64)]

Build [all]

Run CPack

Run Workflow

UTF-8 LF

C++ Win32

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RU... No Config

VARIABLES

Locals

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__for_range = "AdaLovelace"

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Registers

WATCH

CALL STACK

Qt GUI Event Loop [1] PAUSED

[5] PAUSED

[6] PAUSED

[7] PAUSED

Student main... PAUSED ON STEP

nameHash(std::string first

studentMain() NameHa...

std::__invoke_impl<int, in

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std::__invoke_impl<int, Qt

BREAKPOINTS

NameHash.cpp x

int nameHash(string first, string last){

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* prime. These numbers were chosen because their product is less than

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static const int kLargePrime = 15485863;

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int hashVal = 0;

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* name, updating the hash at each step.

*/

for (char ch: first + last) {

/* Convert the input character to lower case. The numeric values of

* lower-case letters are always less than 127.

*/

ch = tolower(ch);

}

ret

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

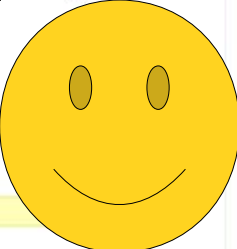
PORTS

Filter (e.g. text, !exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66

66 ch = tolower(ch);

Let's see what's changed.



File Edit Selection View Go Run Terminal ... NameHash.cpp - Assignment 0 Debugger Tutorial - Visual Studio Code

RU... No Config

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OUTPUT

DEBUG CONSOLE

TERMINAL

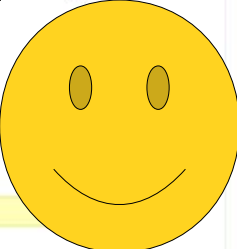
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TERMINAL

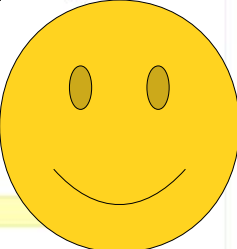
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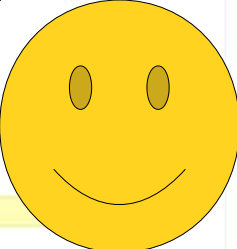
PORTS

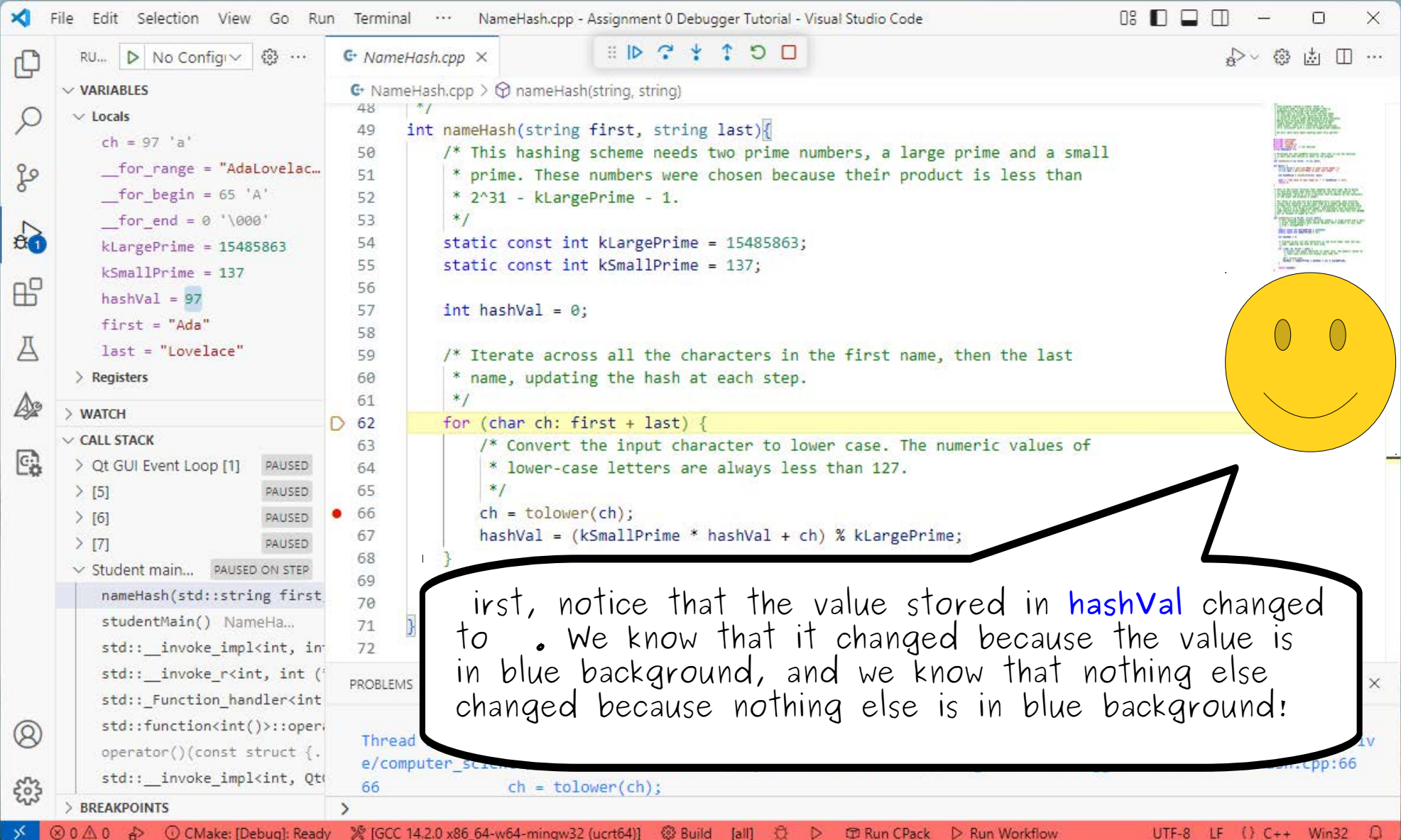
Filter (e.g. text, !exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66

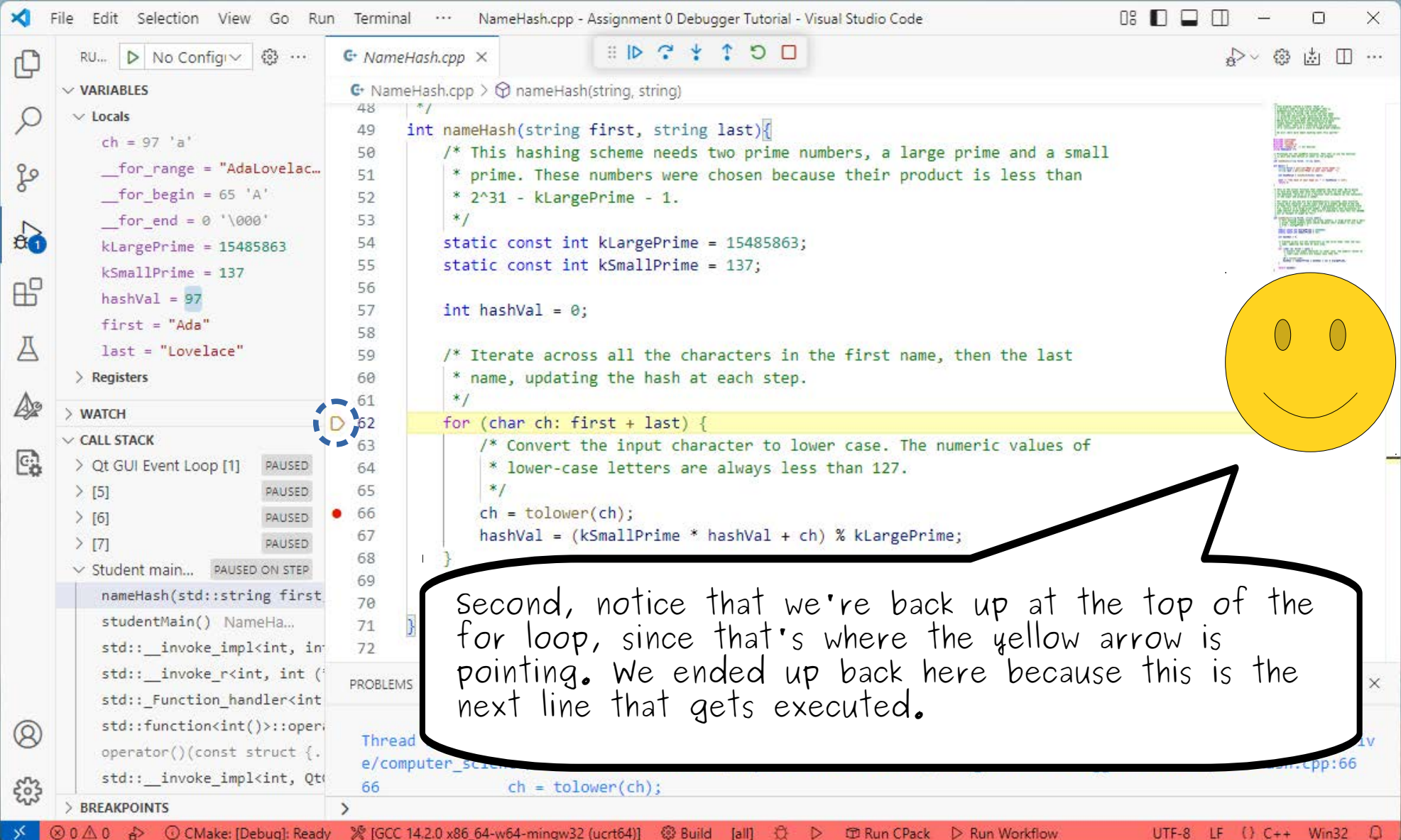
66 ch = tolower(ch);

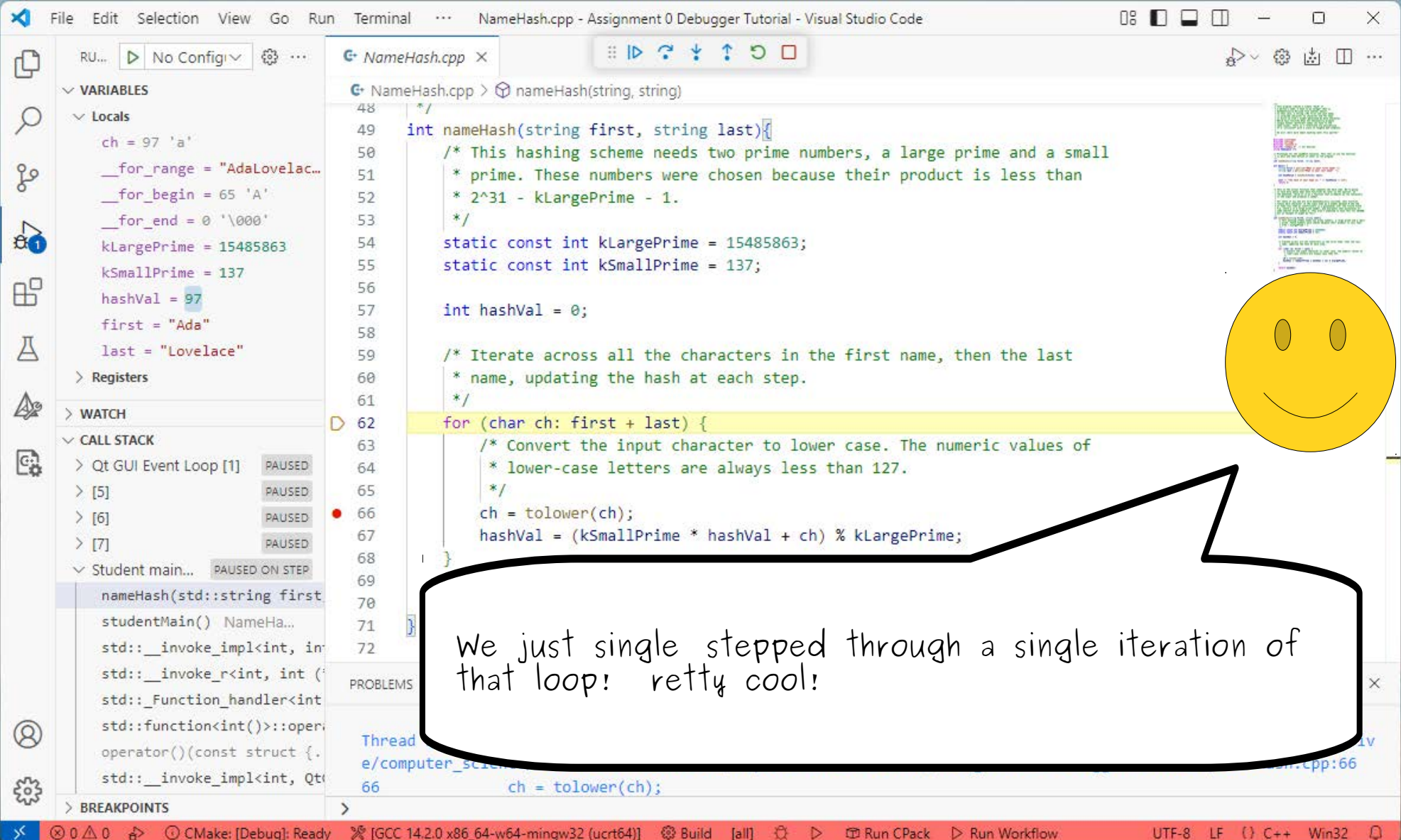
Let's see what's changed.

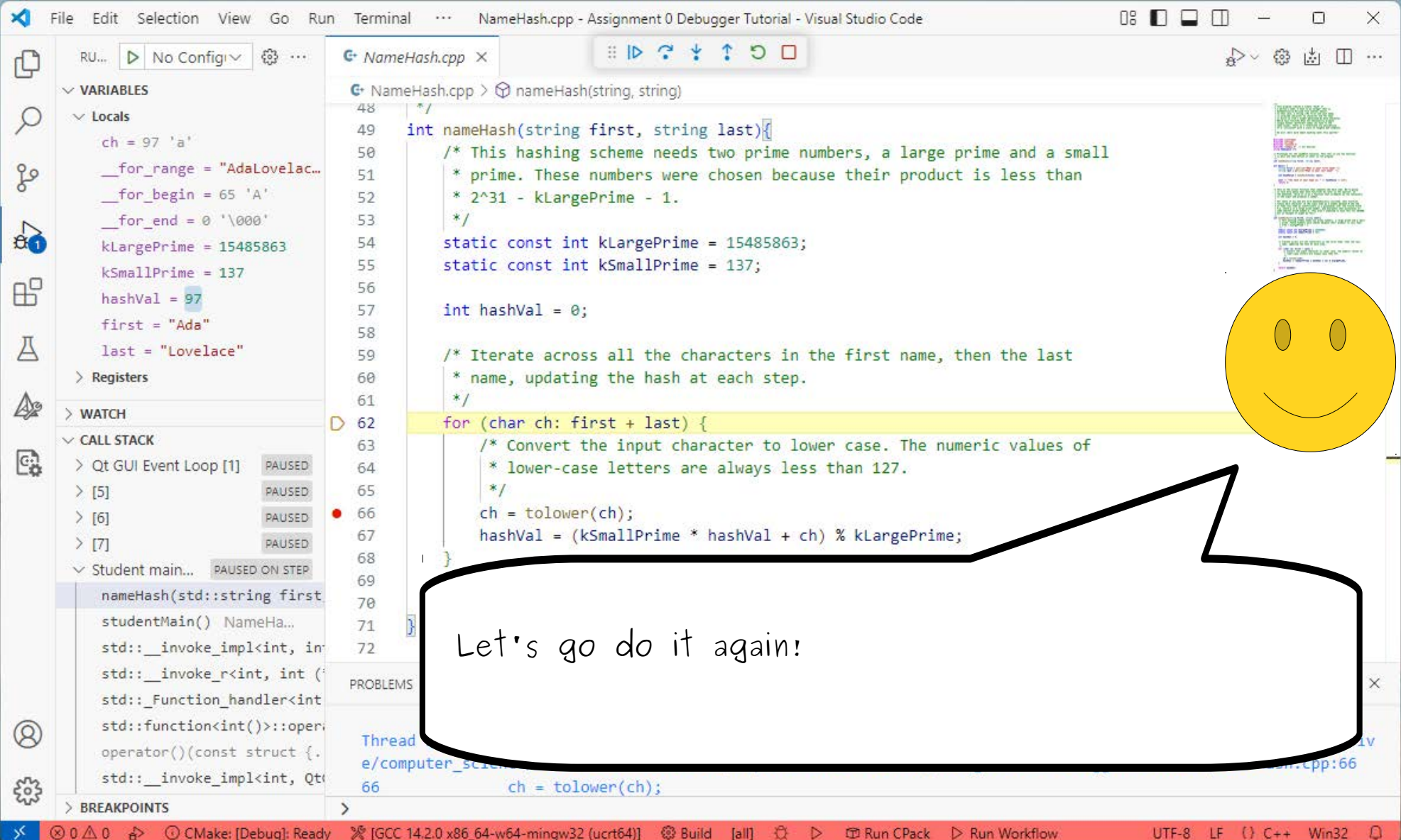


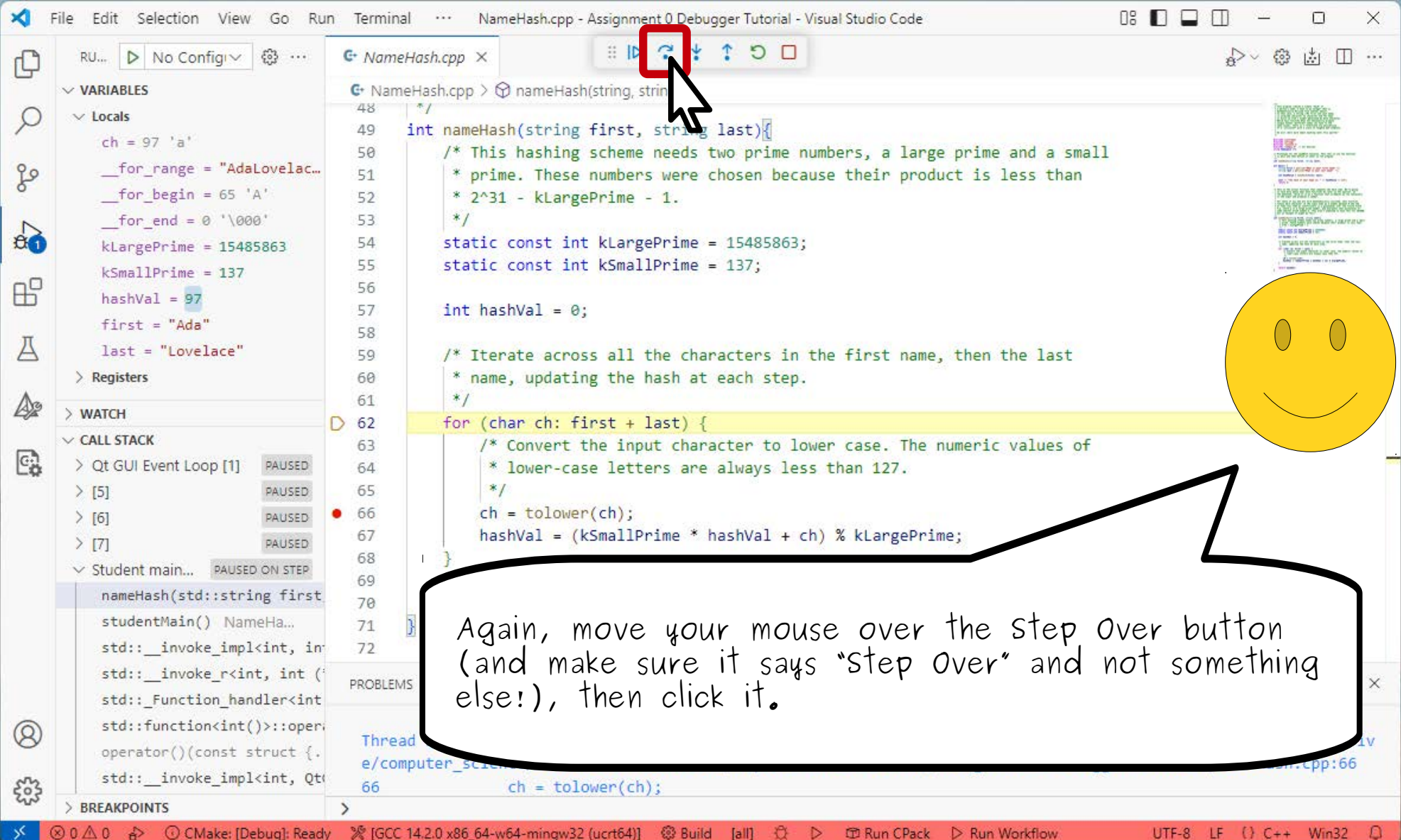


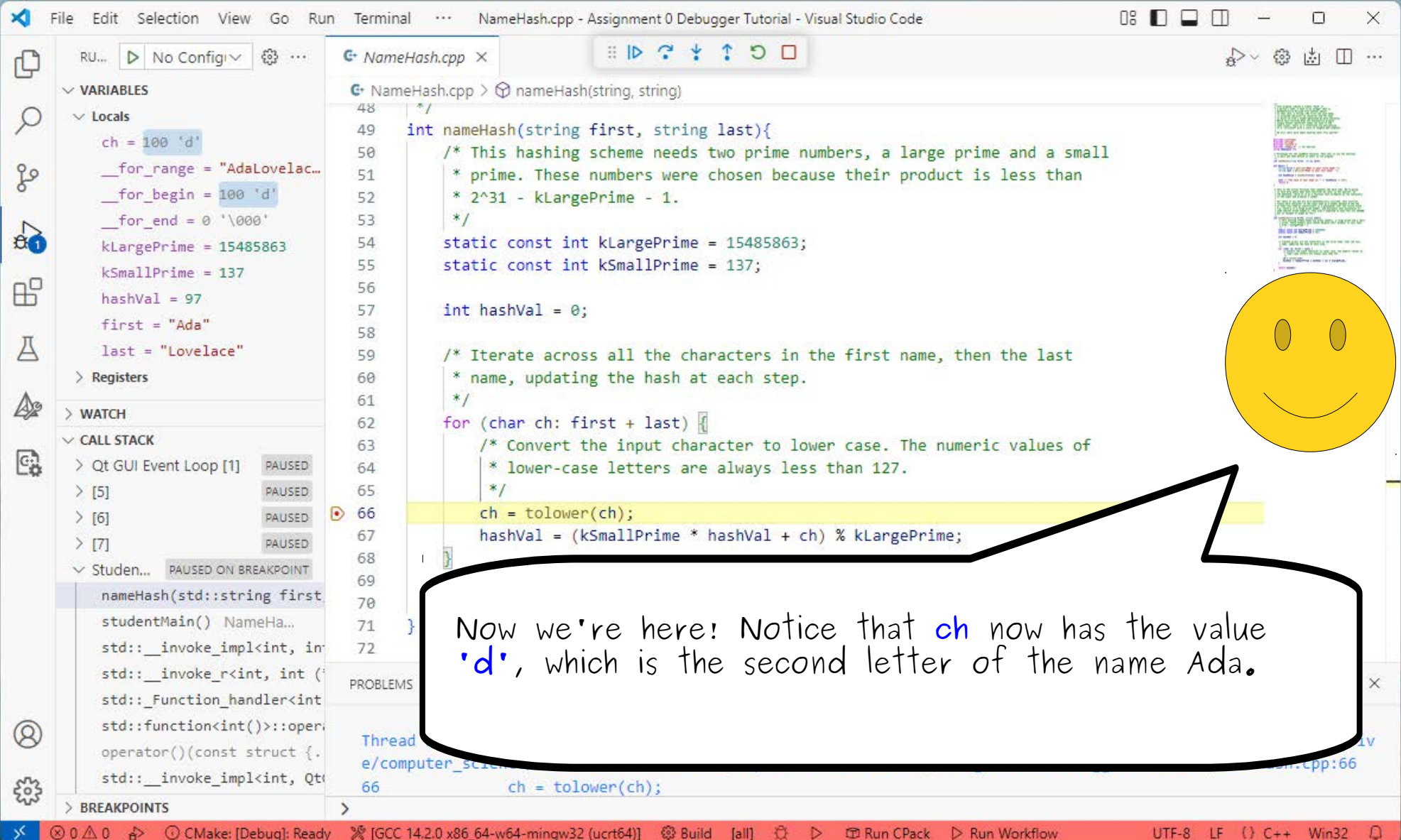
irst, notice that the value stored in `hashVal` changed to `97`. We know that it changed because the value is in blue background, and we know that nothing else changed because nothing else is in blue background!

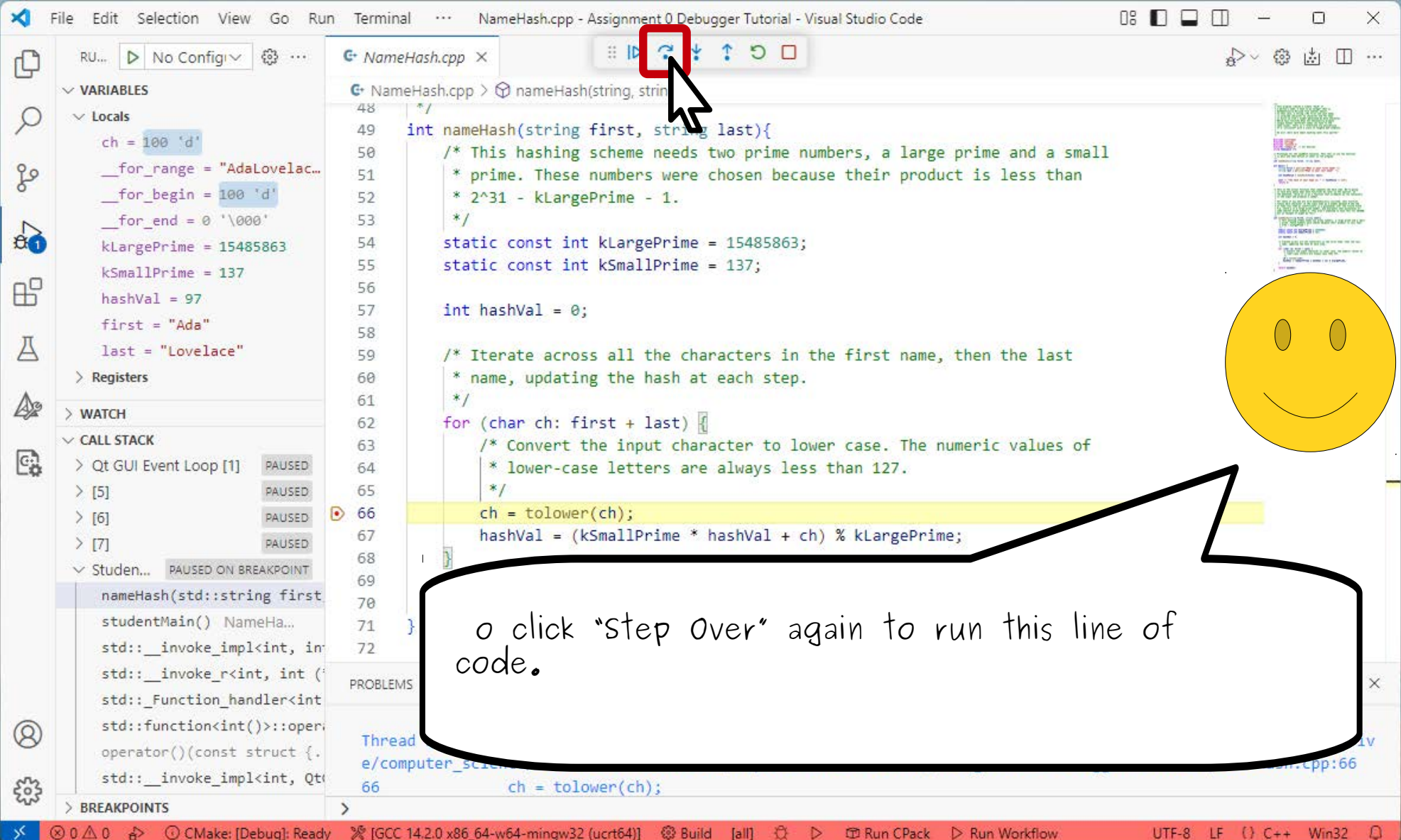


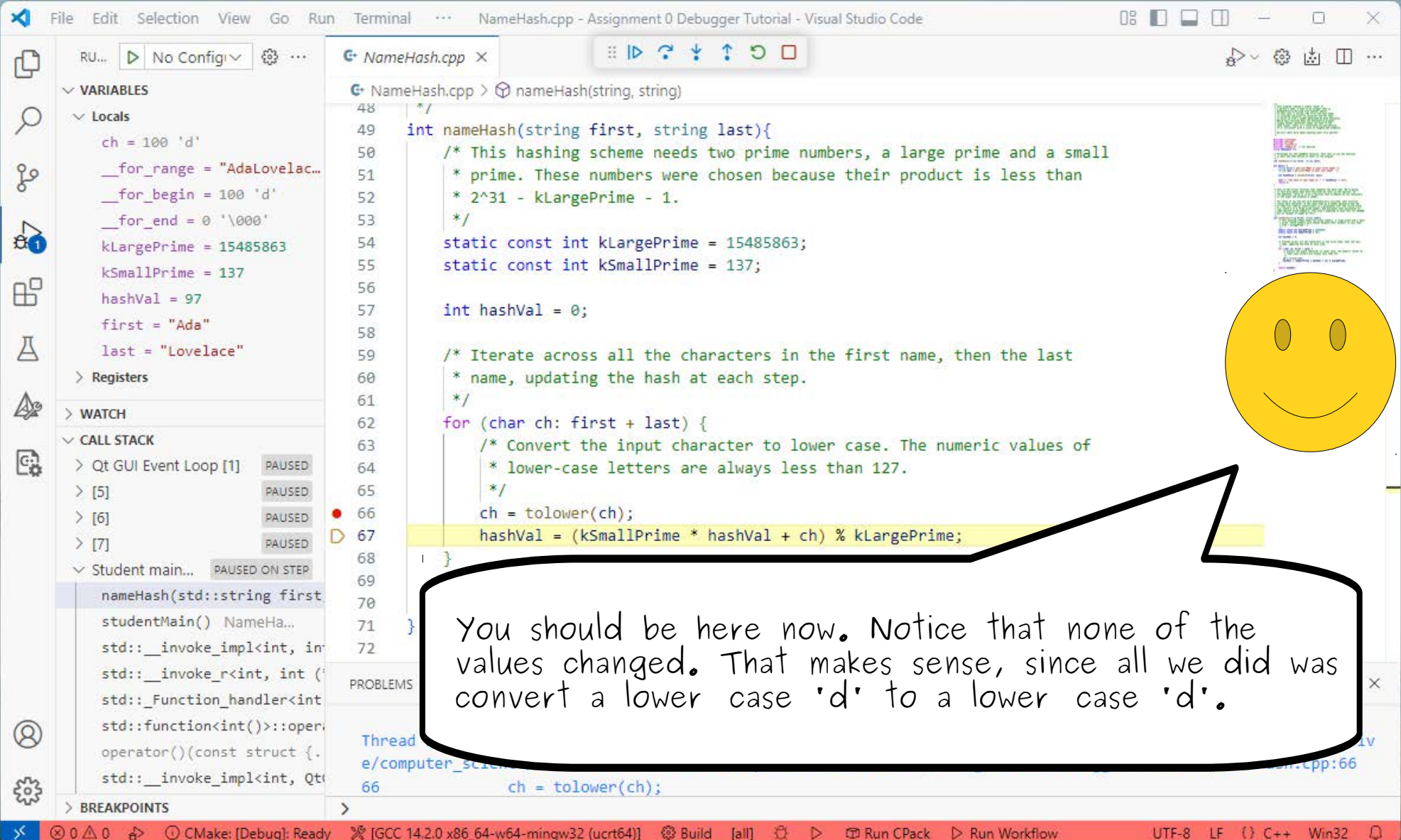


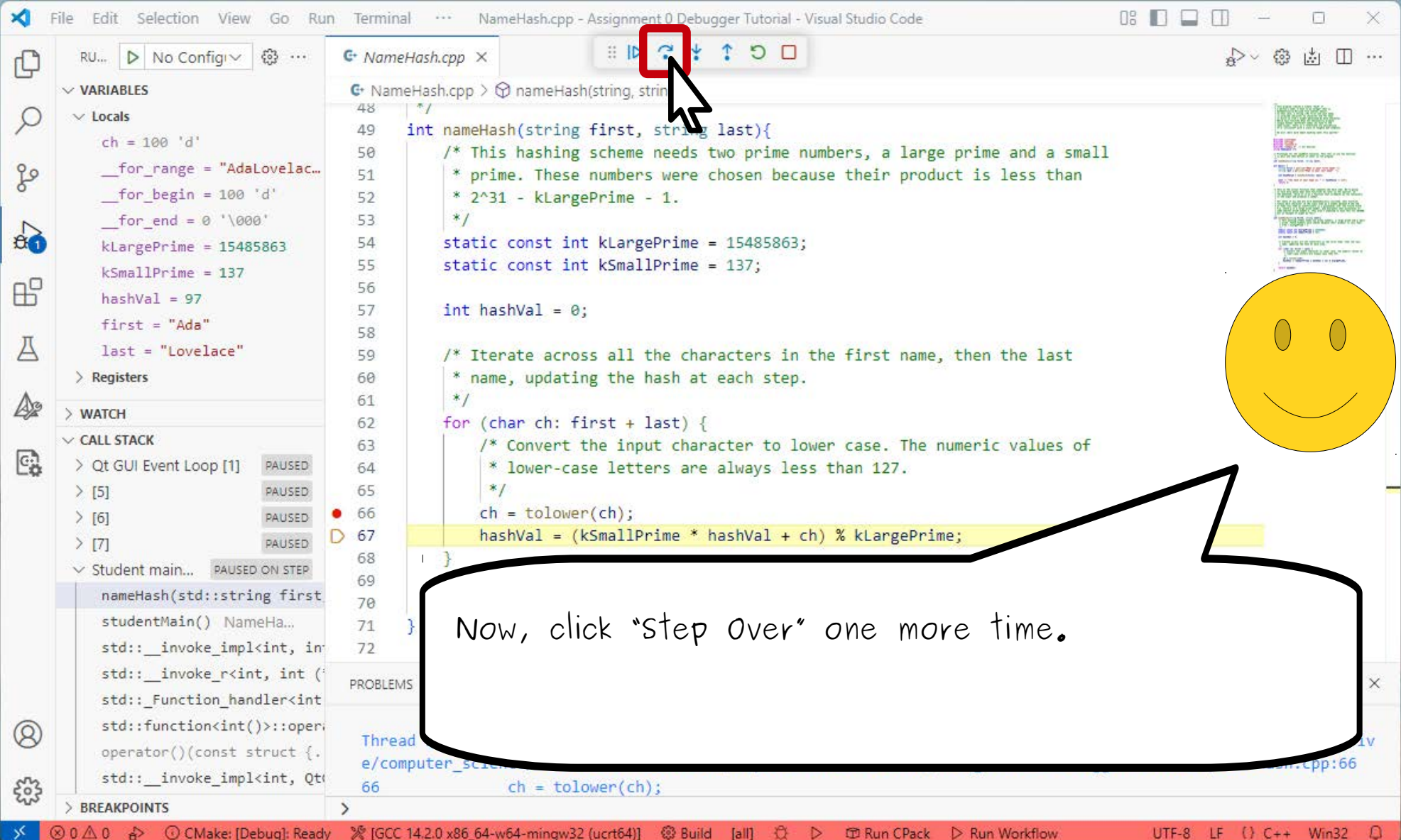


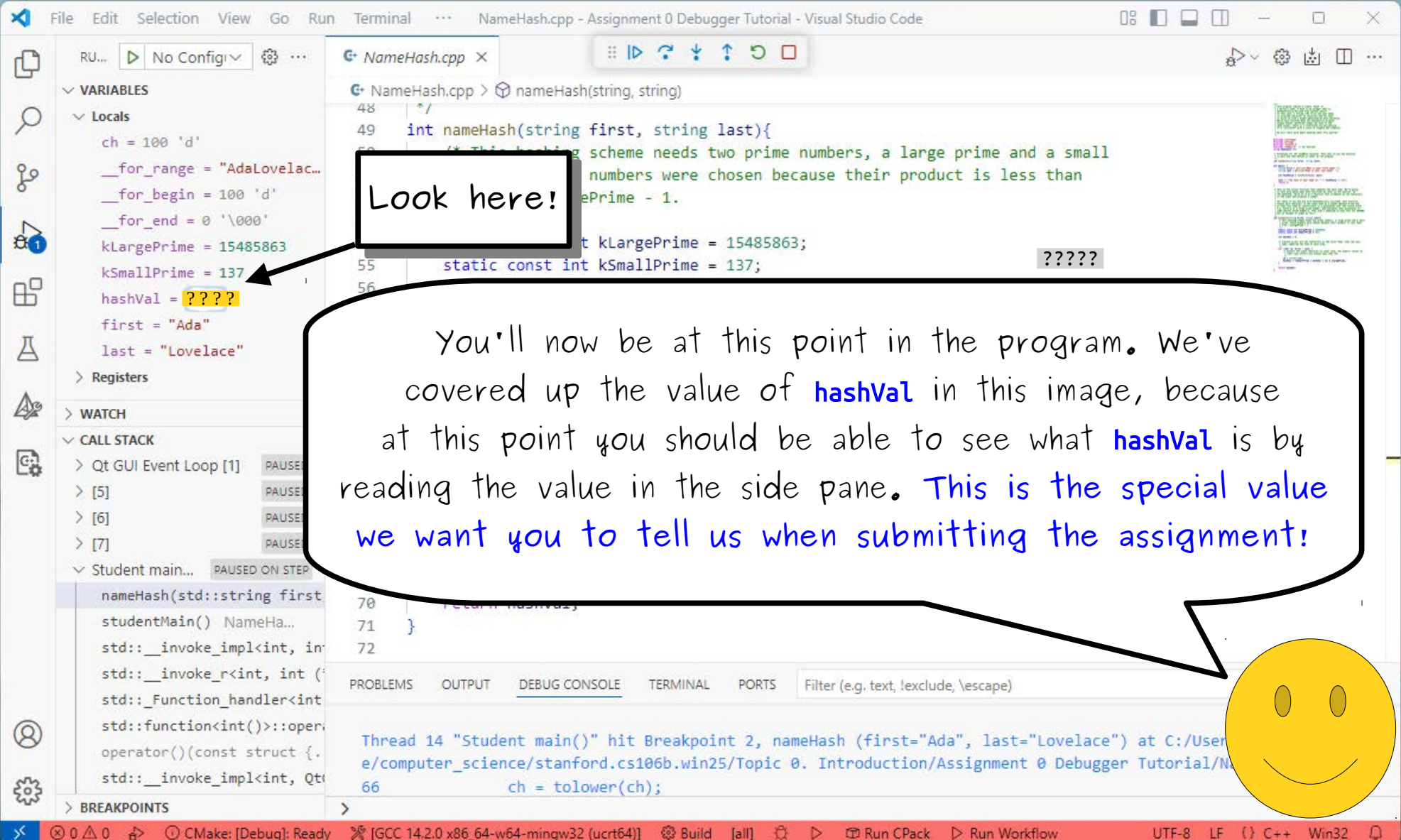


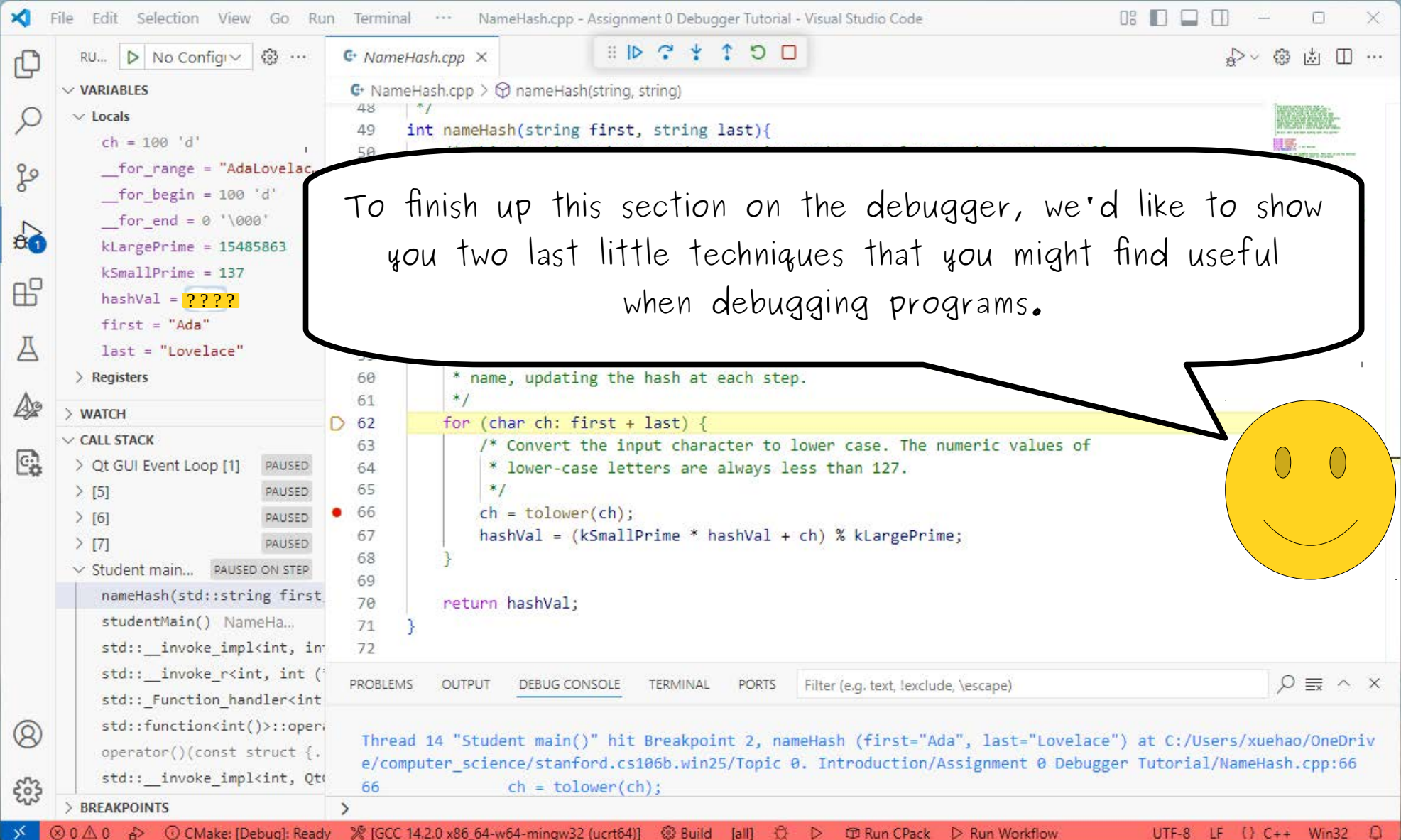




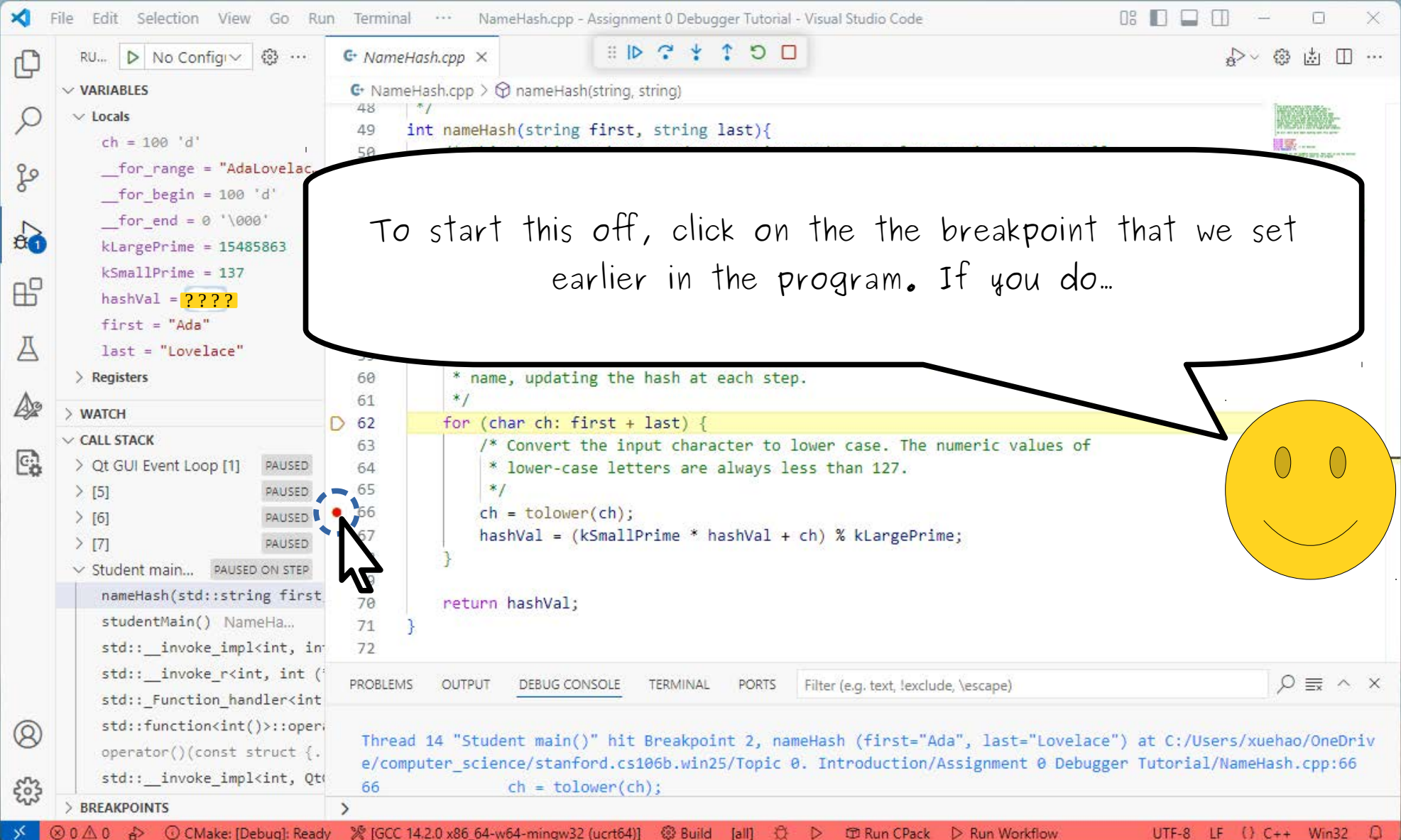




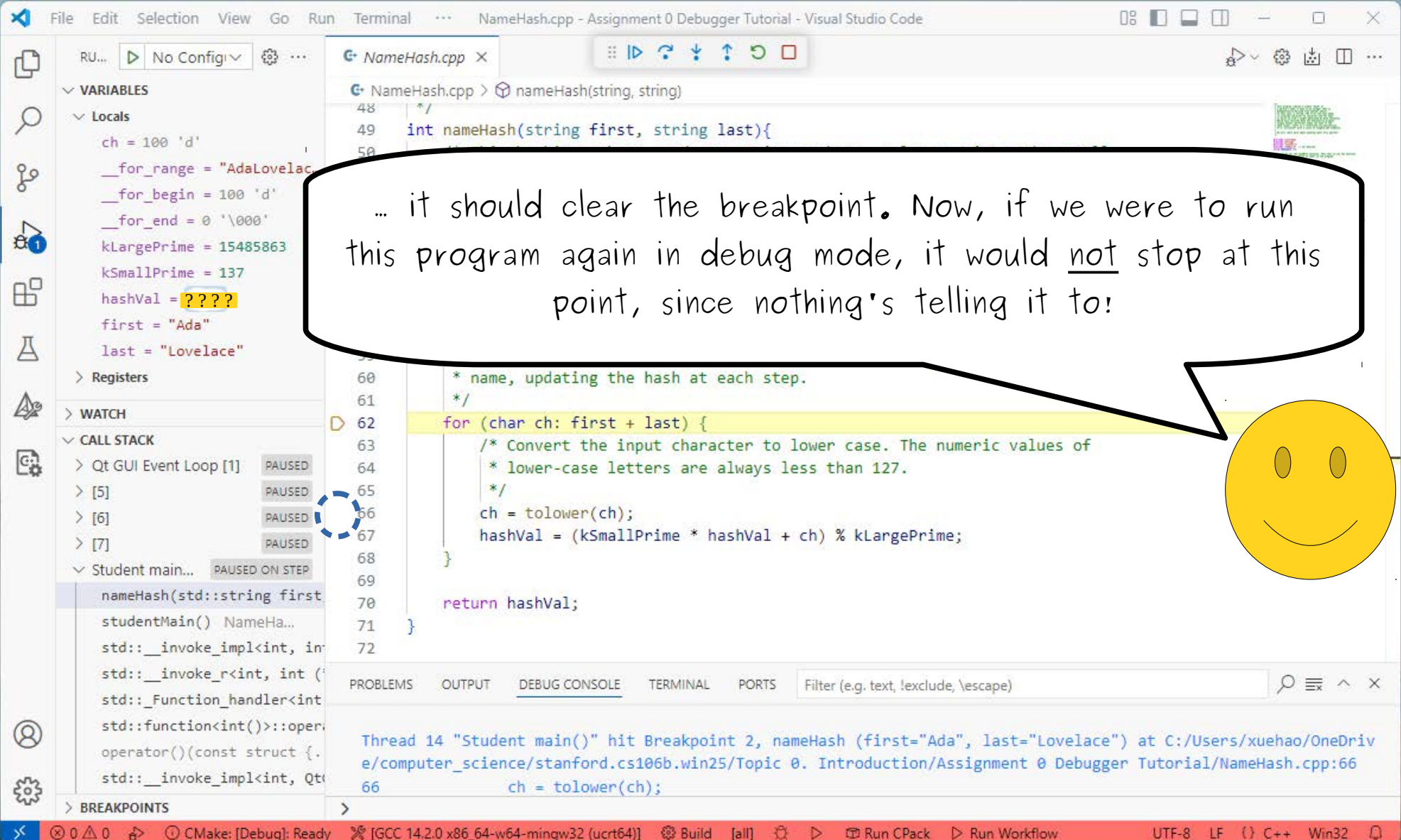




To finish up this section on the debugger, we'd like to show you two last little techniques that you might find useful when debugging programs.



To start this off, click on the the breakpoint that we set earlier in the program. If you do...



... it should clear the breakpoint. Now, if we were to run this program again in debug mode, it would not stop at this point, since nothing's telling it to!



RU... No Config

Variables

Locals

Registers

WATCH

CALL STACK

BREAKPOINTS

NameHash.cpp

48

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*/

int nameHash(string first, string last){

/* This hashing scheme needs two prime numbers, a large prime and a small

* prime. These numbers were chosen because their product is less than

* 2^31 - kLargePrime - 1.

*/

static const int kLargePrime = 15485863;

static const int kSmallPrime = 137;

int hashVal = 0;

/* Iterate across all the characters in the first name, then the last

* name, updating the hash at each step.

*/

for (char ch: first + last)

/*

ch = tolower(ch);

hashVal = (hashVal * kLargePrime + kSmallPrime) % (kLargePrime * kSmallPrime);

*/

return hashVal;

NAMEHASH.CPP

nameHash(string, string)

???

?????

???

Now, take a look back at these buttons.

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, \exclude, \escape)

Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66

66 ch = tolower(ch);

0 0 0 0 0

CMake: [Debug]: Ready

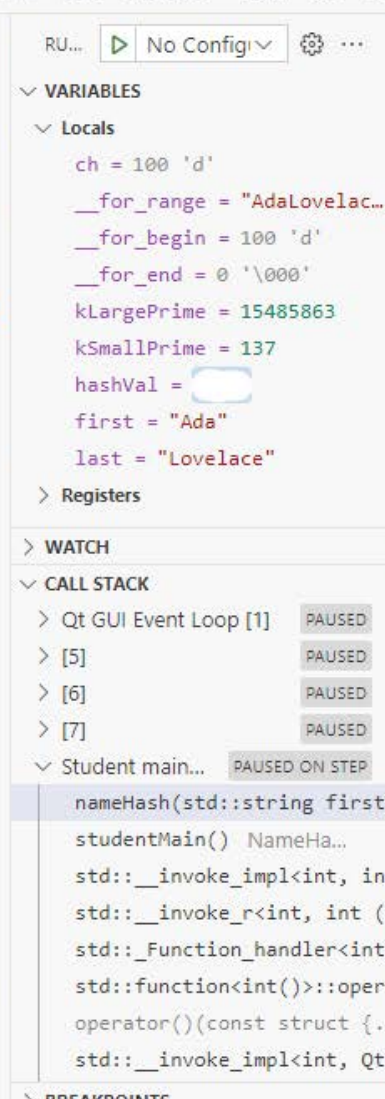
[GCC 14.2.0 x86_64-w64-mingw32 (ucrt64)]

Build [all]

Run CPack

Run Workflow

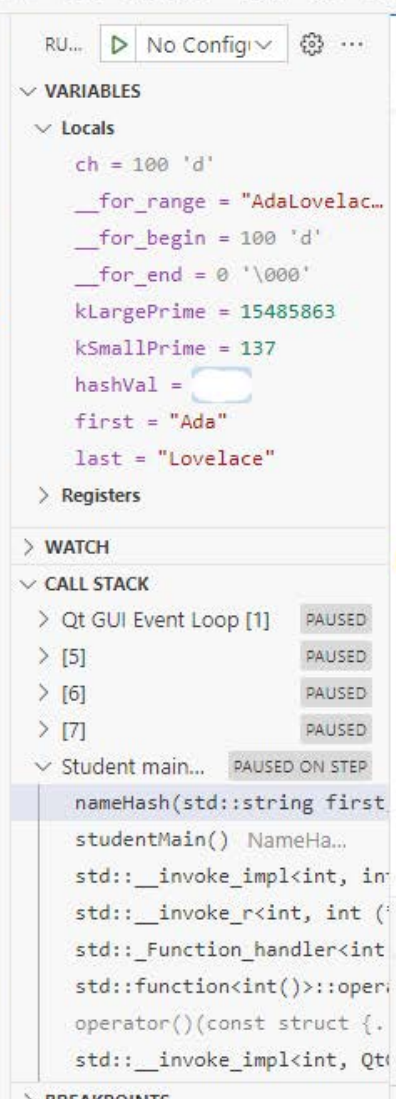
UTF-8 LF C++ Win32



100



```
Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66
66         ch = tolower(ch);
```

100

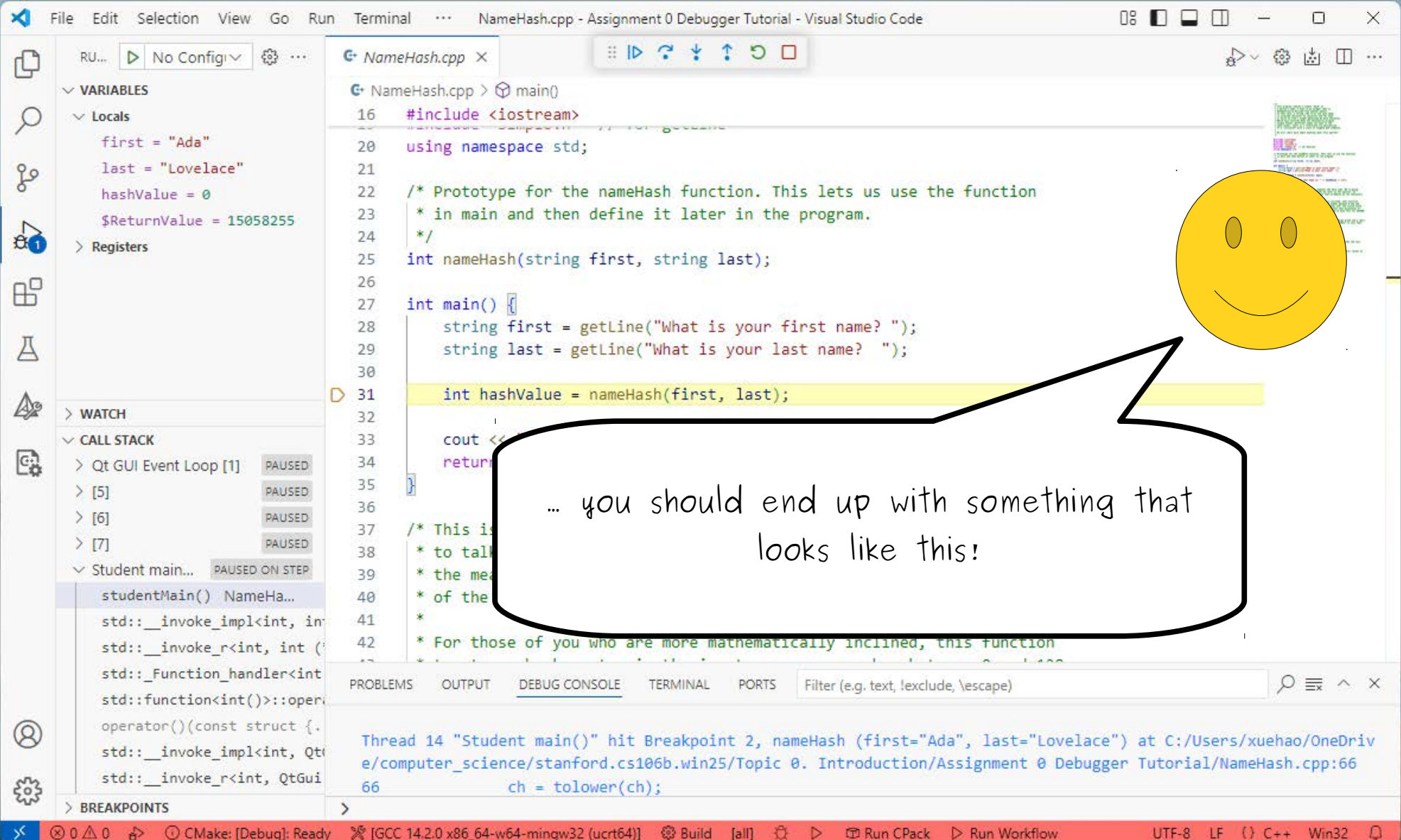
?????

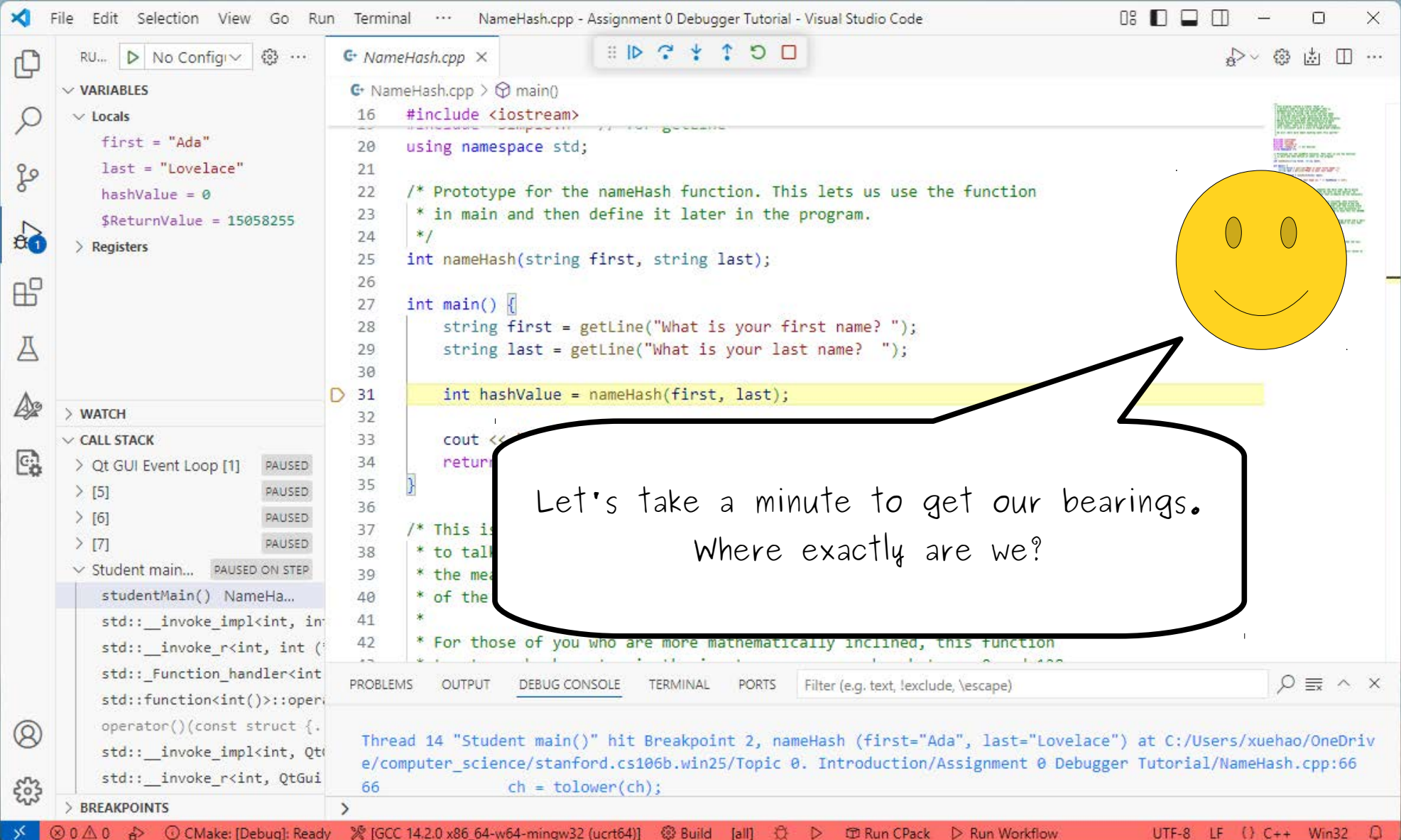


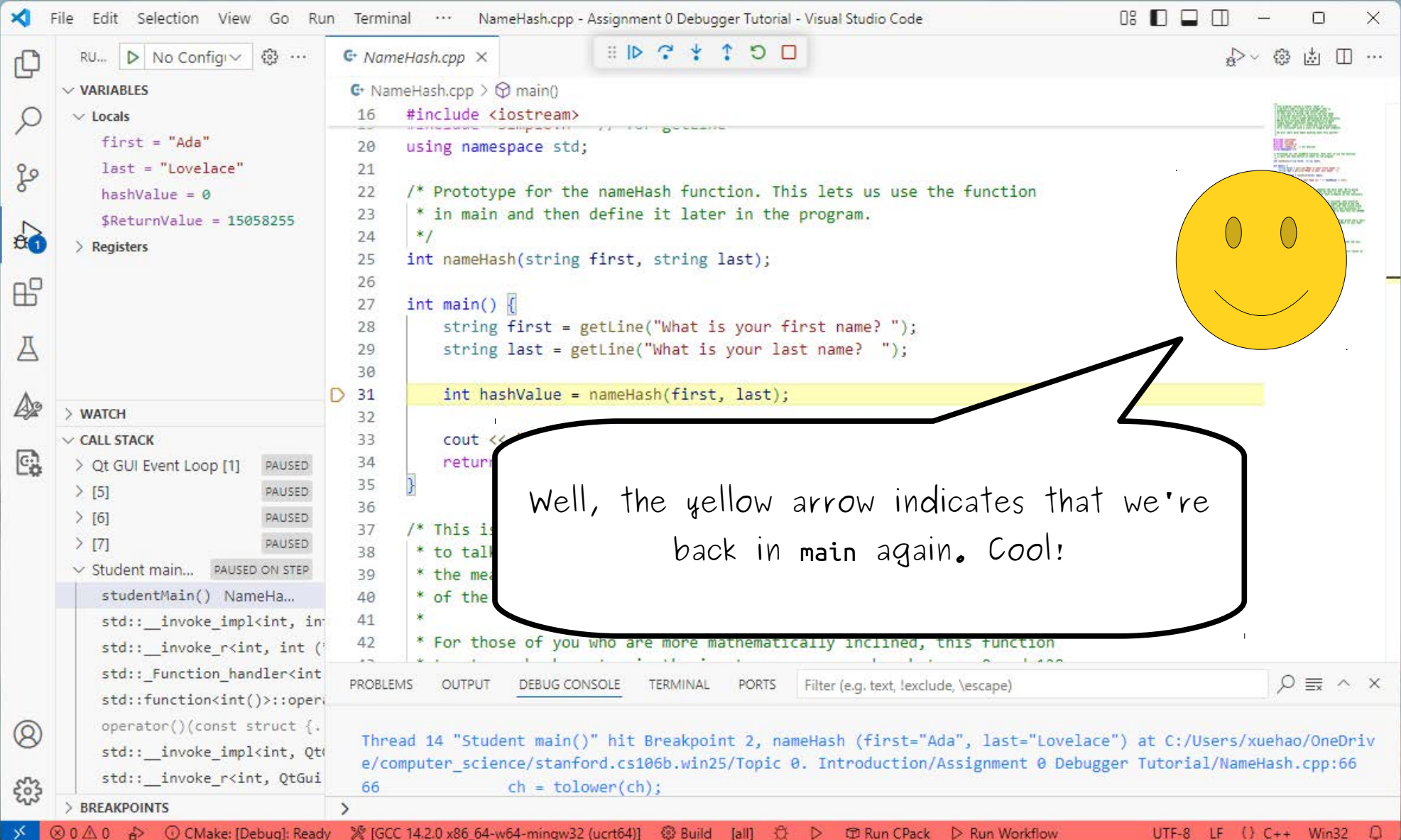
click,
ash
ns.

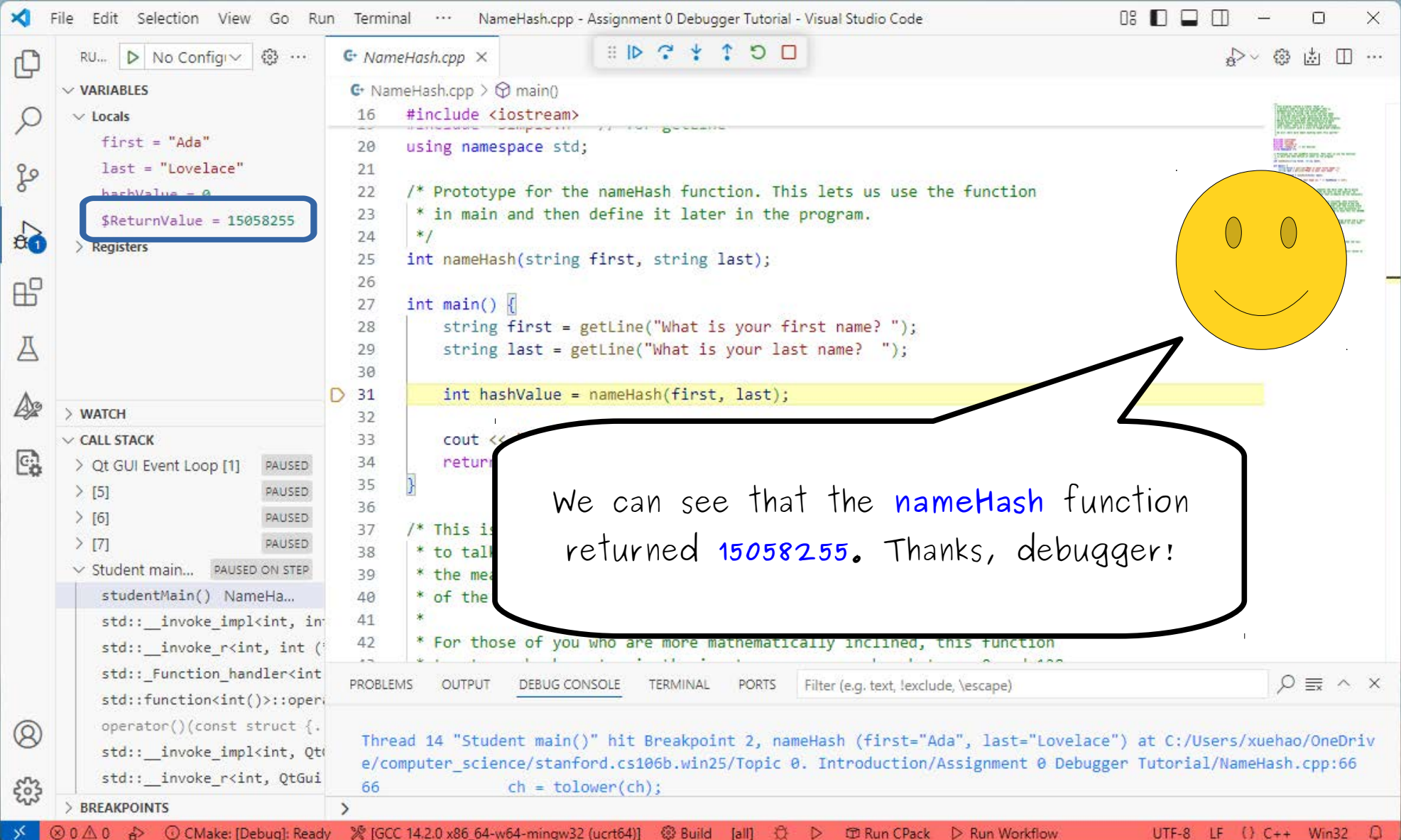


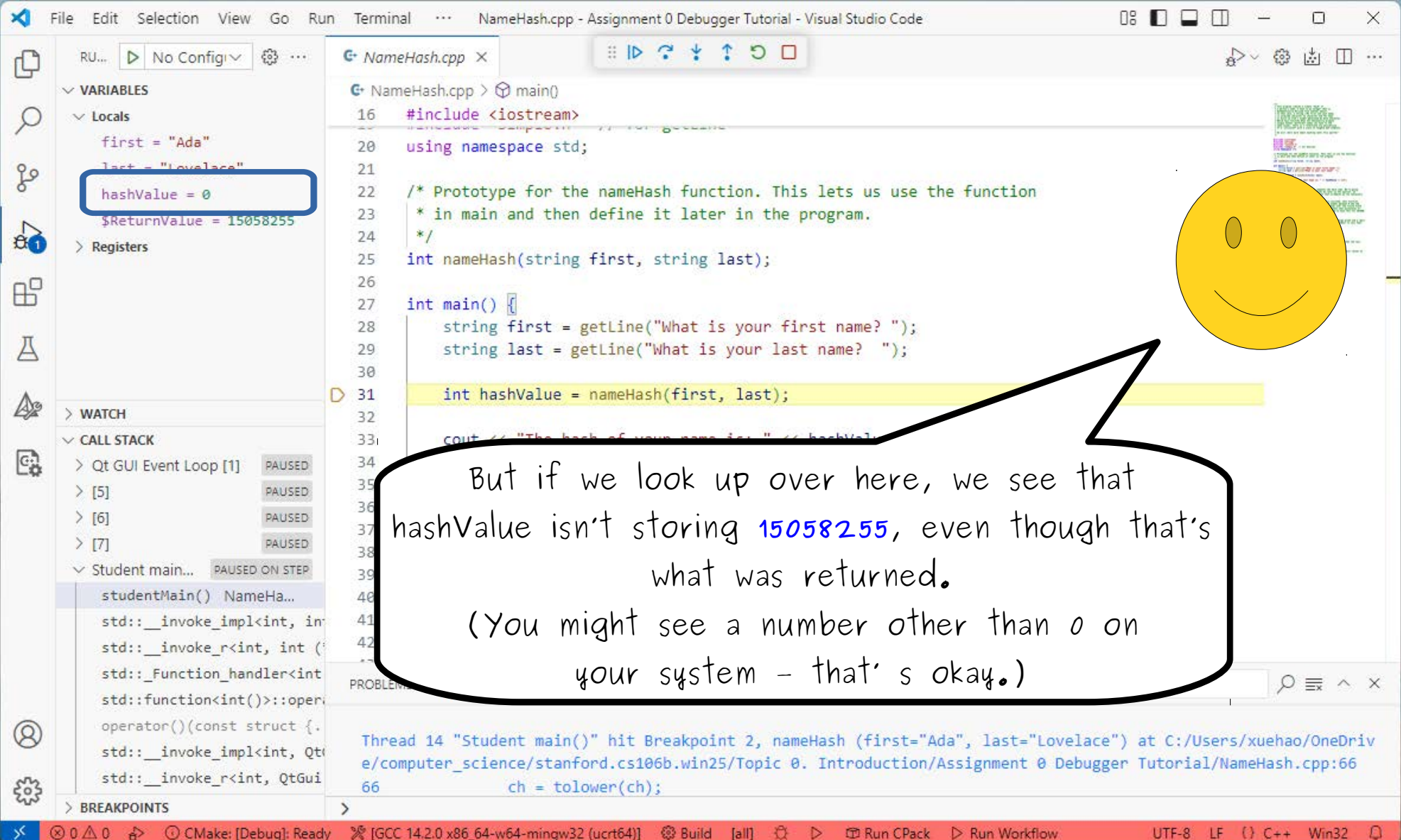
```
Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66
66         ch = tolower(ch);
```

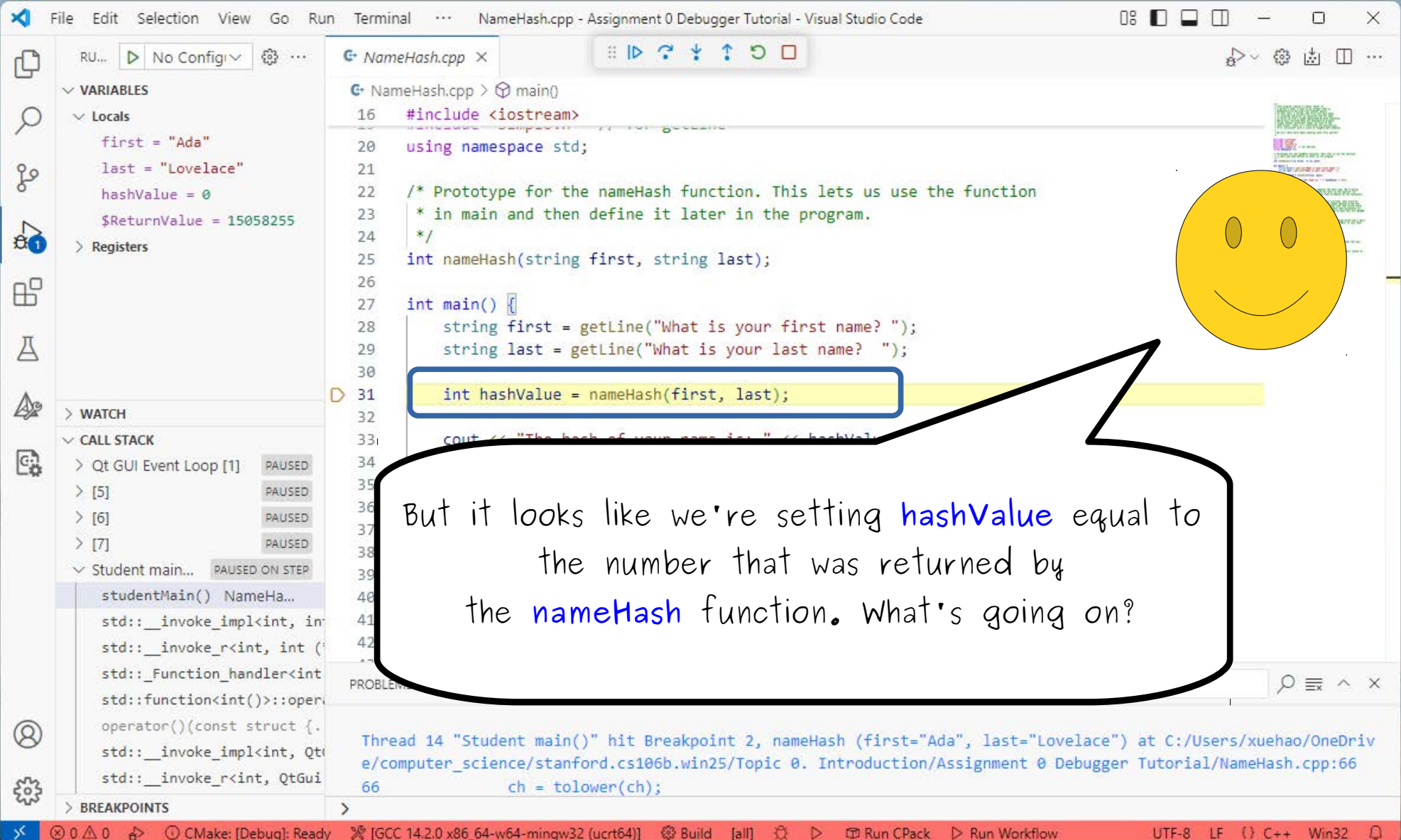









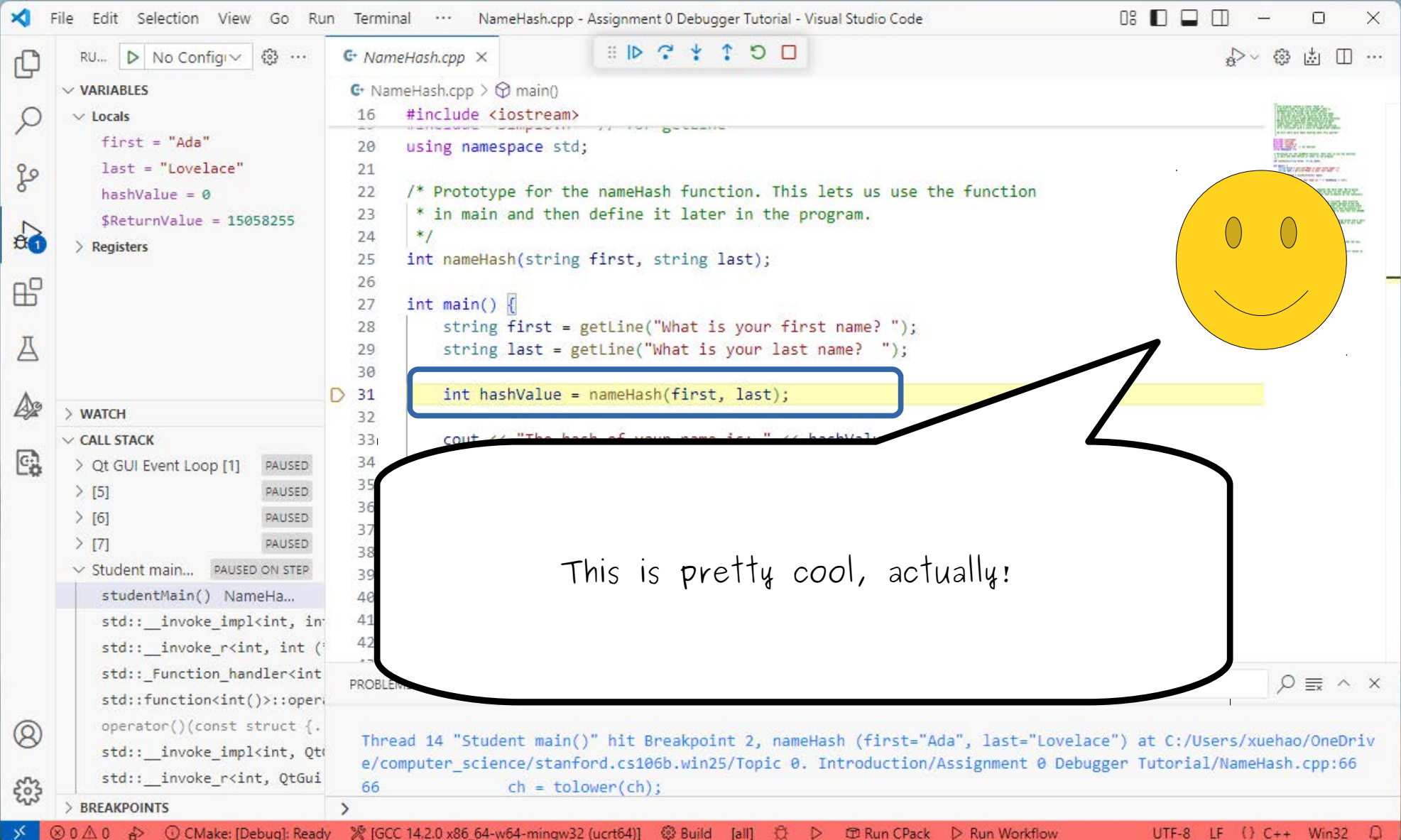


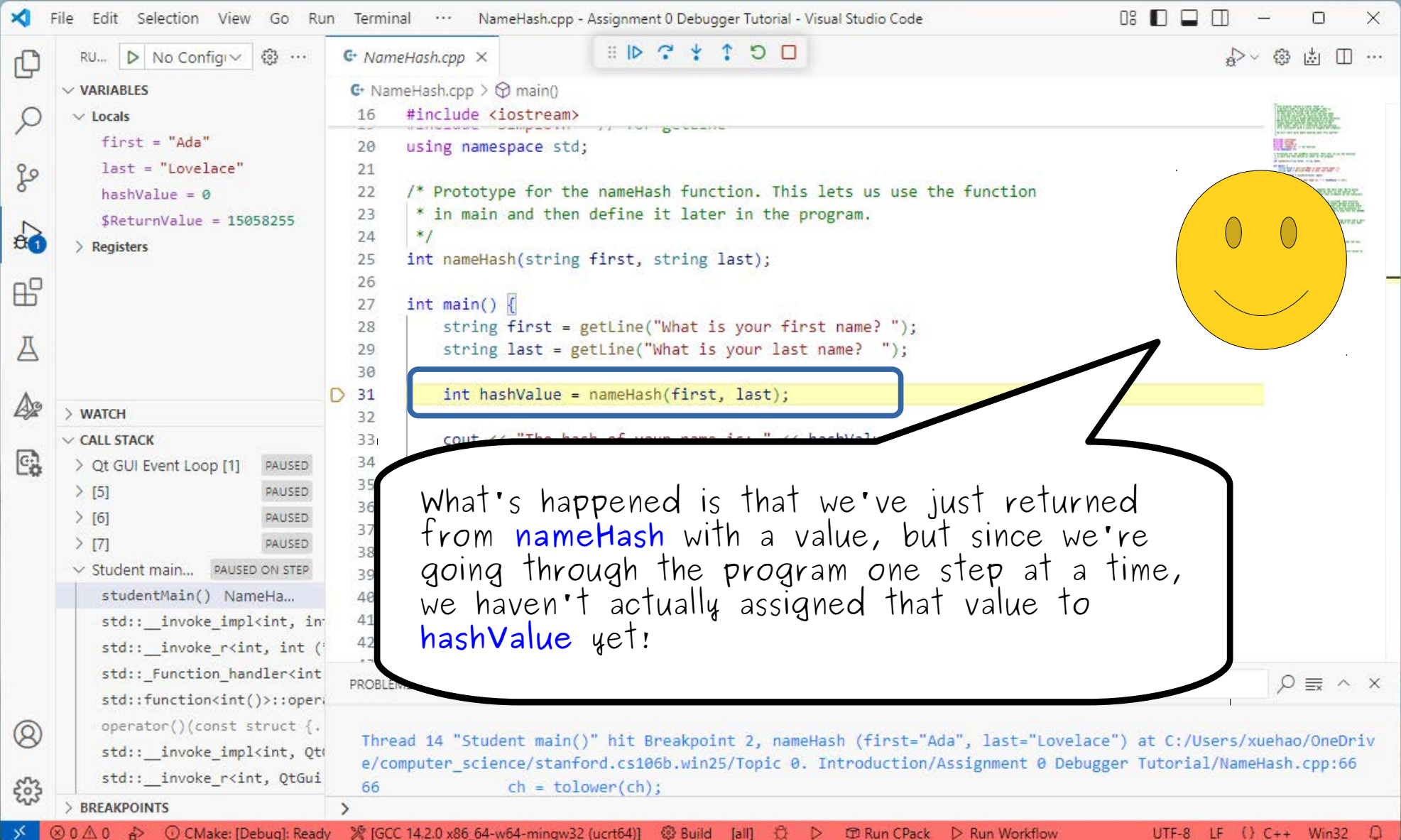


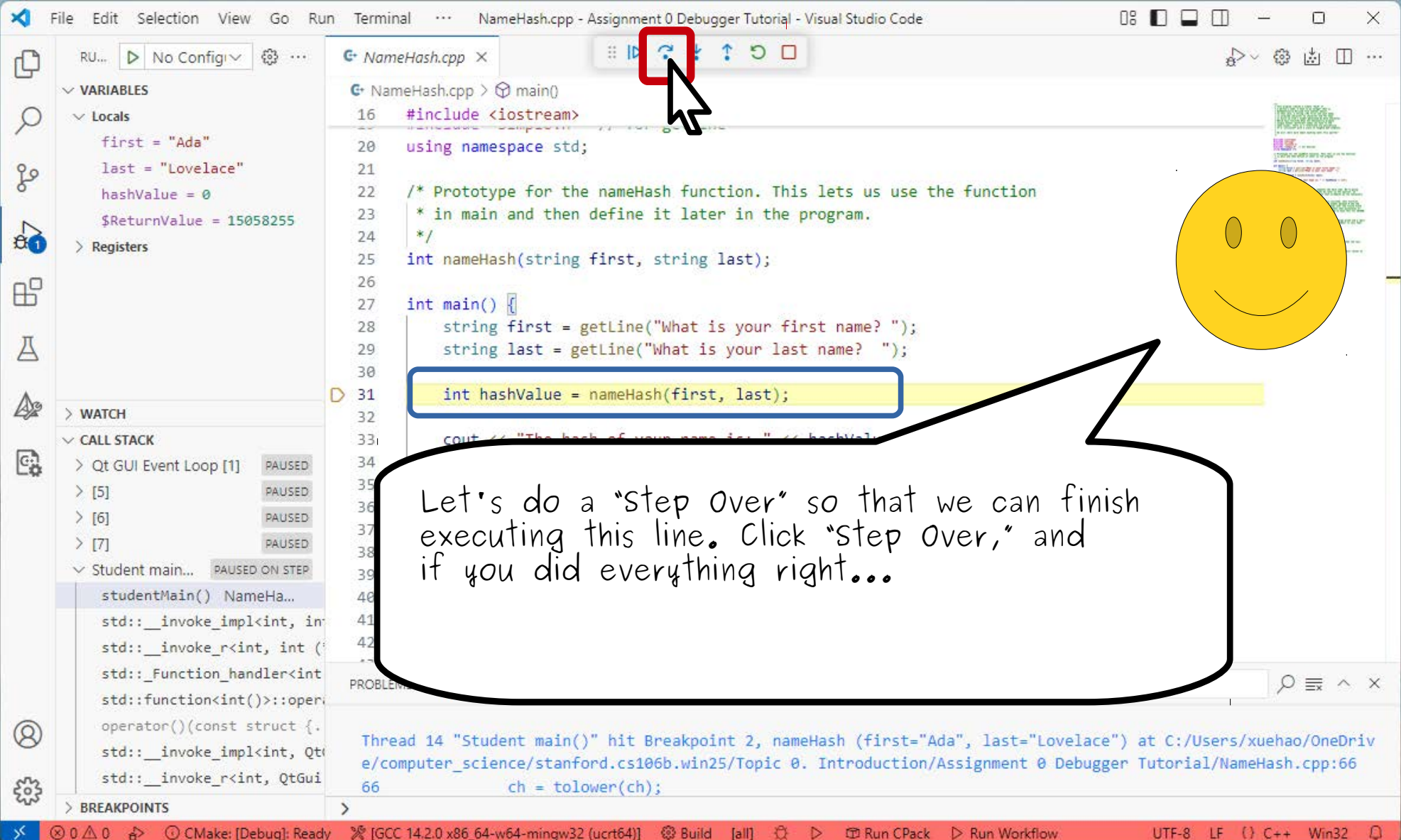
```
NameHash.cpp > main()
16 #include <iostream>
17 using namespace std;
18
19 /* Prototype for the nameHash function. This lets us use the function
20 * in main and then define it later in the program.
21 */
22 int nameHash(string first, string last);
23
24 int main() {
25     string first = getLine("What is your first name? ");
26     string last = getLine("What is your last name? ");
27
28     int hashValue = nameHash(first, last);
29
30     cout << "The hash of your name is: " << hashValue << endl;
31 }
```

But it looks like we're setting `hashValue` equal to the number that was returned by the `nameHash` function. What's going on?

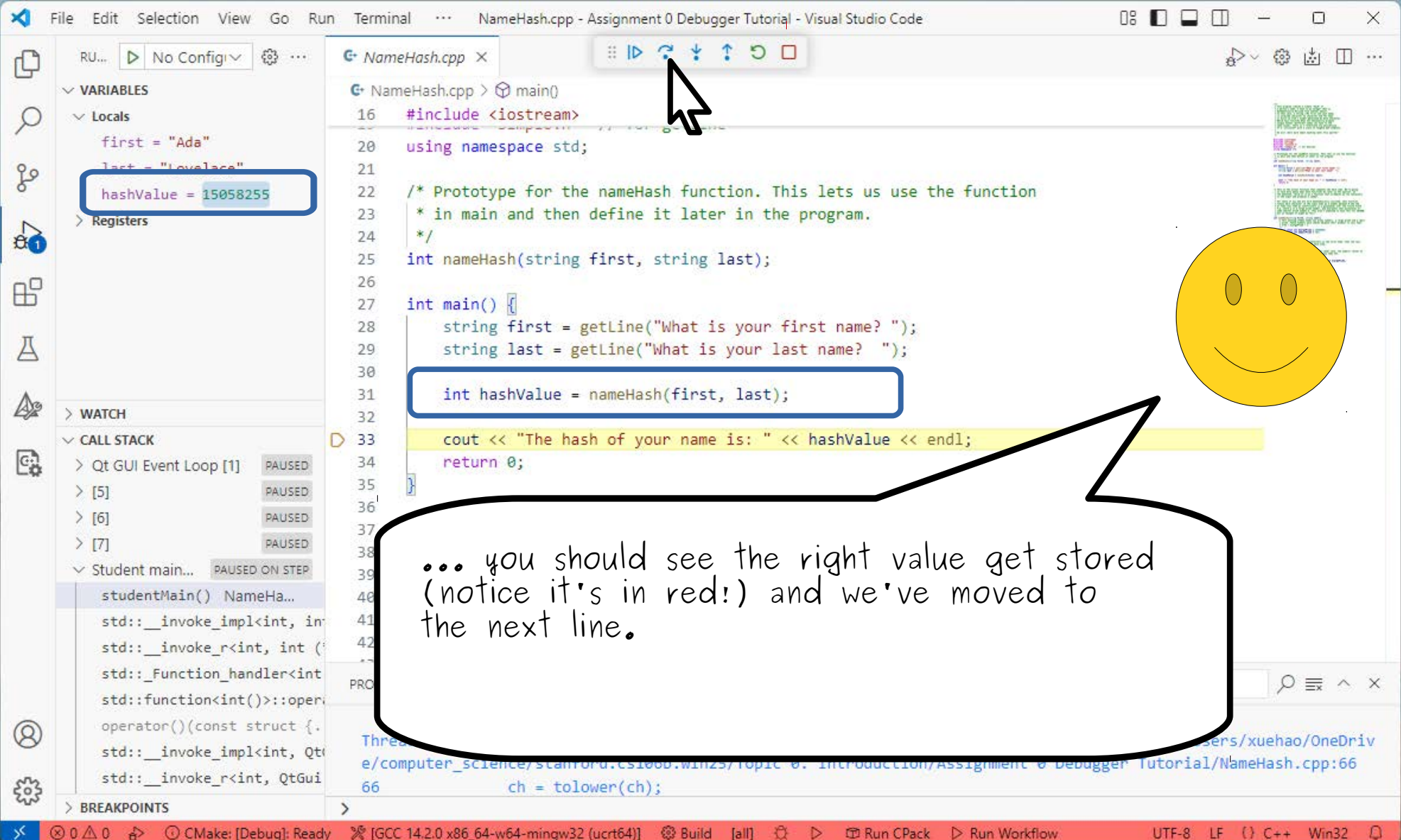
Thread 14 "Student main()" hit Breakpoint 2, nameHash (first="Ada", last="Lovelace") at C:/Users/xuehao/OneDrive/computer_science/stanford.cs106b.win25/Topic 0. Introduction/Assignment 0 Debugger Tutorial/NameHash.cpp:66
66 ch = tolower(ch);

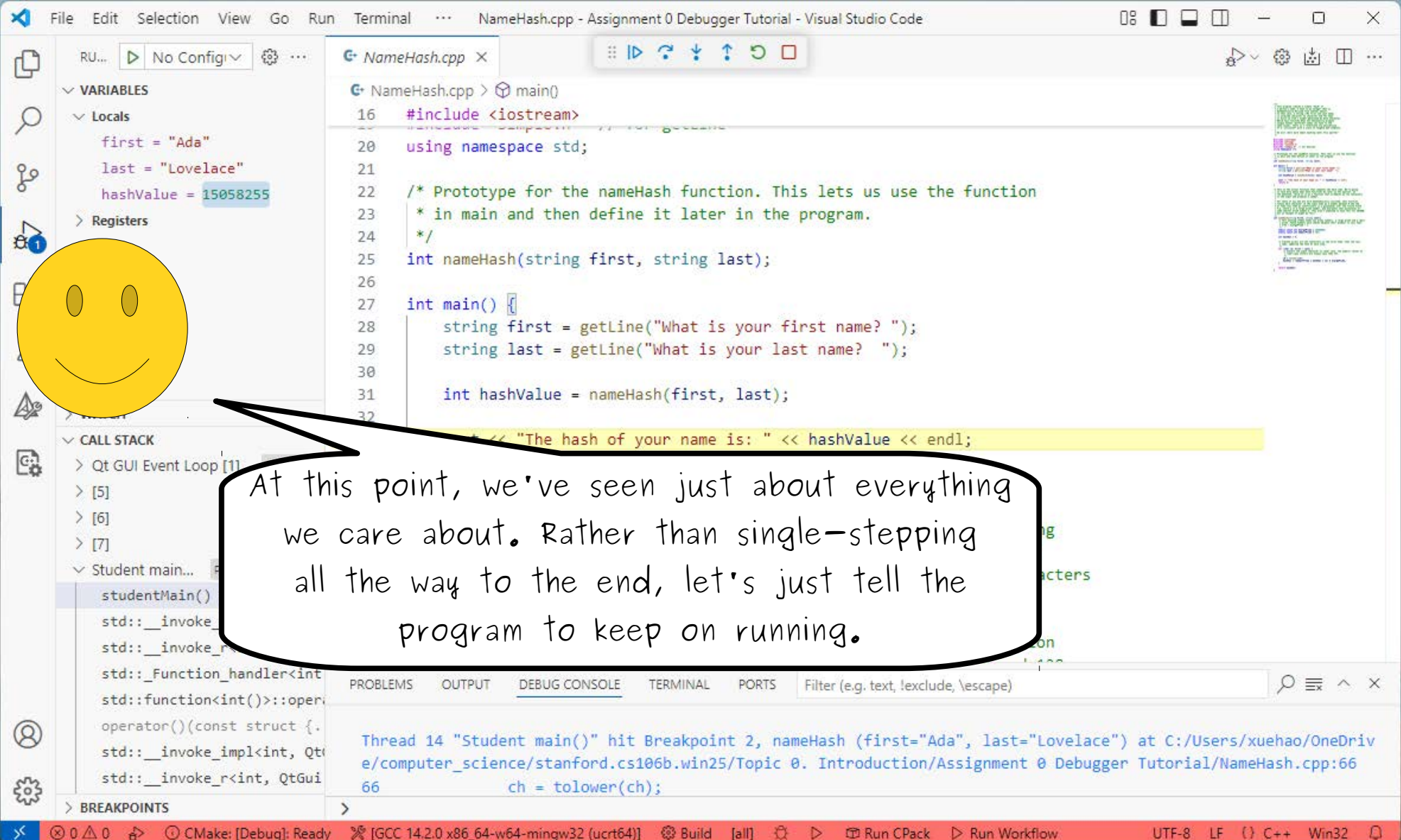


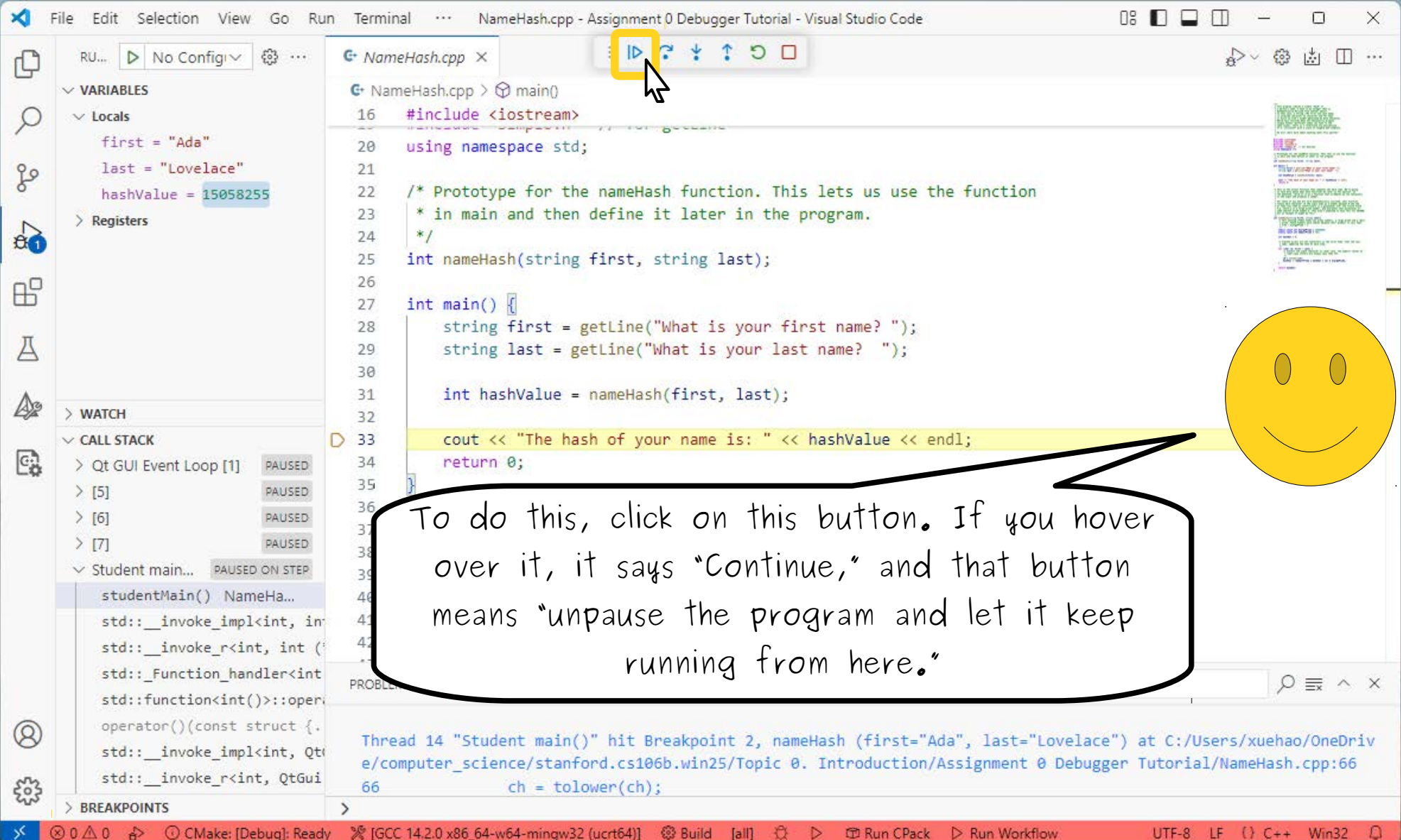


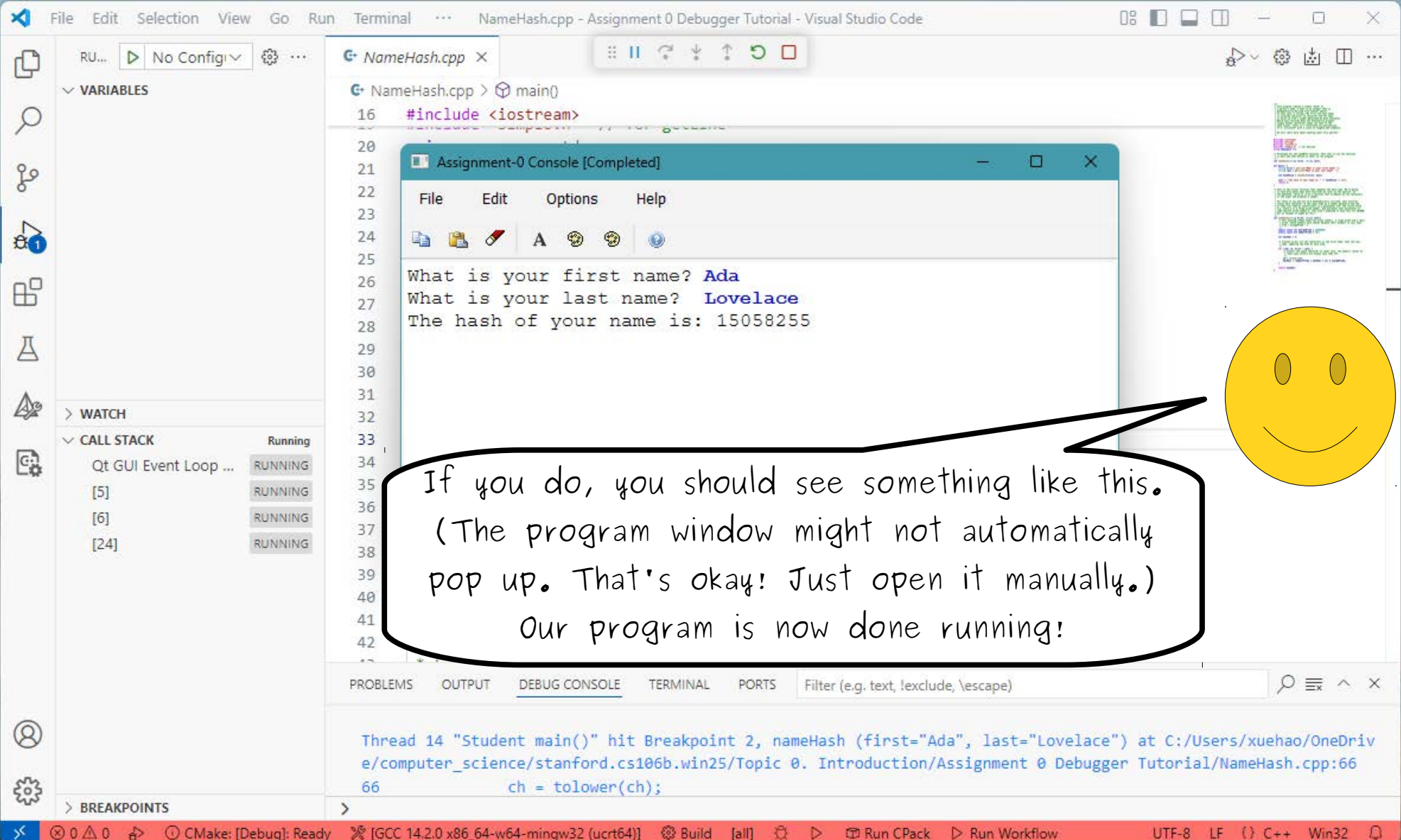


Let's do a "Step Over" so that we can finish executing this line. Click "Step Over," and if you did everything right...

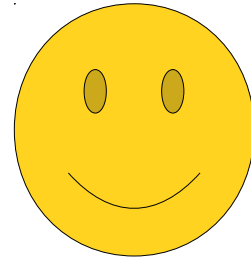




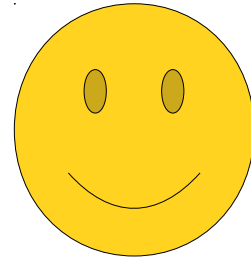




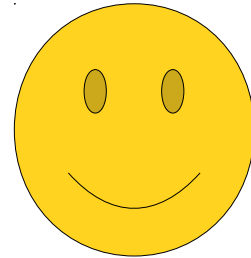
So there you have it! You've now gotten more familiar with the debugger!



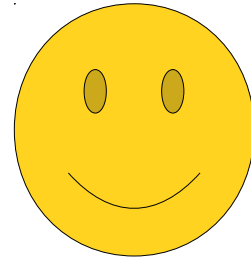
You know how to set a breakpoint to pause the program at a particular point.



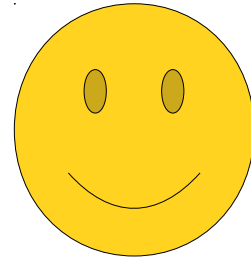
You know how to read the call stack and to
see the values of local variables.



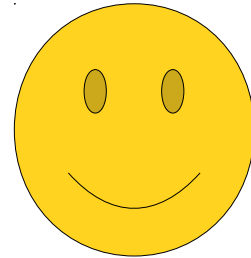
You know how to single-step the program and
see what values change.



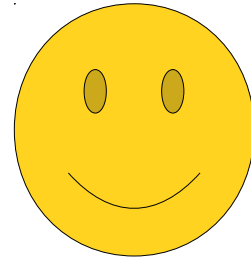
You know how to run a function to completion,
and how to let the program keep on running.



As you write more and more complicated programs this quarter, you'll get a lot more familiar using the debugger and seeing how your programs work.



And, if you continue to build larger and larger pieces of software, you'll find that knowing how to use a debugger is a surprisingly valuable skill!



Hope this helps, and welcome to CS101B!

